

RZ/N2L Group

Quick Start Guide: EtherCAT Slave Software

Introduction

This document explains Sample Program setup procedures for EtherCAT® slave functionalities with the adapted EtherCAT Stack Code for Renesas RZ/N2L platform. This describes steps to confirm slave behavior and stack features using TwinCAT® Master Configuration tool.

Target Device

RZ/N2L

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1. Overview

This document describes how to run EtherCAT on the RZ/N2L Group.

EtherCAT(Ethernet for Control Automation Technology) is an Ethernet based fieldbus system, developed by Beckhoff Automation. Development of EtherCAT was to apply Ethernet for automation applications (e.g., for motion control, I/O, sensors) requiring short data update times with low communication jitter and reduced hardware costs.

Tool to generate EtherCAT Slave Stack Code (SSC Tool) is available to the ETG members free of charge. This can be downloaded from the ETG website. SSC tool can be used to generate customized stack, device description files (ESI) and individual source code documentation to suit the developer's own needs.

This document describes the procedure for testing the EtherCAT slave function using EtherCAT stack code compatible with the Renesas RZ/N2L platform. Scope of the documentation is limited to explaining how to use the SSC tool for EtherCAT slave stack code generation and testing its behavior against TwinCAT masters and test applications.

1.1 Abbreviations/Definitions

Table 1.1 Abbreviations/Definitions

Index	Abbreviations /Definitions	Description
1	CoE	CAN application protocol over EtherCAT
2	EEPROM	Electrically Erasable Programmable Read-Only Memory
3	ESC	EtherCAT Slave Controller
4	ESI	EtherCAT Slave Information
5	FoE	File Access Over EtherCAT
6	I2C	Inter-Integrated Circuit
7	MB	Mail Box
8	PDO	Process Data Object
9	SSC	Slave Stack Code
10	EoE	Ethernet Over EtherCAT

1.2 Reference

Technical information about EtherCAT is available via ETG member site, and information about RZ/N2L is available via Renesas.

Table 1.2 Technical Inputs

Index	Document No.
1	r20ut4984 (RZ/N2L Group Renesas Starter Kit+ for RZ/N2L User's Manual)
2	r01uh0955 (RZ/N2L Group User's Manual: Hardware)
3	r01ds0397 (RZ/N2L Group Datasheet)
4	r01an6434 (RZ/T2, RZ/N2 Getting Started with Flexible Software Package)

2. Features

EtherCAT slave stack code generated by SSC Tool provides the functionality of EtherCAT slave controller.

Includes the following features :

- ESM (EtherCAT State Machine)
- Mailbox protocols:
 - CoE (CAN application protocol over EtherCAT)
- Synchronization Modes:
 - Free Run
 - Sync Manager Synchronization
 - DC Synchronization
- I/O function:
 - I/O Input DIP SW
 - I/O Output LED
- CiA402 Drive Profile:
 - csp Mode
 - csv Mode
- File Access over EtherCAT Profile:
- Semi Device Profile:
 - Common Device Profile



EtherCAT is a registered trademark and patented technology, licensed by Beckhoff Automation GmbH, Germany.

About the old documents

This application note is the document which integrated the following old documents.

- RZ/N2L Group Quick Start Guide: EtherCAT Slave Software (Document No. r01an6178)
- RZ/N2L Group Quick Start Guide: EtherCAT CiA402 Slave Software (Document No. r01an6522)
- RZ/N2L Group Quick Start Guide: File Access over EtherCAT Slave Software (Document No. r01an6890)

The old documents will no longer be updated, this application note will be updated.

3. Project Setup

3.1 Requirements

Table 3.1 Requirements

Item	Description
Board	Renesas Electronics - Renesas Starter Kit+ for RZ/N2L - CPU clock is 400MHz.
IDE	IAR Systems - IAR Embedded Workbench® for ARM Version 9.50.1 Renesas Electronics - RZ/N2 FSP with e² studio 2024-01 (FSP versions of 2.0.0) - RZ/N2 FSP Smart Configurator (FSP versions of 2.0.0)
Emulator	IAR Systems - I-jet SEGGER - J-Link
SSC Tool	Slave Stack Code (SSC) Tool Version 5.13
Software PLC	Beckhoff Automation - TwinCAT3

Note). EEPROM capacity of the following applicable products is different.

If applicable, refer to "9.4 Appendix D : EEPROM capacity setting change" and change the setting.

Product name: Renesas Starter Kit+ for RZ/N2L

Product model name: RTK9RZ/N2L0S00000BE

Serial number other than "241783~241882, 243167~243266"

3.2 Hardware

This document describes the major hardware. Please refer to board user's manual and schematic for more board details.

Device connection between EtherCAT master and slave is shown below.

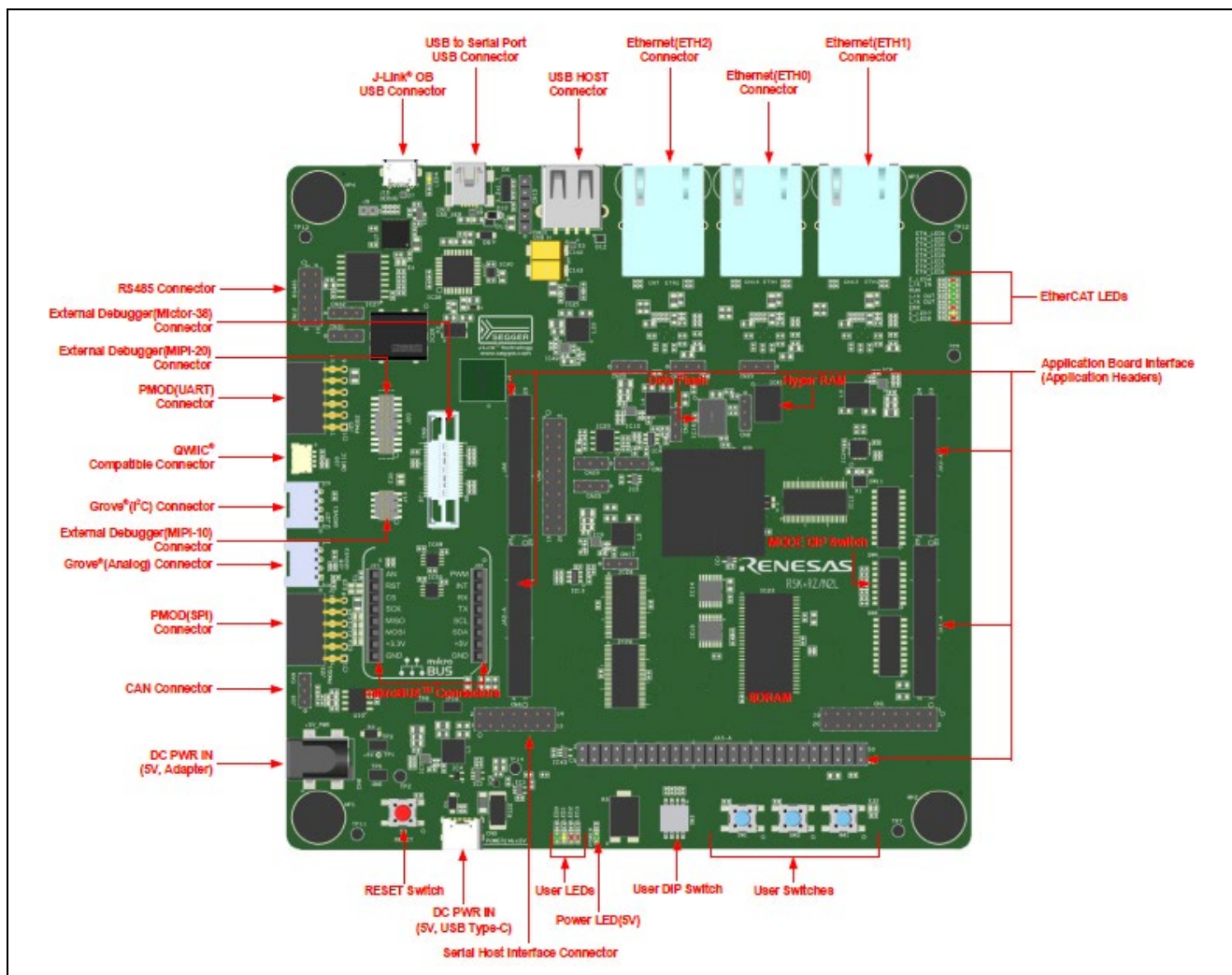


Figure 3.1 : RZ/N2L RSK board layout

Table 3.2 Jumper Pin settings

Reference	Jumper Position	Description
CN8	Short 2-3	Use QSPI Serial Flash
CN17	Short 2-3	VCC1833_2 Use power supply at 1.8V
CN20	Short 1-2	Use 3 ports in same PHY mode
CN21	Short 1-2	Use 3 ports in same PHY mode
CN22	Short 1-2	Use 3 ports in same PHY mode
CN24	Short 2-3	VCC1833_2 Use power supply at 1.8V
CN29	Short 1-2	Use the USB serial signal.

Table 3.3 Switch4 settings

SW4	Setting	Description
SW4-1	ON	xSPI0 boot mode (x1 boot Serial flash)
SW4-2	ON	* Refer to the Document No."r20ut4984" and set according to the mode to be used.
SW4-3	ON	
SW4-4	ON	MDD=0, JTAG Authentication by Hash is disabled.
SW4-5	OFF	Not used
SW4-6	OFF	Valid signals other than trace signals
SW4-7	ON	Signals other than the external bus are valid
SW4-8	OFF	SW3 is valid

Table 3.4 Switch8 settings

SW8	Setting	Description
SW8-1	OFF	Enable the "LED_GREEN" signal.
SW8-2	ON	
SW8-3	OFF	
SW8-4	ON	Enable the "LED5" signal.
SW8-5	OFF	

Table 8.5 Switch11 settings

SW11	Setting	Description
SW11-1	ON	Enable the "LED_RED2" signal.
SW11-2	OFF	
SW11-3	OFF	

Note). The settings which is not written is not used in this sample.
There is no problem with default settings.

Build and run the sample code on the RZ/N2L RSK board by following the steps below.
Both loading into RAM and flash can be done using IAR Embedded Workbench or e² studio.

1. Connect the decoder to the header "J20" on the RZ/N2L RSK board."

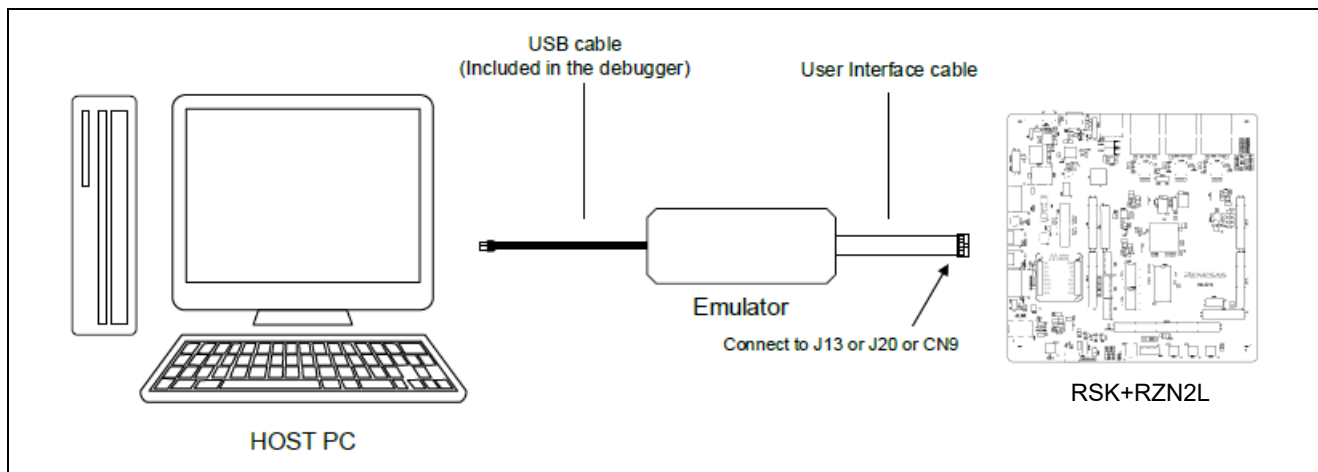


Figure 3.2: RZ/N2L RSK board debug connection diagram

When using J-Link OB, connect the USB cable to the header "J10" and set "J9" to open.

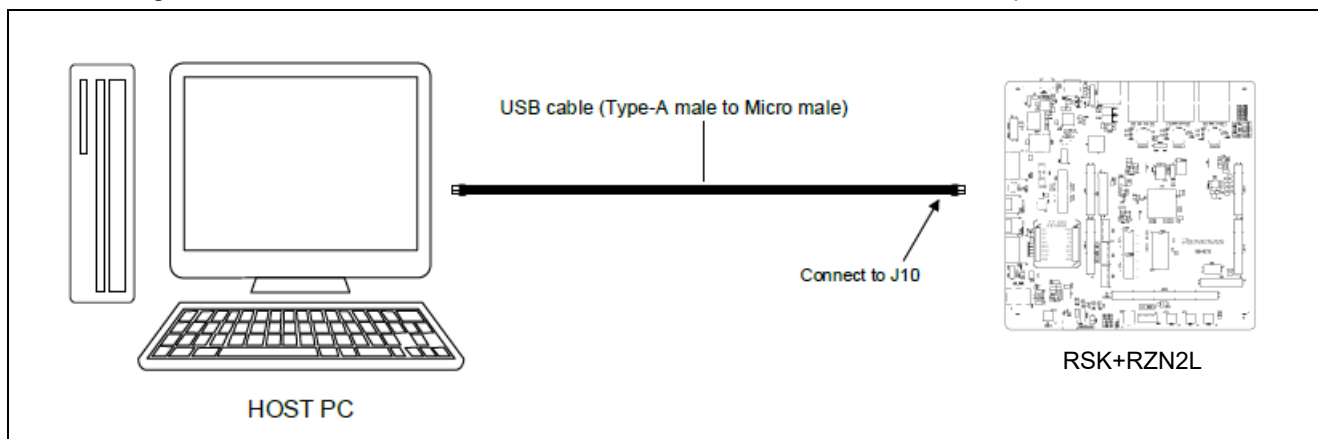


Figure 3.3: RZ/N2L RSK board debug connection diagram (J-Link OB)

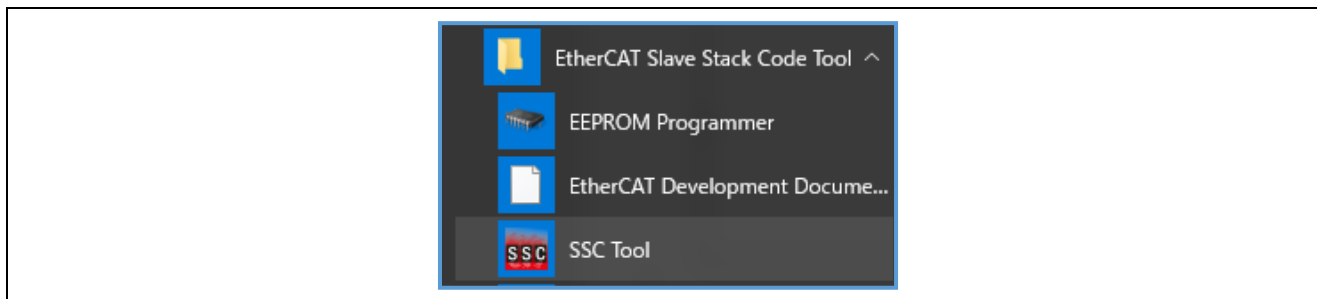
2. Power is supplied using a USB cable (Type-C) or an AC / DC adapter. When using a USB cable (Type-C), connect it to the USB connector "CN5" of the RZ/N2L RSK board. When connecting the AC/DC adapter, connect it to the connector "CN6" of the RZ/N2L RSK board.
3. Connect host PC and RZ/N2L RSK board using an ethernet cable. When using an ethernet cable, connect the ethernet cable into "ETH0" connector of the RZ/N2L RSK board.

3.3 Generating the Slave Stack Code

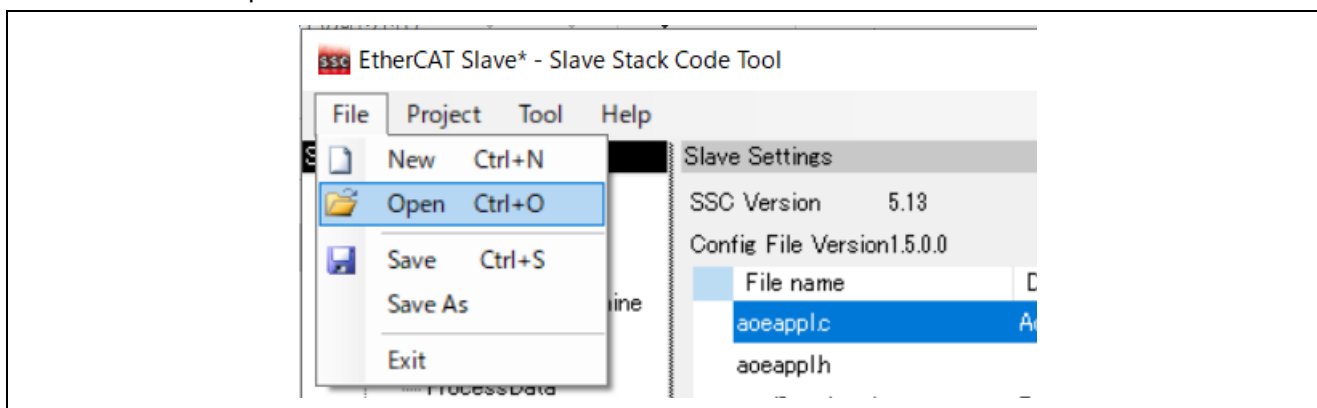
SSC Tool is used for generating the slave stack code.

Note). Replace the folder name in the following description according to the tool to be used.

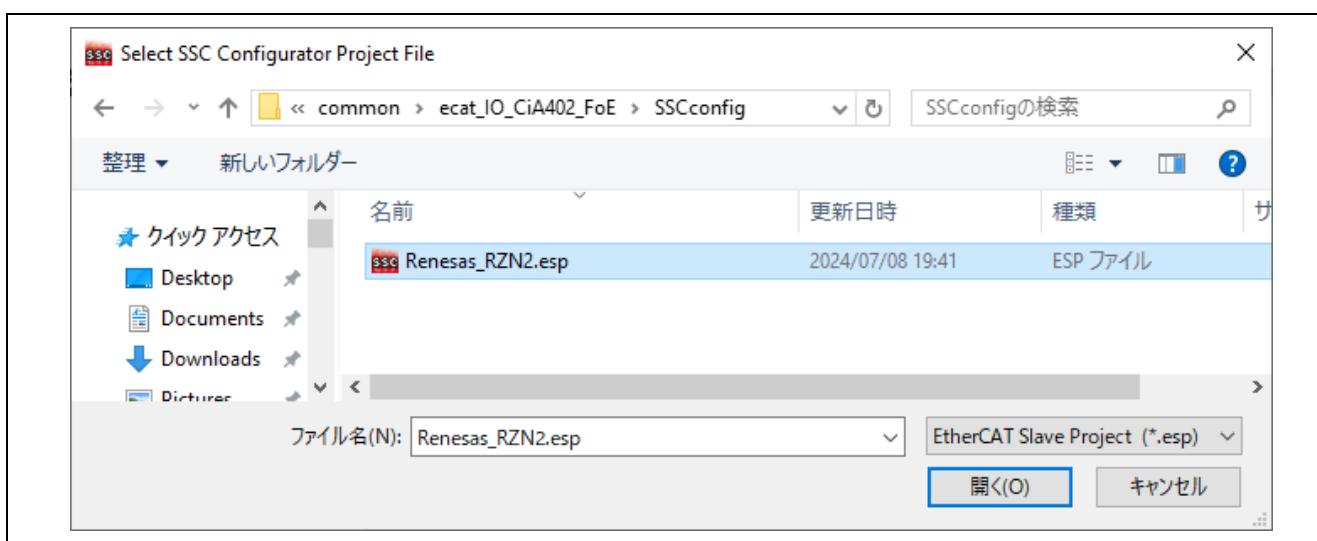
1. Start the SSC Tool from the Window Start menu.



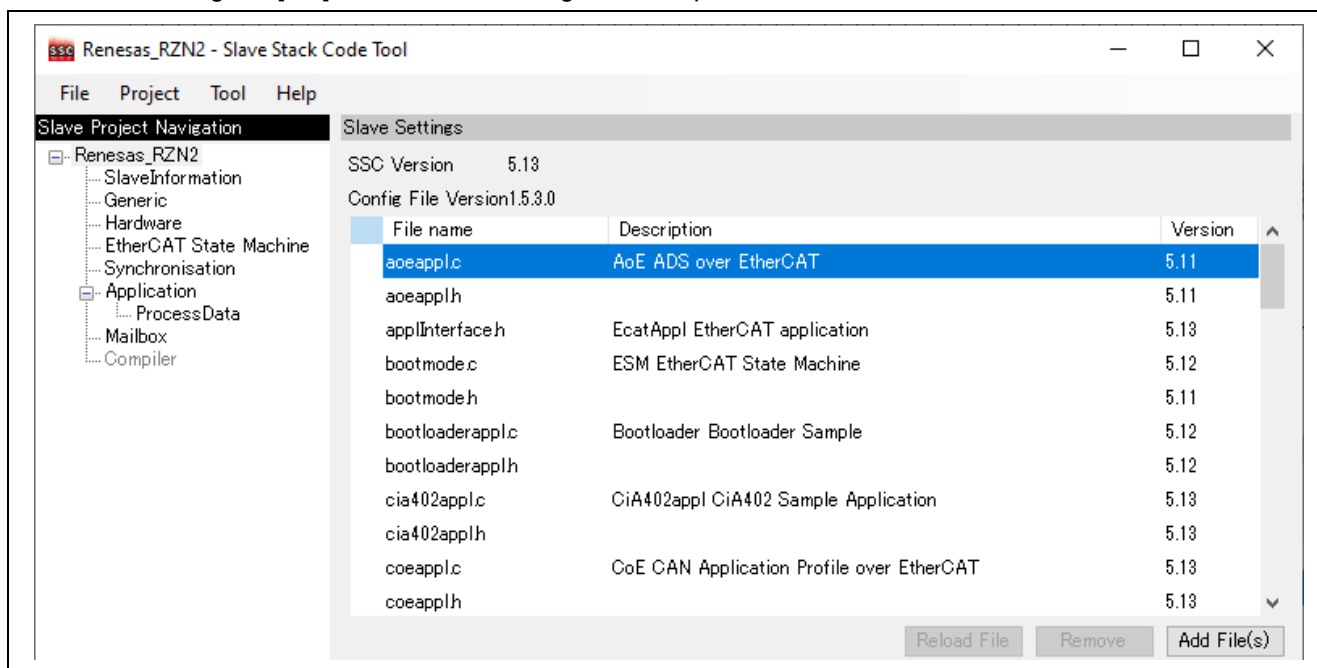
2. Select File > Open.



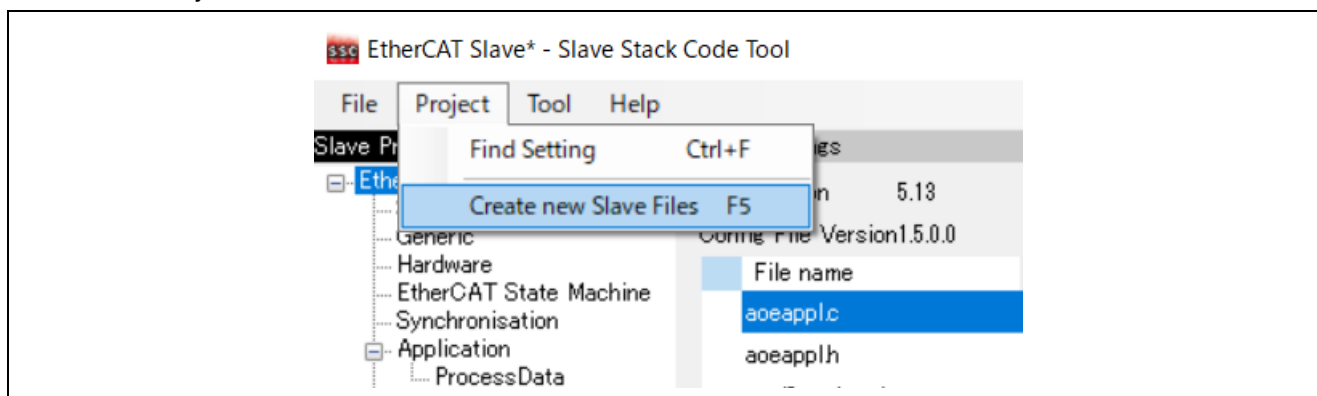
3. Select the following file,
 “RZN2L_EtherCAT_RSK_rev0200\common\ecat_IO_CiA402_FoE\SSCconfig\Renesas_RZN2.esp”



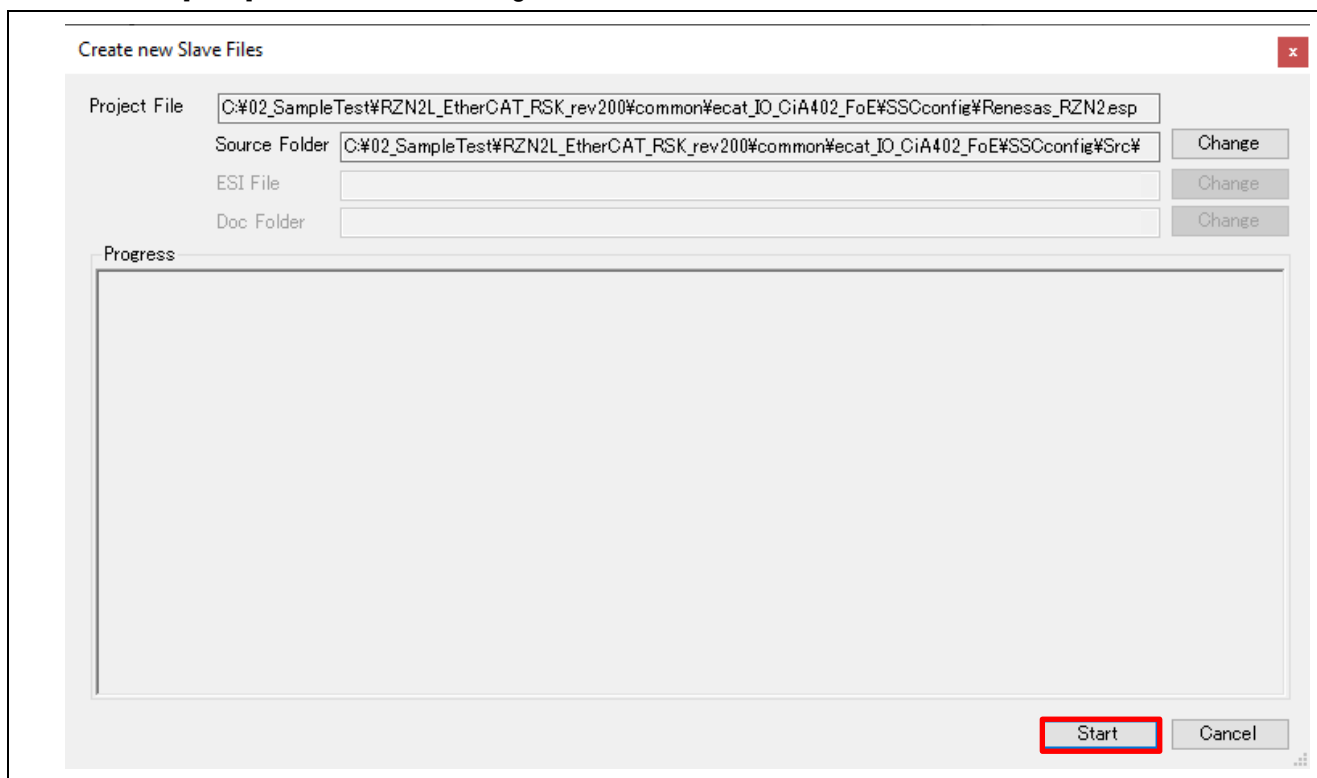
4. After clicking the [OK] button, the following window opens.



5. Select Project > Create new Slave Files.

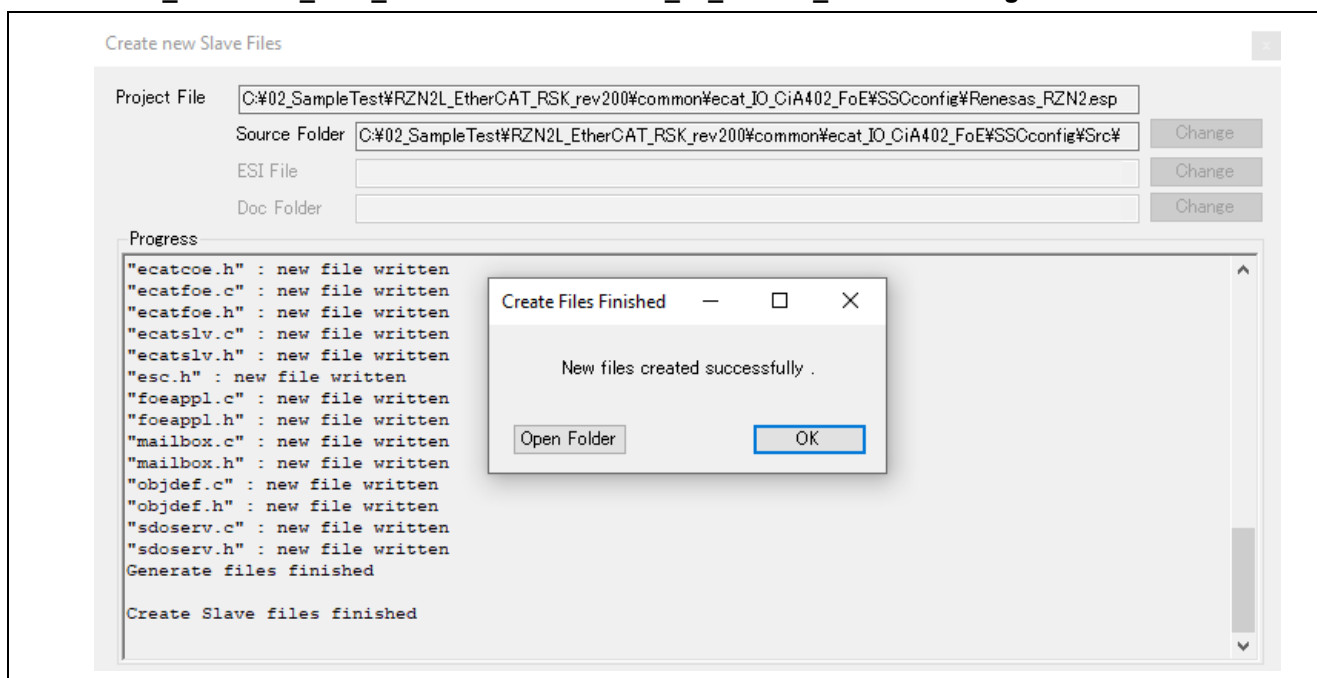


6. Click the [Start] button to start creating the EtherCAT Slave Stack Code.



7. When a message "New file created successfully" appears, the creation processing is completed, and the source files are located in the following folder.

"RZN2L_EtherCAT_RSK_rev0200\common\ecat_IO_CiA402_FoE\SSCconfig\Src"



Note), If an error occurs during generation, uncheck "Create device description" in [Tool]-> [Option]-> [Create Files] of the SSC tool.

8. Double-click on the bat file below.

The patch file contains modifications to make the SSC source file suitable for the RZ/N2L.

[ewarm]

"RZN2L_EtherCAT_RSK_rev0200\common\ecat_IO_CiA402_FoE\Patch\apply_patch_ewarm.bat"

[e² studio]

"RZN2L_EtherCAT_RSK_rev0200\common\ ecat_IO_CiA402_FoE\Patch\apply_patch_e2studio.bat"

Note). If you right-click on the bat file and select [Run as administrator], please open the ewarm or e²studio as administrator. Otherwise, the building will be a failure.



```

C:\WINDOWS\system32\cmd.exe
--- Move SSC Src folder ---
1 dir(s) moved.
--- Patching process start ---
patching file Src/bootmode.c
patching file Src/bootmode.h
patching file Src/cia402appl.c
patching file Src/cia402appl.h
patching file Src/coeappl.c
patching file Src/coeappl.h
patching file Src/ecat_def.h
patching file Src/ecatappl.c
patching file Src/ecatappl.h
patching file Src/ecatfoe.c
patching file Src/ecatfoe.h
patching file Src/ecatslv.c
patching file Src/ecatslv.h
patching file Src/foeappl.c
patching file Src/mailbox.h
patching file Src/objdef.c
patching file Src/objdef.h
patching file Src/sdoserv.h
--- Patching process end ---
--- Move patched Src folder ---
Press any key to continue . . .

```

After execution of the patch, the modified source file is stored in the following folder.

[ewarm]

"RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2\ecat_IO_CiA402_FoE\ewarm\ xSPI0\src\ethercat\beckhoff"

[e² studio]

"RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2\ecat_IO_CiA402_FoE\e2studio \xSPI0\src\ethercat\beckhoff"

Note). **If the patch command is not installed on your PC, you will need to install 64bit patch.exe. This patch command has been confirmed using patch.exe in MinGW. When installing it, refer to "9.5 Appendix E : How to install patch". If it is already installed, skip this step.**

4. Setting up a TwinCAT3

4.1 Copying the ESI Files

Before starting TwinCAT, copy the ESI files that are included in the release folder to TwinCAT destination
 “\TwinCAT\3.x\Config\IO\EtherCAT”

ESI file for current release available at,

“RZN2L_EtherCAT_RSK_rev0200\common\ecat_IO_CiA402_FoE\ESI\Renesas EtherCAT RZN2 CiA402 FoE CDP.xml”

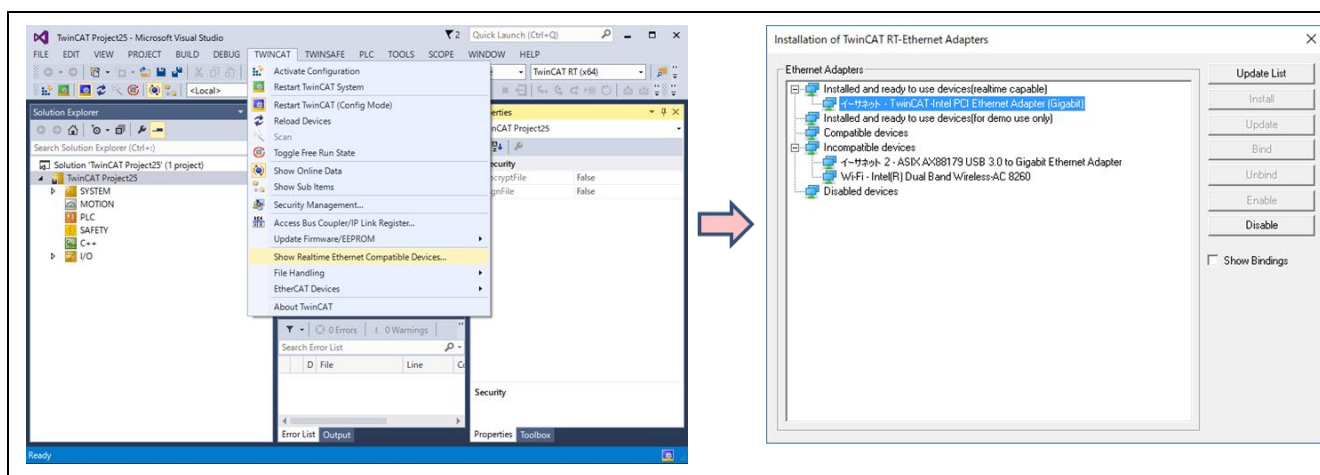
4.2 Add Driver

Add the Ether driver for TwinCAT. (First time only)

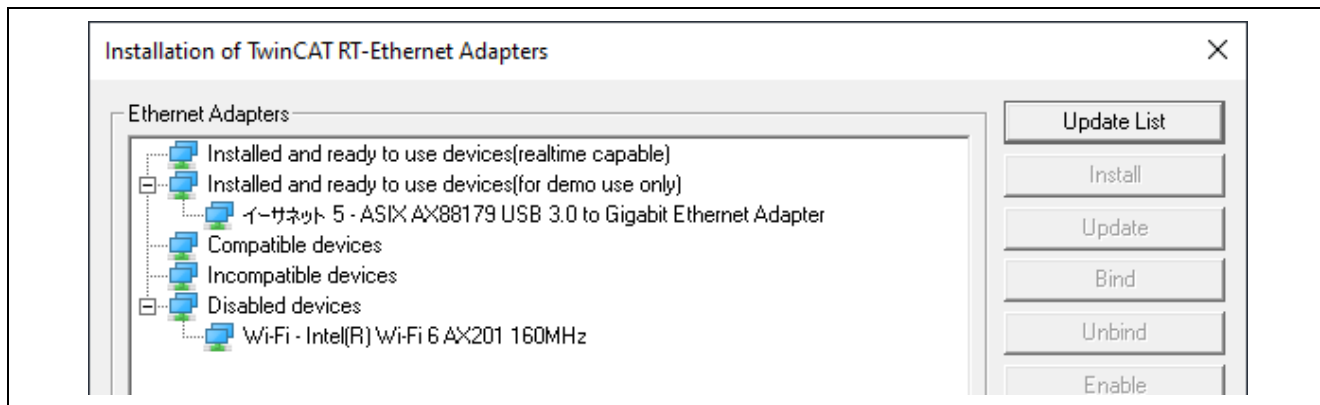
From the start menu, select [TwinCAT3] → [Show Realtime Ethernet Compatible Devise].

Select the connected Ether port from the communication ports and install it.

If the selected ethernet adapter is moved in [Installed and ready to use devices(realtime capable)], there is no problem.



Note). If you don't use the NIC made by Intel, ethernet adapter is moved in [Installed and ready to use devices(for demo use only)].



5. Running the sample application

5.1 Setting sample code for EWARM

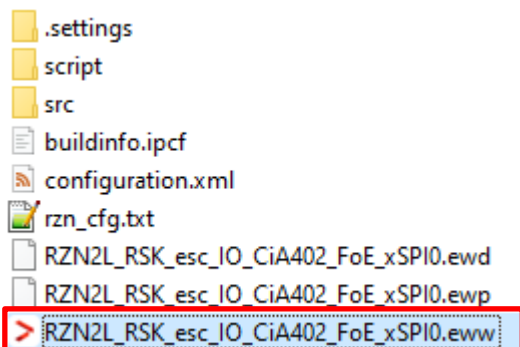
Build the sample code and load it into the flash memory using IAR Embedded Workbench.

Note). Please install FSP Smart Configurator in advance.

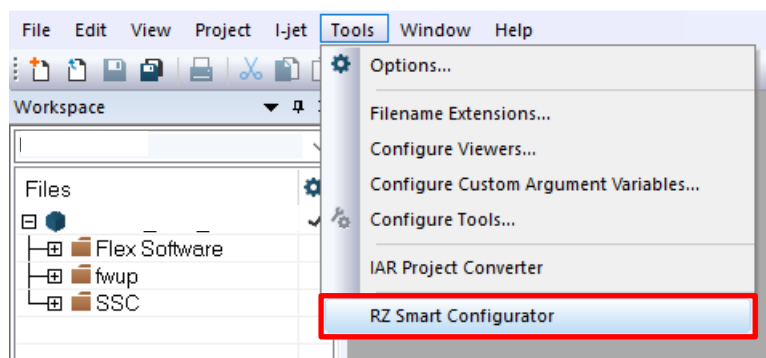
Refer to the latest getting started guide. (Refer to the Table 1.2 for the Document No.)

1. Open the sample project.

"RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2l\ecat_IO_CiA402_FoE\ewarm\xSPI0\RZN2L_RSK_esc_IO_CiA402_FoE_xSPI0.eww"



2. Open the Smart Configurator supporting FSP v2.0.0.
(For example, if menu text is "RZ Smart Configurator", press it.)



3. Prepare for bank 1 code generation. After selecting the BSP tab, change the setting value of BSP property [RSK+RZN2L (xSPI0 x1 boot mode)] → [Parameter information for the loader] → [LDR_ADDR_NML].
0x6010005C → 0x6020005C

Board Support Package Configuration

Generate Project Content
Restore Defaults

Device Selection

FSP version: 2.0.0
Board: RSK+RZN2L (xSPI0 x1 boot mode)
Device: R9A07G084M04GBG
Core: CR52_0
RTOS: No RTOS

Board Details
Renesas Starter Kit+ for RZ/N2L CPU Board (xSPI0 x1 boot mode)

Summary **BSP** Clocks Pins Interrupts Event Links Stacks Components

Properties Problems

RSK+RZN2L (xSPI0 x1 boot mode)

Property	Value
Cortex-R52 CPU core	CPU0
RSK+RZN2L (xSPI0 x1 boot mode)	
Parameter information for the loader	
CACHE_FLG	0x00000000
WRAPCFG_V	0x00000000
COMCFG_V	0x00000000
BMCFG_V	0x00000000
xSPI_FLG	0x00000000
LDR_ADDR_NML	0x6020005C
LDR_SIZE_NML	0x00006000
DEST_ADDR_NML	0x00102000
CSSCTL_V	0x0000003F
LIOCFGCS0_V	0x00070000
ACCESS_SPEED	0x00000006
CHECK_SUM	Auto Calculate.

Console
No consoles to display at this time.

4. Generate the code with "Generate Project Content".

Board Support Package Configuration

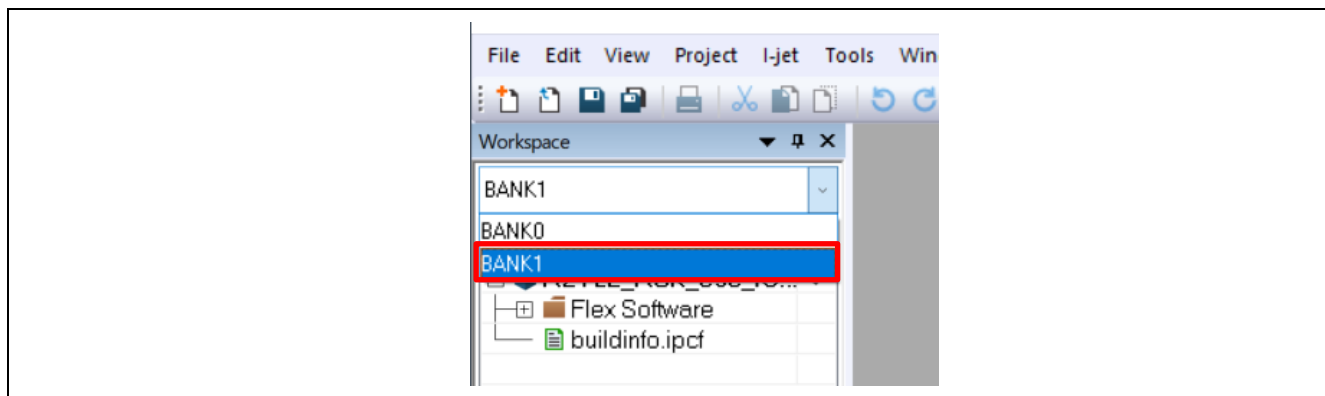
Generate Project Content
Restore Defaults

Device Selection

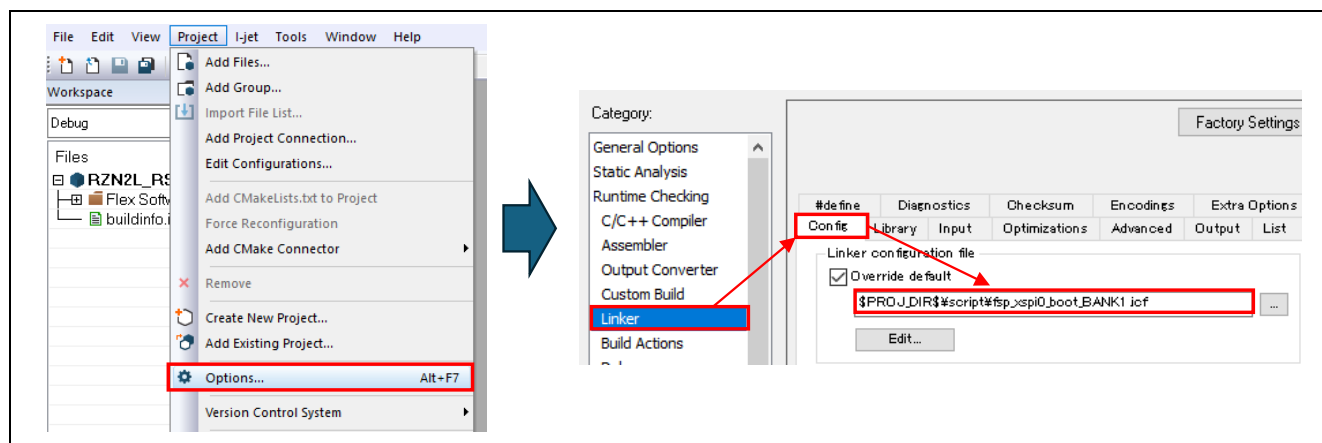
FSP version: 2.0.0
Board: RSK+RZN2L (xSPI0 x1 boot mode)
Device: R9A07G084M04GBG
Core: CR52_0
RTOS: No RTOS

Board Details
Renesas Starter Kit+ for RZ/N2L CPU Board (xSPI0 x1 boot mode)

5. Select "BANK1" of the project.



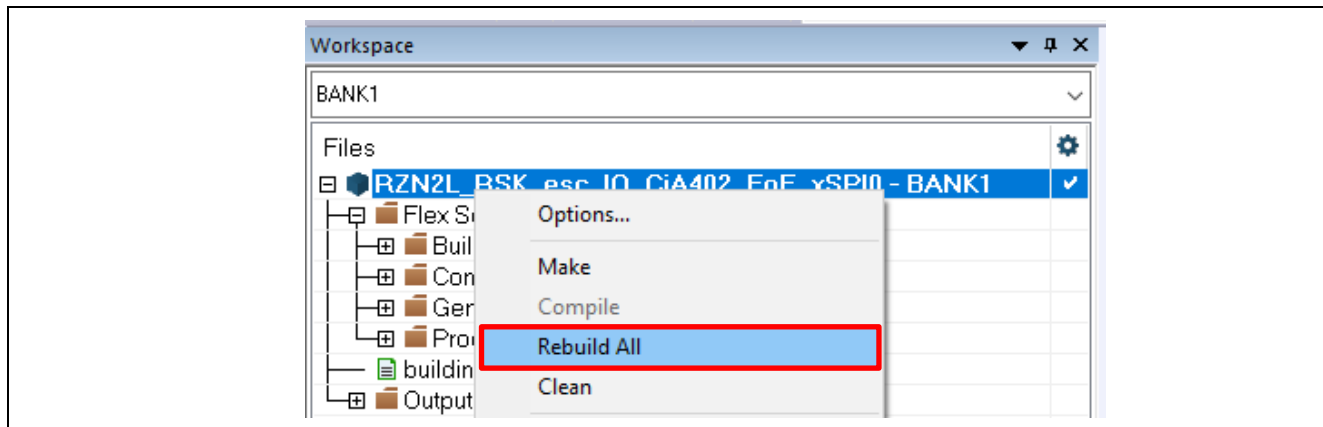
6. Select the "Options..." item for the "Project" menu, open the "Config" tab in the "Linker" category. Verify "Linker configuration file" is "fsp_xspi0_boot_BANK1.icf." If it's wrong, change it.



7. When Debugging with xSPI0 boot, the program cannot be stopped at the beginning of the user program (loader program), and program will move to Default_Handler(). Insert the following loop part into startup_core.c before building project. Please refer to FSP getting started for details.

```
BSP_TARGET_ARM void system_init (void)
{
    #if 1 // Software loops are only needed when debugging.
        __asm volatile (
            " mov r0, #0          \n"
            " movw r1, #0xf07f    \n"
            " movt r1, #0x2fa     \n"
            "software_loop:      \n"
            " adds r0, #1         \n"
            " cmp r0, r1          \n"
            " bne software_loop   \n"
            ::: "memory");
    #endif
}
```


8. Execute the build of BANK1.

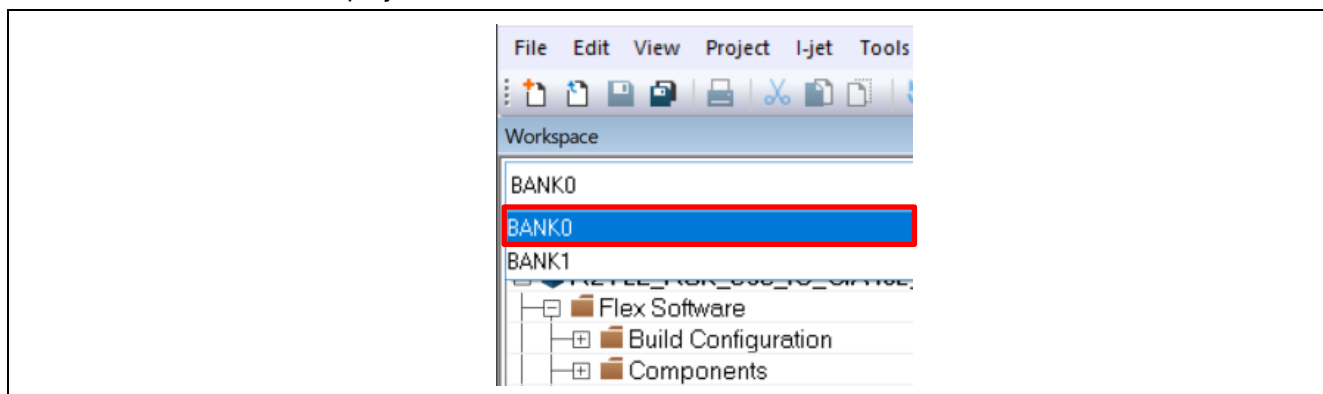


9. Prepare for bank 0 code generation. After selecting the BSP tab, change the setting value of BSP property [RSK+RZN2L(xSPI0 x1 boot mode)] → [Parameter information for the loader] → [LDR_ADDR_NML].

0x6020005C → 0x6010005C

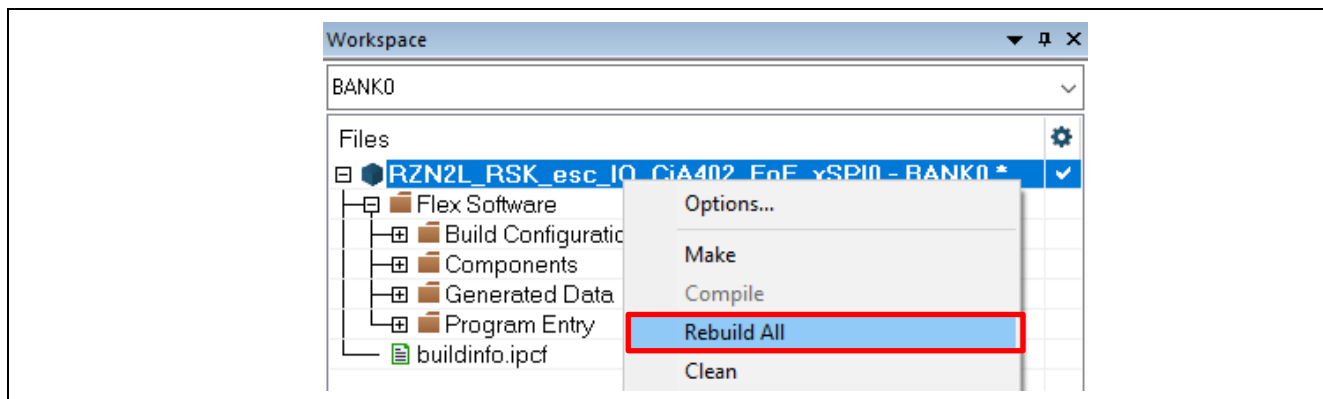
10. Generate the code with "Generate Project Content".

11. Select "BANK0" of the project.

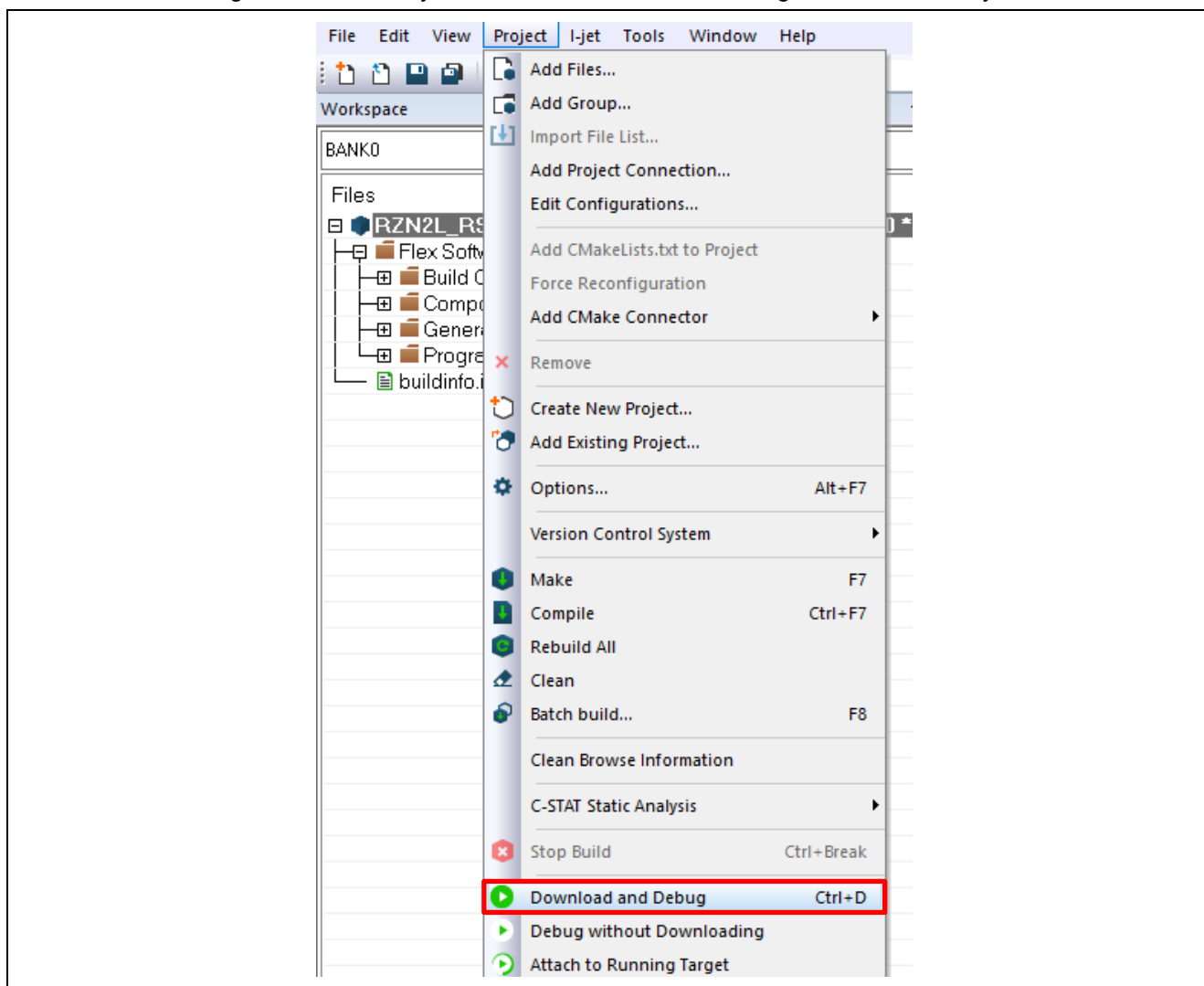


11. Select the "Options..." item for the "Project" menu, open the "Config" tab in the "Linker" category. Verify "Linker configuration file" is "**fsp_xspi0_boot_BANK0.icf**." If it's wrong, change it.

12. Execute the build of BANK0.



13. After connecting the board and I-jet, click the Download and Debug button on the Project toolbar.



14. Confirm that the program works.

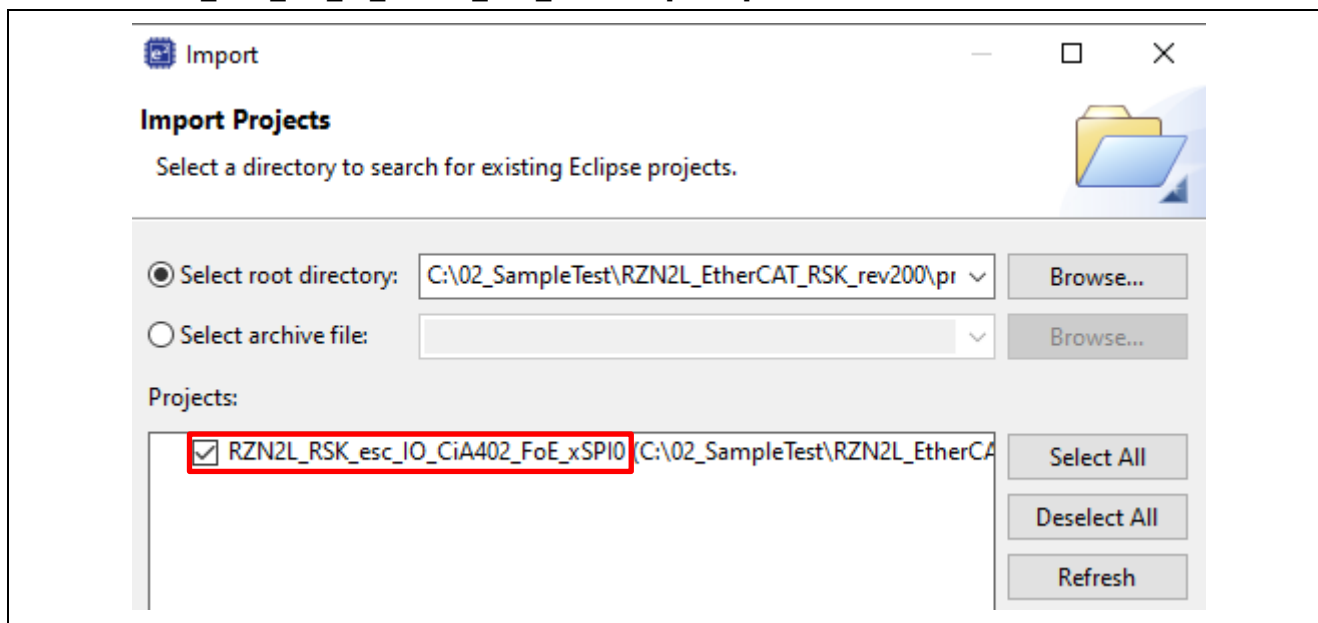
5.2 Setting sample code for GCC

Build the sample code and load it into RAM using Renesas Electronics e² studio.

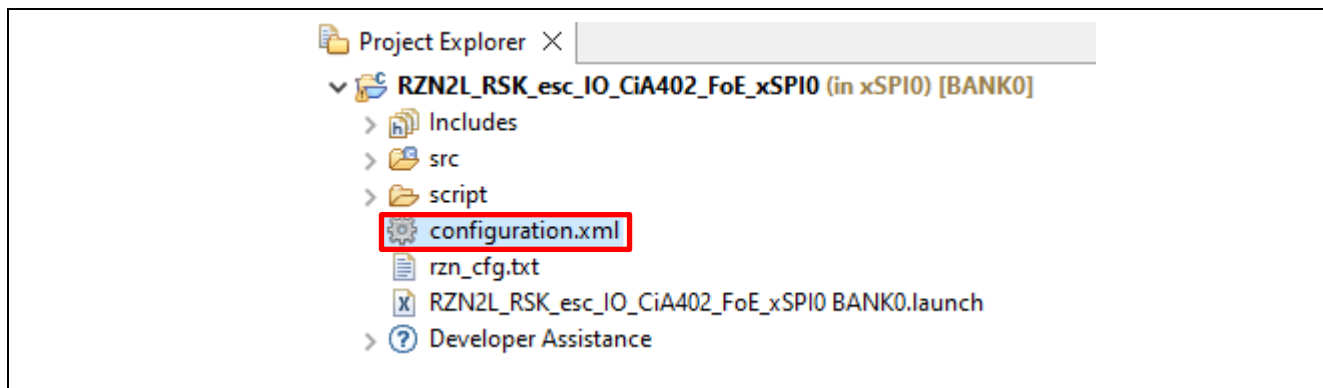
Note). Please install FSP with e² studio 2024-01 (FSP versions of 2.0.0) in advance.

Refer to the latest getting started guide. (Refer to the Table 1.2 for the Document No.)

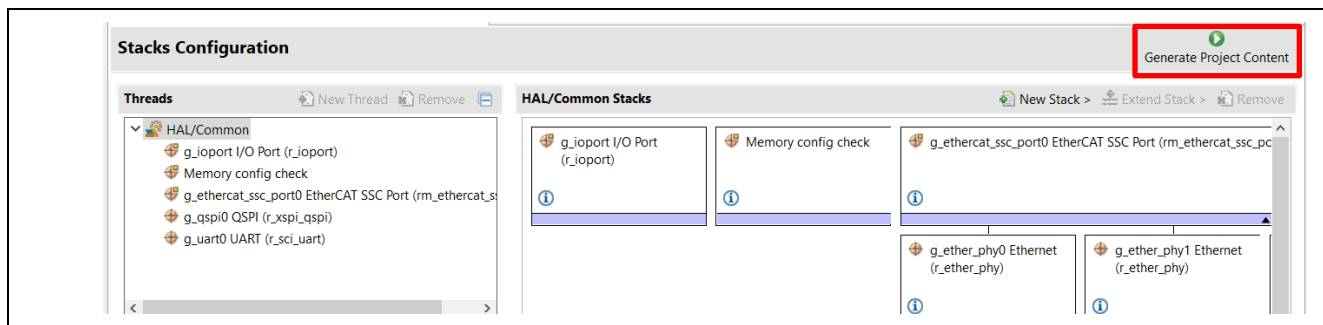
1. Import the sample project. After the program is started, by selecting [File] → [Import] → [Existing Projects into Workspace]. Check the "select root directory" and select "RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2l\ecat_IO_CiA402_FoE\e2studio\xSPI0" folder → "RZN2L_RSK_esc_IO_CiA402_FoE_xSPI0" → [Finish].



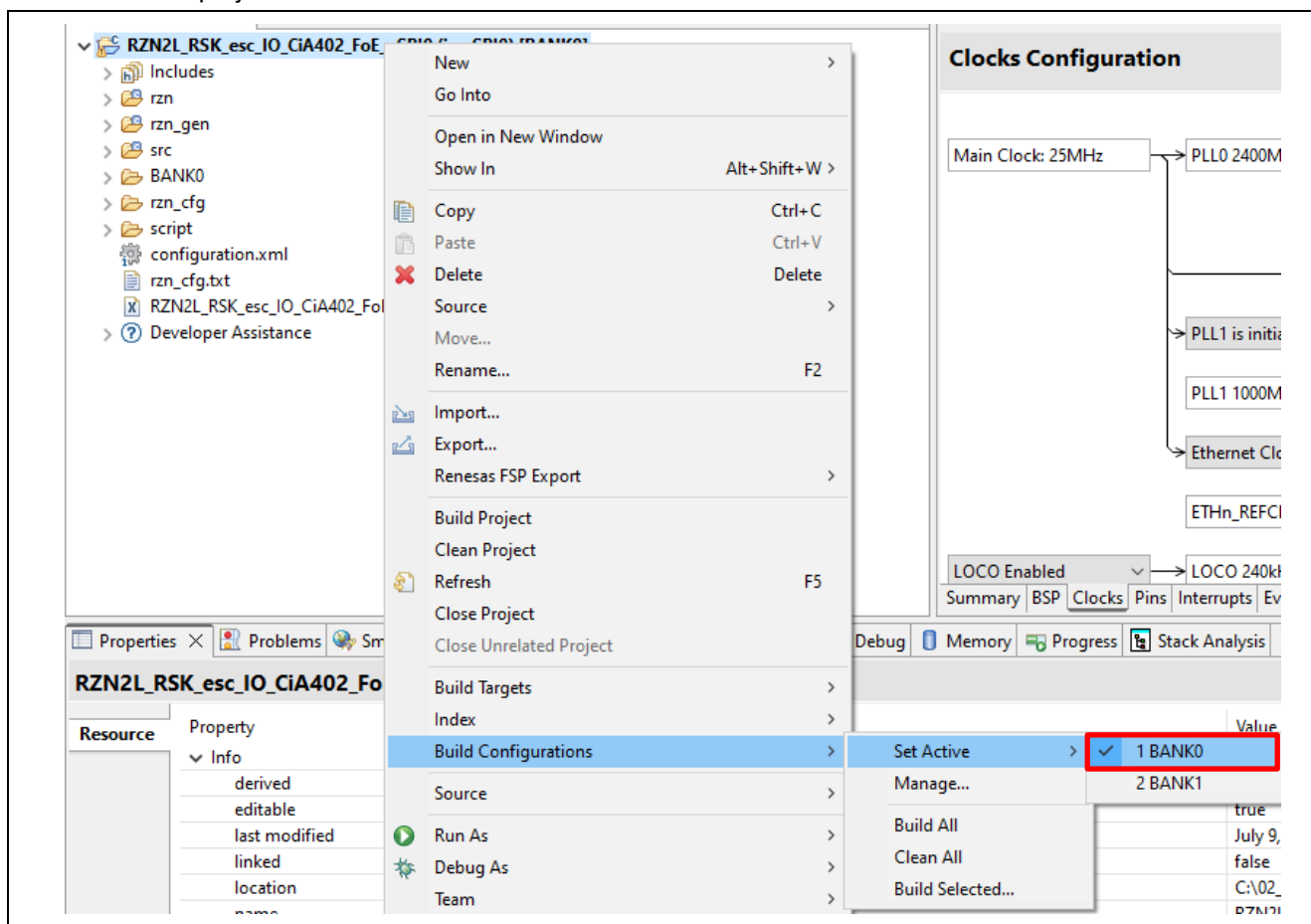
2. Open "configuration.xml" in the "RZN2L_RSK_esc_IO_CiA402_FoE_xSPI0"



3. Generate the code with "Generate Project Content".



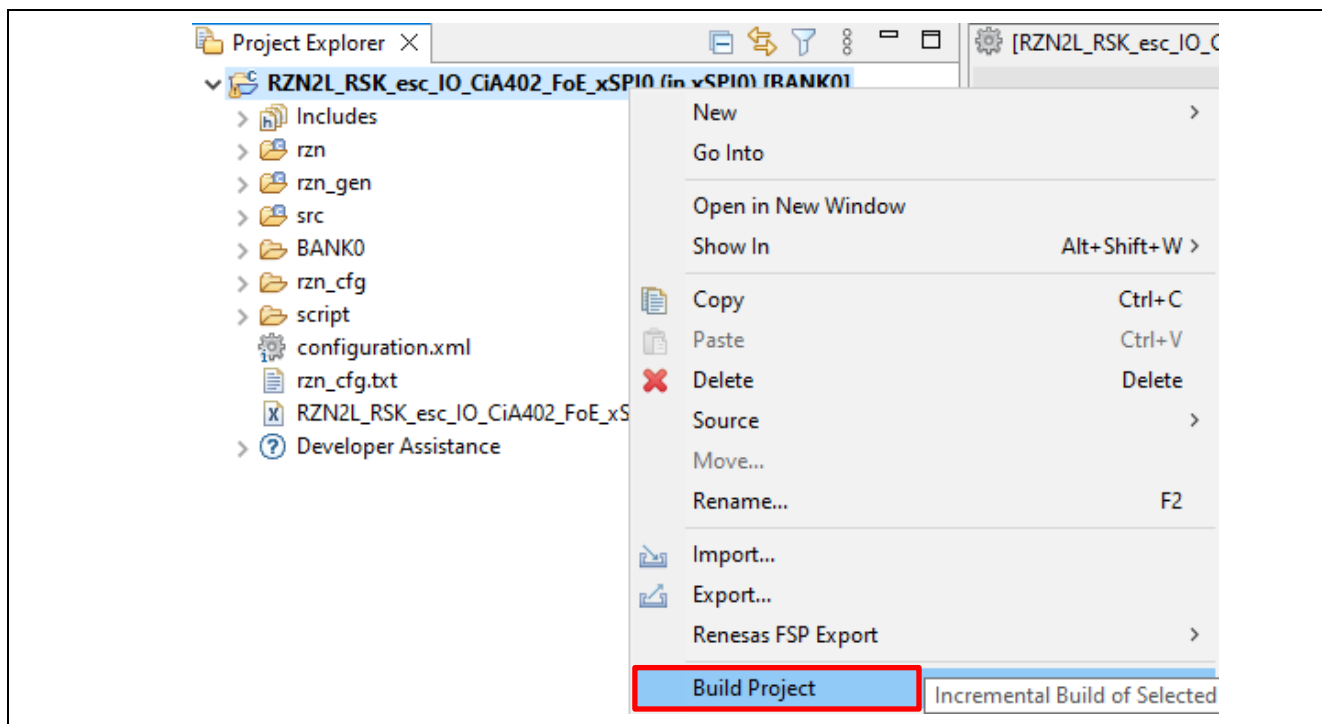
4. Select the project and confirm bank 0 is active.



5. When Debugging with xSPI0 boot, the program cannot be stopped at the beginning of the user program (loader program).
 Insert the following loop part into startup_core.c before building project.
 Please refer to FSP getting started for details.

```
BSP_TARGET_ARM void system_init(void)
{
  #if 1 // Software loops are only needed when debugging.
    __asm volatile (
      " mov r0, #0          \n"
      " movw r1, #0xf07f    \n"
      " movt r1, #0x2fa     \n"
      "software_loop:       \n"
      " adds r0, #1         \n"
      " cmp r0, r1          \n"
      " bne software_loop   \n"
      ::: "memory");
  #endif
}
```

6. Execute the build.



7. Prepare for bank 1 code generation. After selecting the BSP tab, change the setting value of BSP property [RSK+RZN2L (xSPI0 x1 boot mode)] → [Parameter information for the loader] → [LDR_ADDR_NML].
0x6010005C → 0x6020005C

Device Selection

FSP version: 2.0.0

Board: RSK+RZN2L (xSPI0 x1 boot mode)

Device: R9A07G084M04GBG

Core: CR52_0

RTOS: No RTOS

Board Details
 Renesas Starter Kit+ for RZ/N2L CPU Board (xSPI0 x1 boot mode)

Summary | **BSP** | Clocks | Pins | Interrupts | Event Links | Stacks | Components

Properties X

RSK+RZN2L (xSPI0 x1 boot mode)

Settings	Property	Value
	R9A07G084M04GBG	
	RSK+RZN2L (xSPI0 x1 boot mode)	
	Parameter information for the loader	
	CACHE_FLG	0x00000000
	WRAPCFG_V	0x00000000
	COMCFG_V	0x00000000
	BMCFG_V	0x00000000
	xSPI_FLG	0x00000000
	LDR_ADDR_NML	0x6020005C
	LDR_SIZE_NML	0x00006000
	DEST_ADDR_NML	0x00102000
	CSSCTL_V	0x0000003F
	LIOCFGCS0_V	0x00070000
	ACCESS_SPEED	0x00000006
	CHECK_SUM	Auto Calculate.

8. Generate the code with "Generate Project Content".

Stacks Configuration

Threads

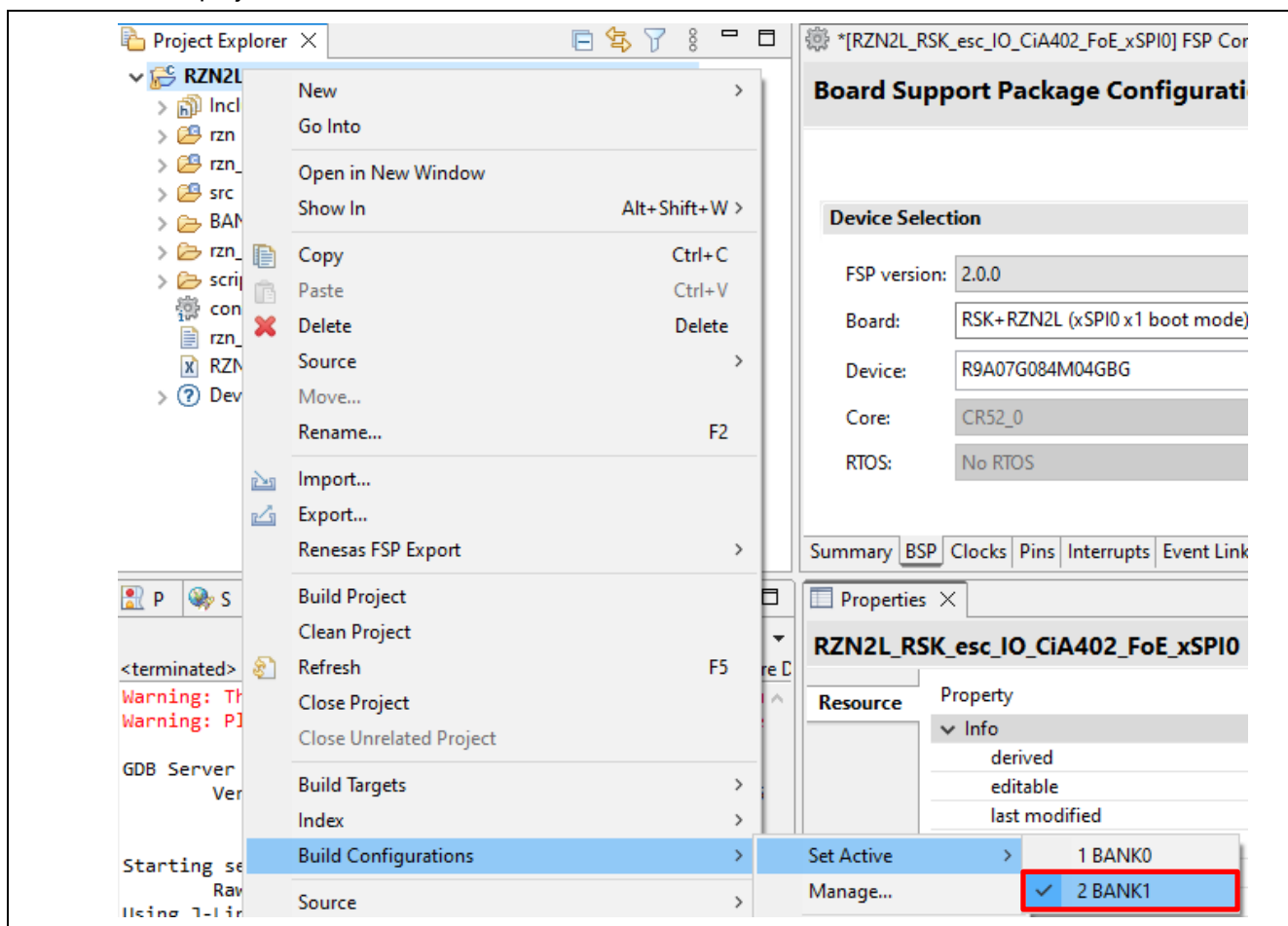
- HAL/Common
 - g_ioport I/O Port (r_ioport)
 - Memory config check
 - g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port0)
 - g_qspi0 QSPI (r_xspi_qspi)
 - g_uart0 UART (r_sci_uart)

HAL/Common Stacks

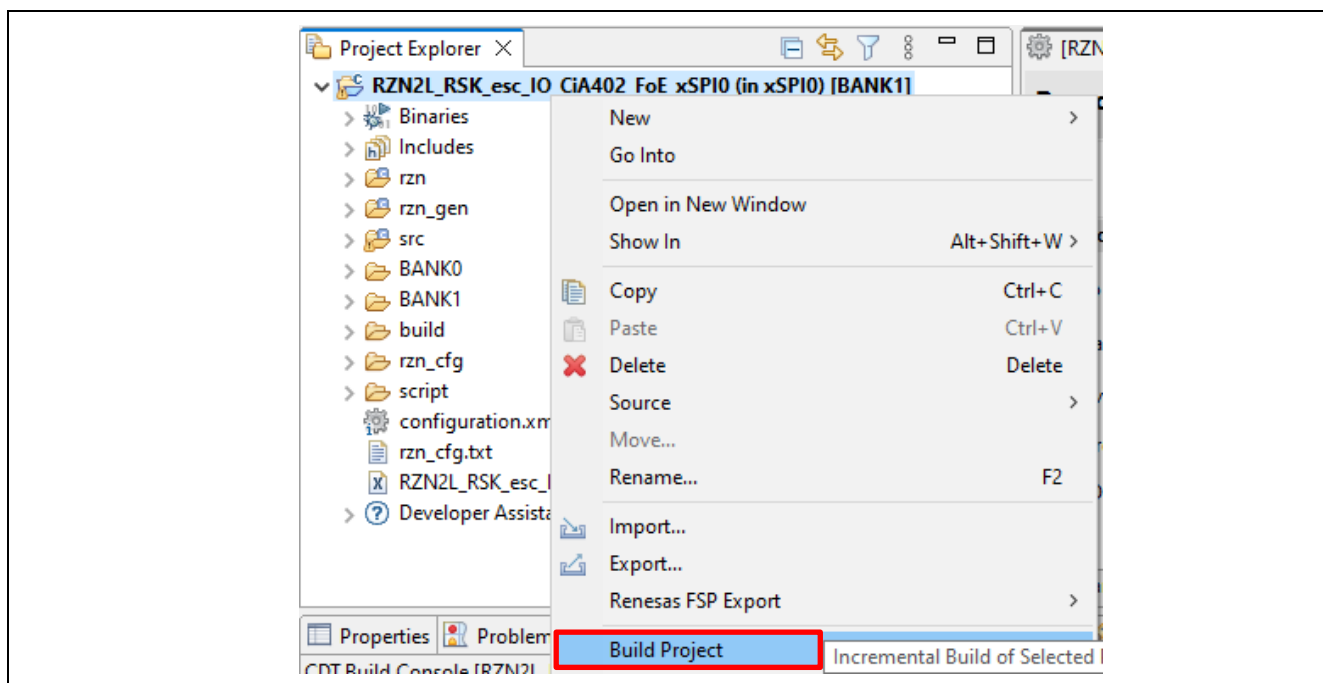
- g_ioport I/O Port (r_ioport)
- Memory config check
- g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port0)
- g_ether_phy0 Ethernet (r_ether_phy)
- g_ether_phy1 Ethernet (r_ether_phy)

Generate Project Content

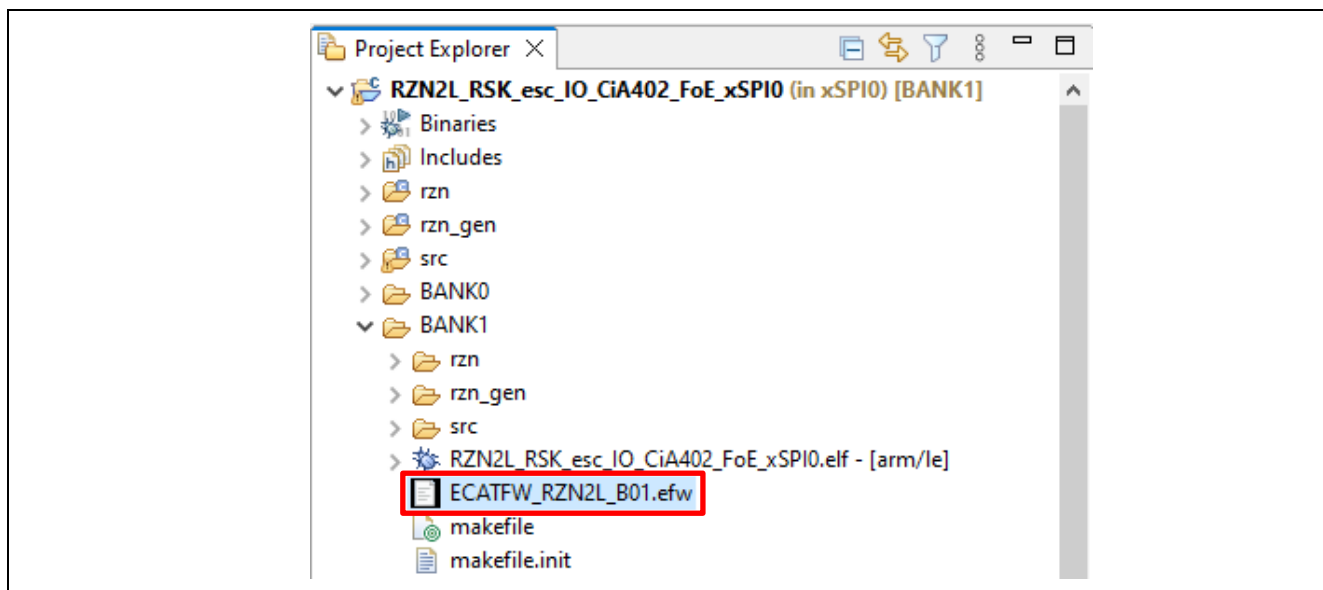
9. Select the project and confirm BANK1 is active.



10. Execute the build.

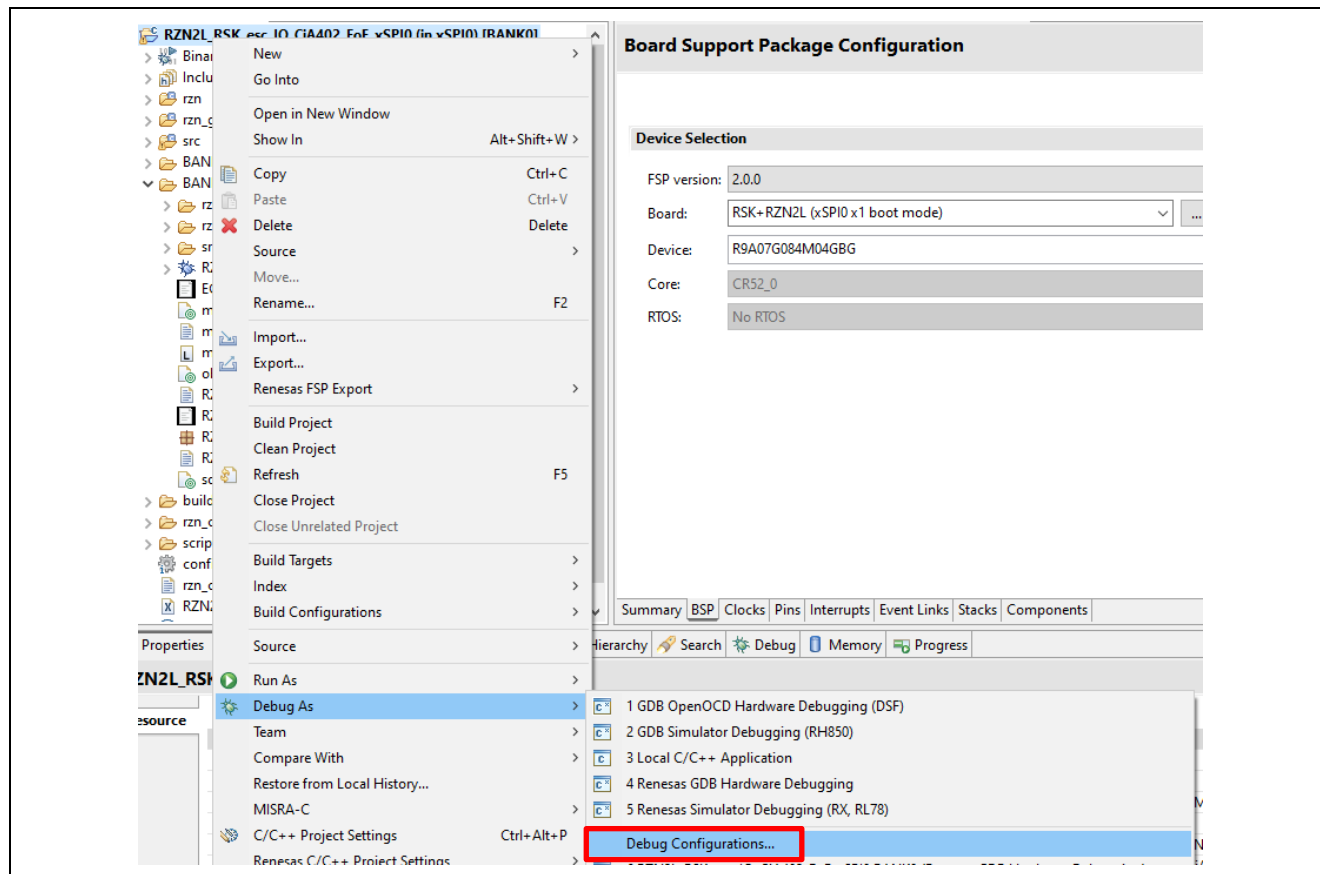


11. Confirm [BANK1]→[ECAFW_RZN2L_B01.efw] exists after the build is finished.

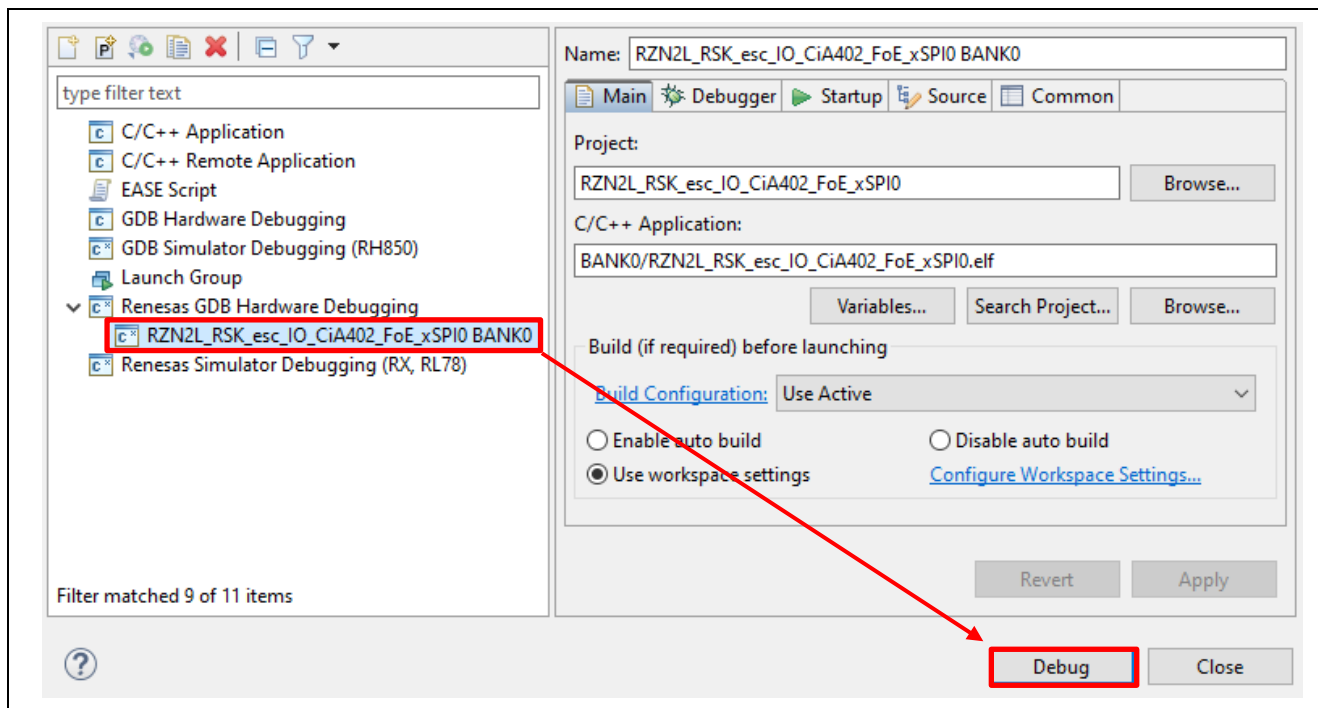


12. Restore the setting value of [LDR_ADDR_NML] so that the Bank0 program will start.
0x6020005C → 0x6010005C

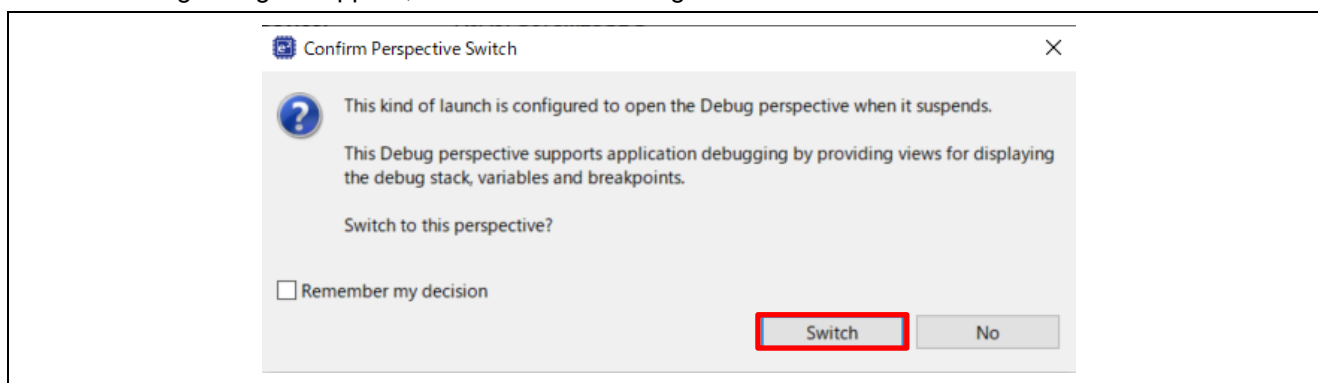
13. After connecting the board and J-Link, start debugging in the following procedure.
 In [Project Explorer] view, right click the node of project to be debugged and select [Debug As] → [Debug Configurations].



[Renesas DBG Hardware Debugging] → [RZN2L_RSK_esc_IO_CiA402_FoE_xSPI0 BANK0] item, then press [Debug].



Following dialog will appear, so switch to the debug screen.



14. Press the "Resume" button.

When debugging CPU0 is started, the program is interrupted at "hal_entry ();" in main.c. Press the "Resume" button again. Program will run.

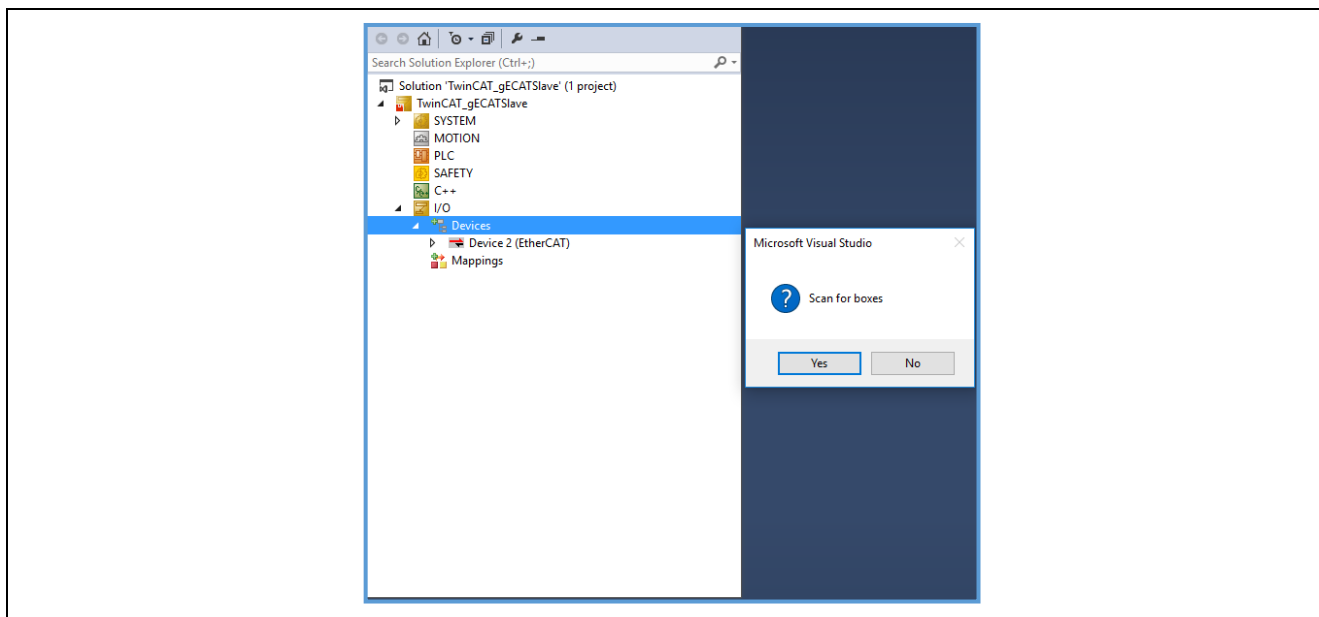
6. Connecting to TwinCAT3

Start TwinCAT3 by using the procedure described below,
From the start menu, select [Beckhoff] → [TwinCAT3] → [TwinCAT XAE].

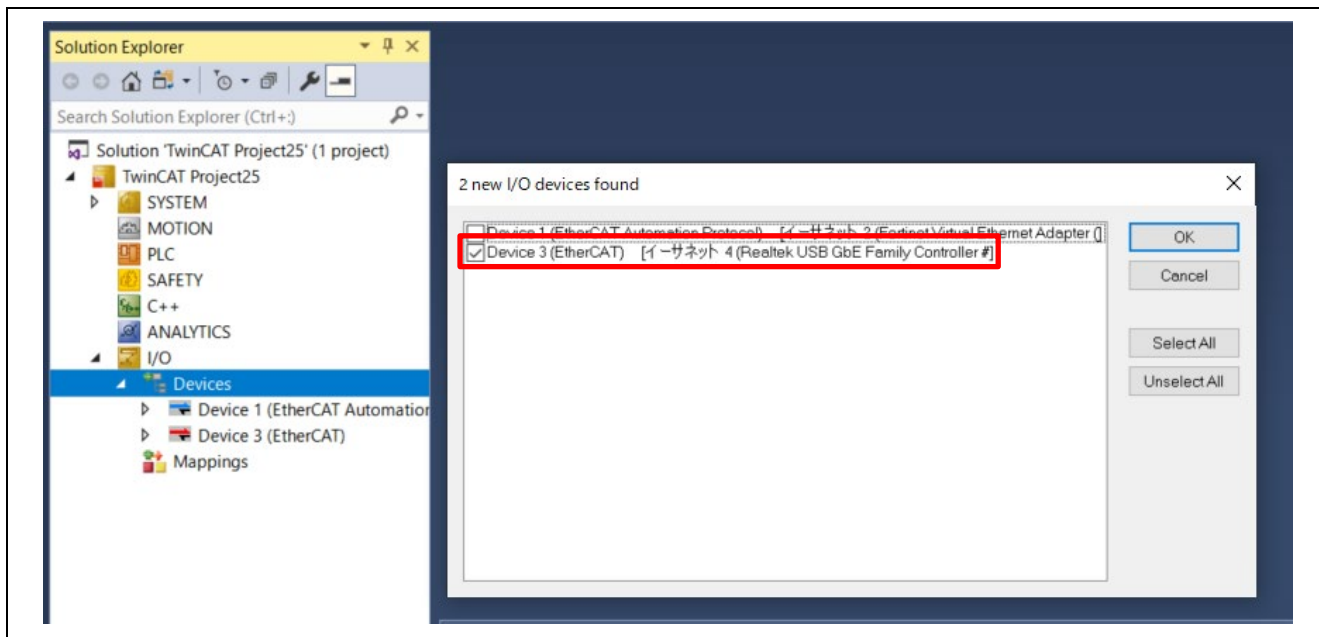
After the program is started, by selecting [File] → [New] → [Project], create a new project of the TwinCAT XAE Project type. The subsequent procedure is described below.

6.1 Scanning I/O Devices

1. Under solution explorer -> I/O -> Devices, select 'Scan' as in Figure below.

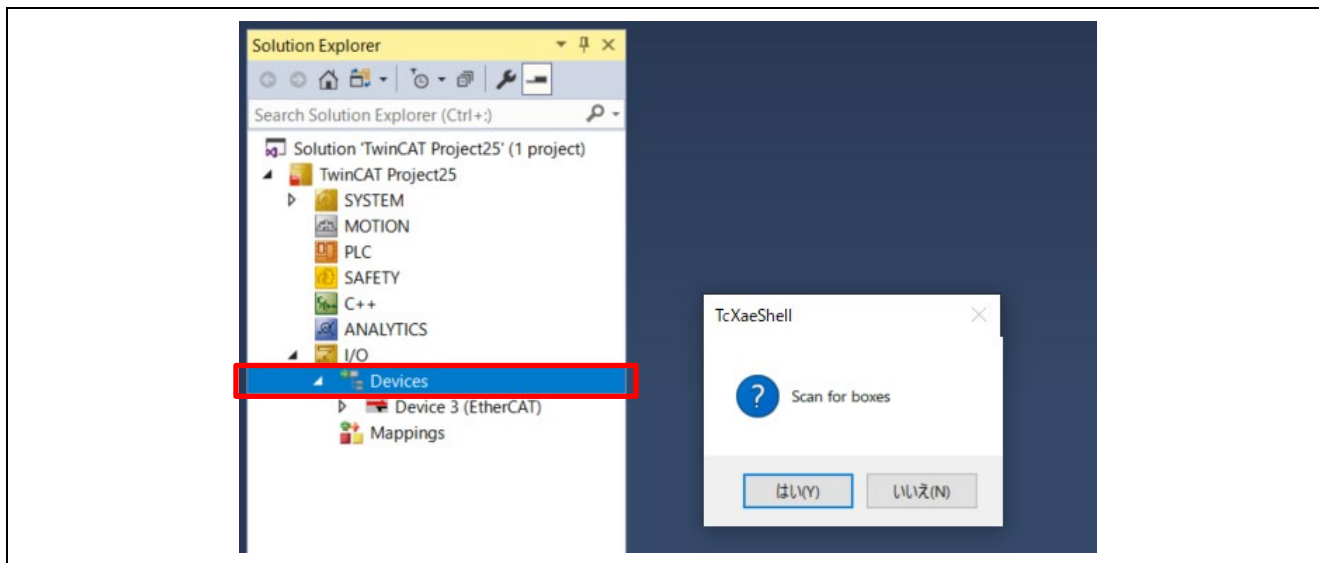


2. EtherCAT port will be displayed as below. Select and press OK.



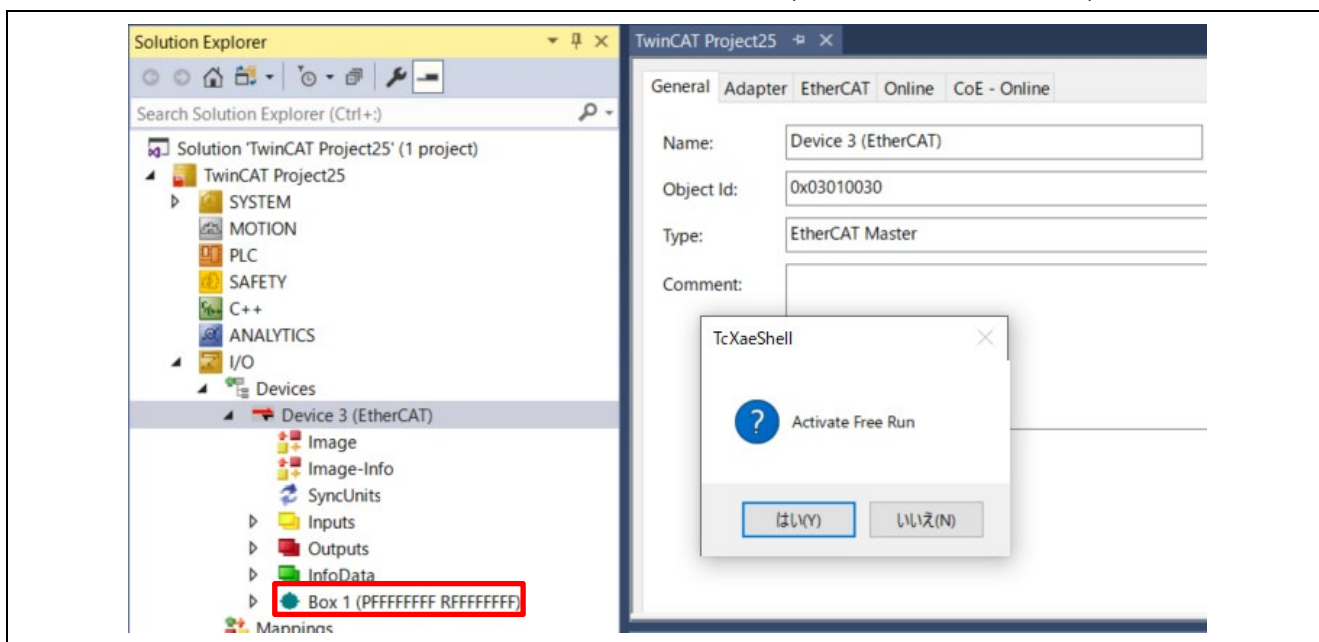
Note). If a valid slave exists in the network, TwinCAT will display the candidate with a checkbox.

3. Start scanning the device.



3. If the detected device is "Box1 (PFFFFFFFF)" as shown in the figure, ESI file has not downloaded, so, proceed to chapter 7.2 without Free Run.

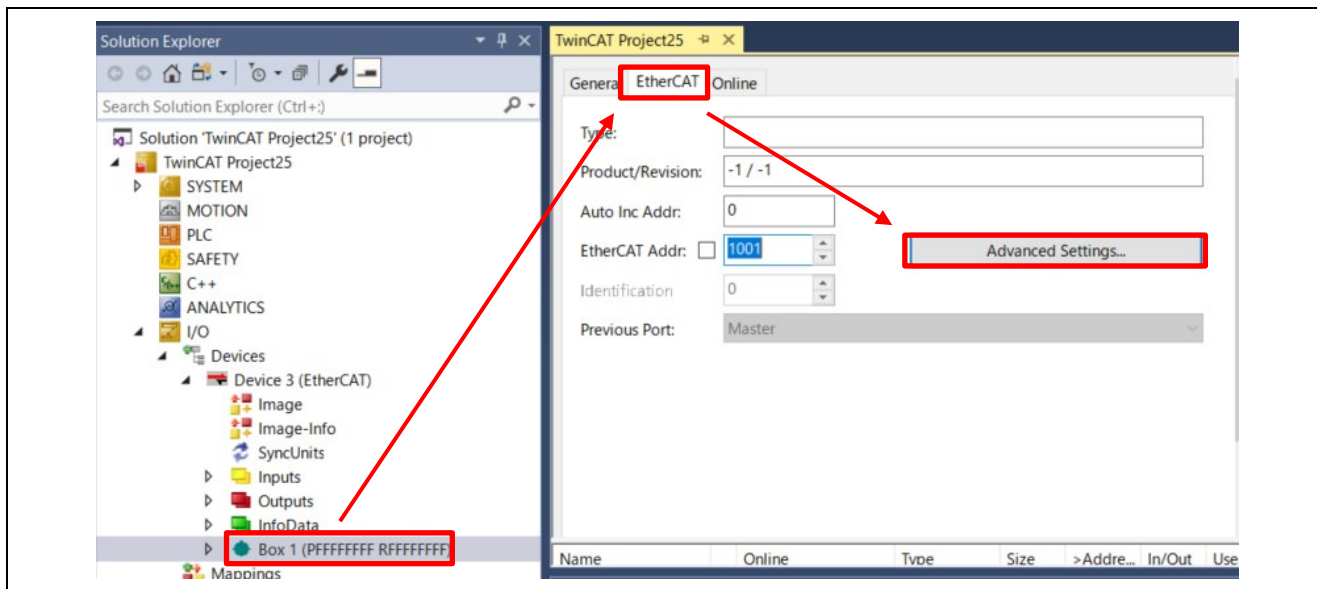
If the detected device shows desired ESI filename like "Box1 (Renasas EtherCAT xxxx)", select Free run.



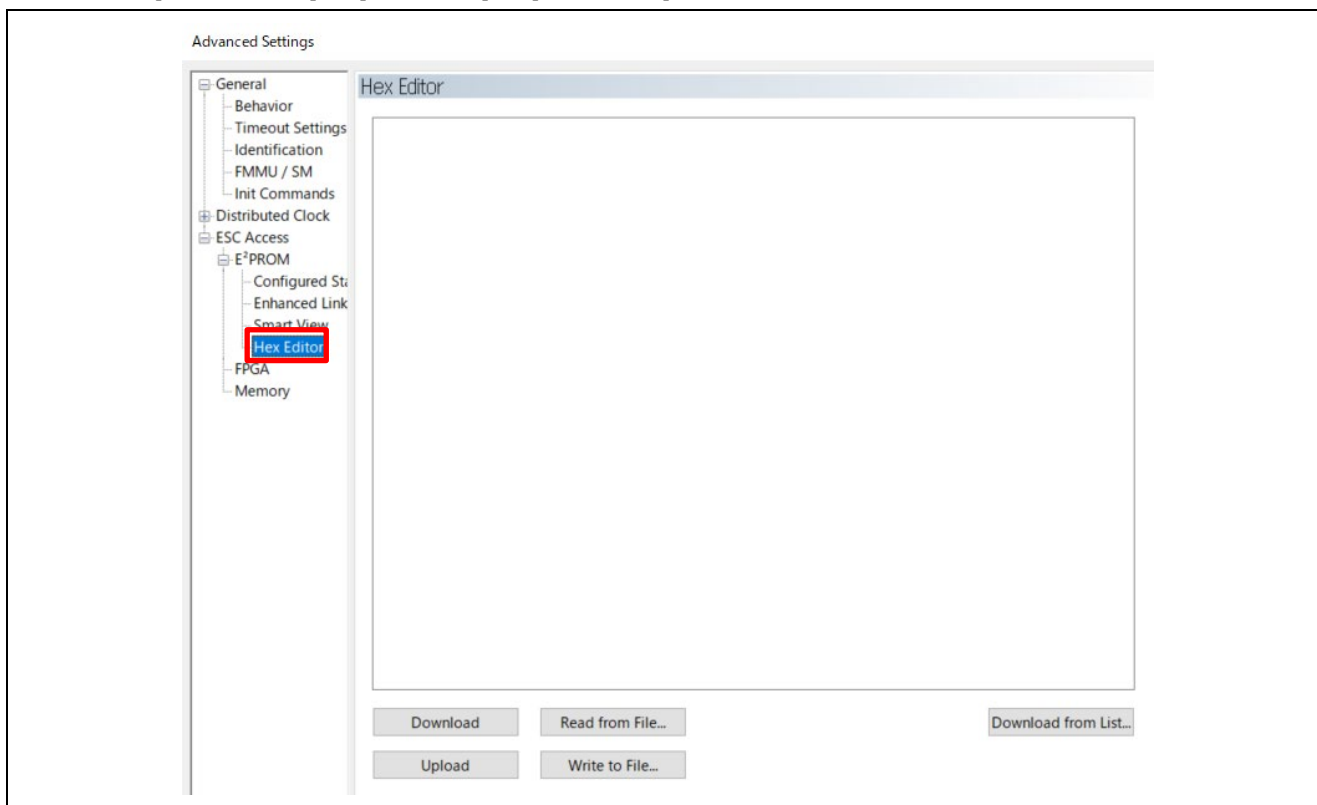
6.2 Updating EEPROM Data

If the data of another application has already been written to the EEPROM, replace the data. The following shows the procedure for replacing the data on the EEPROM:

1. Double-click [Box 1] to display a panel on the right side of the window.
2. Select the [EtherCAT] tab.
3. Click the [Advanced Setting] button.



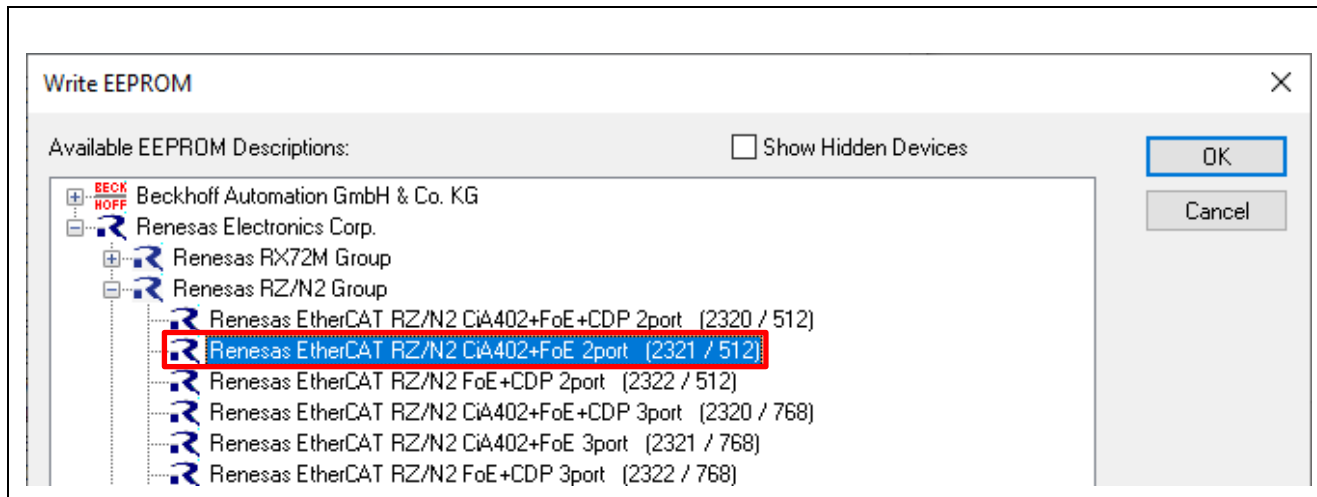
4. Select [ESC Access] → [EEPROM] → [Hex Editor].



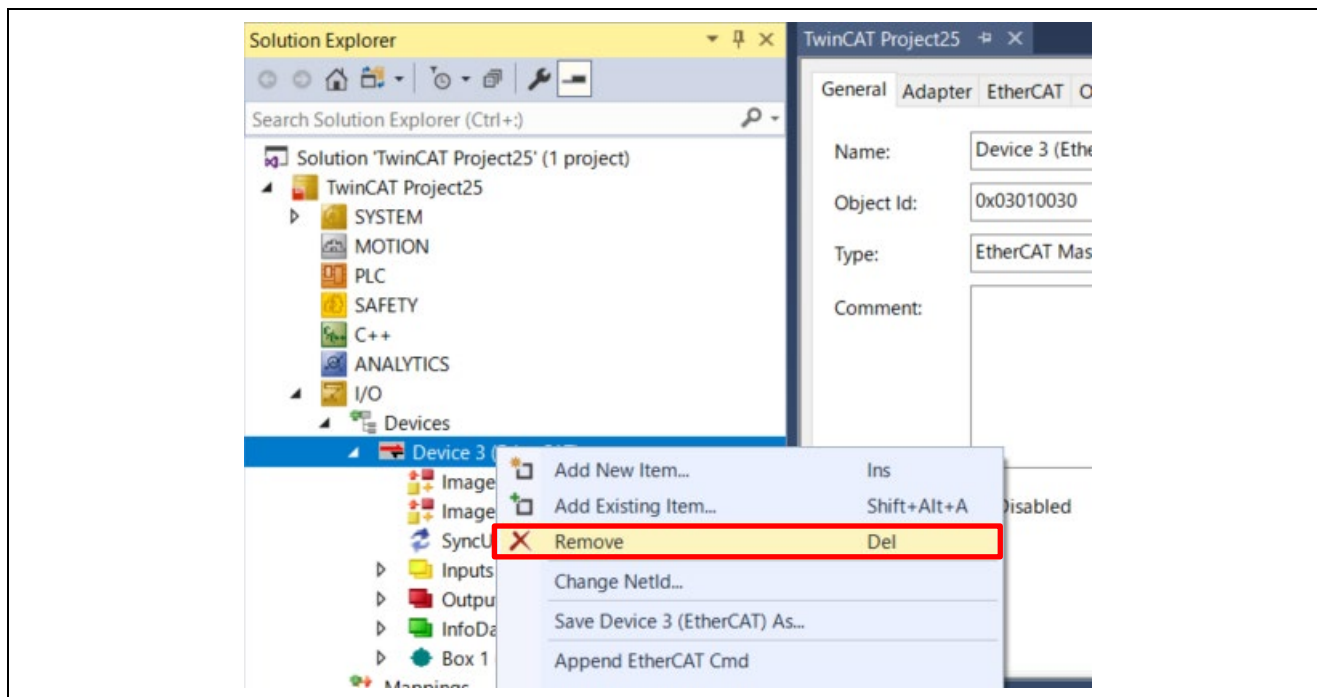
5. Select [Download from List] → Select "Renesas EtherCAT RZN2 CiA402+FoE 2port."

Note). Select "Renesas EtherCAT RZ/N2 CiA402+FoE 2port" or
"Renesas EtherCAT RZ/N2 CiA402+FoE 3port"

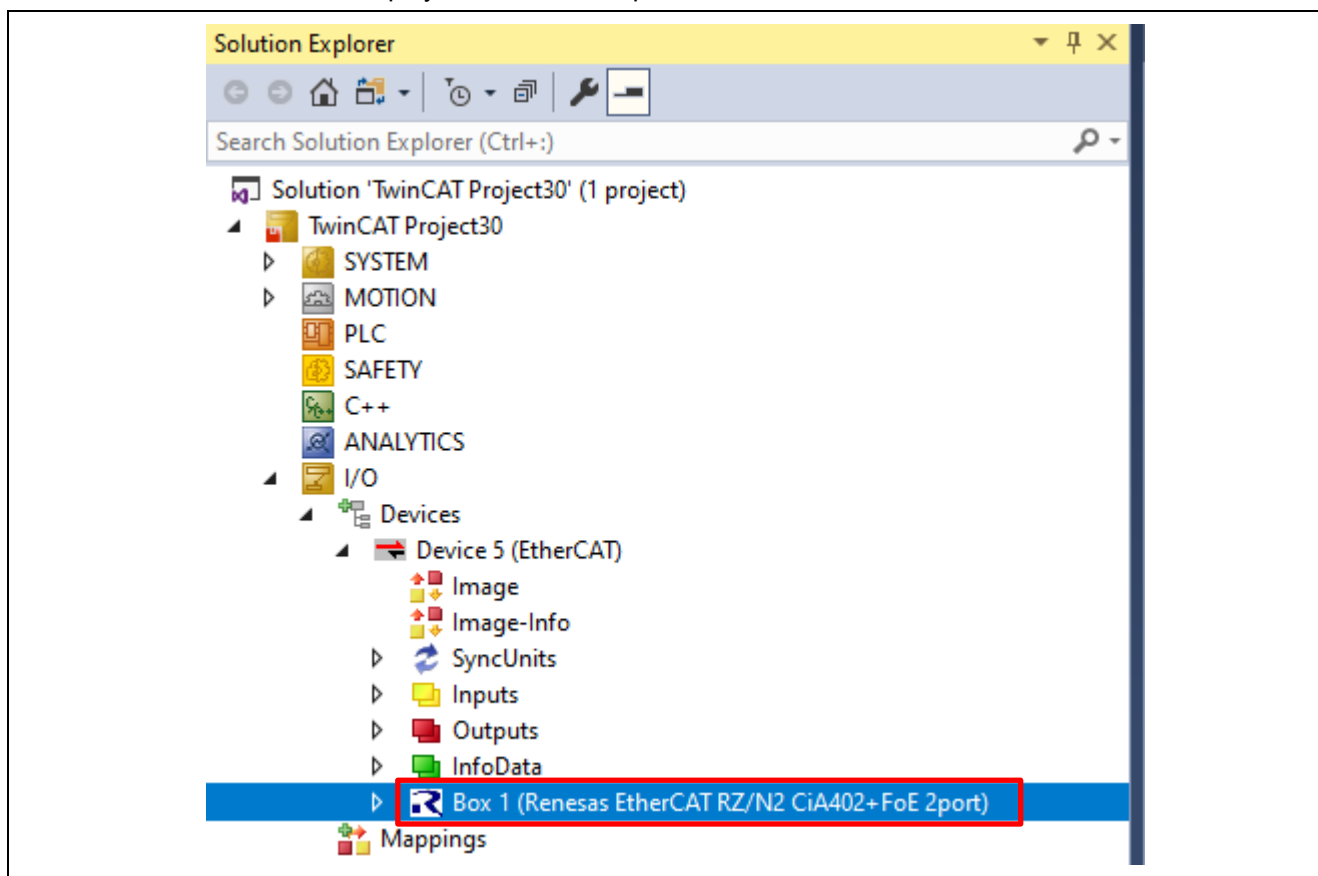
6. OK and Download.



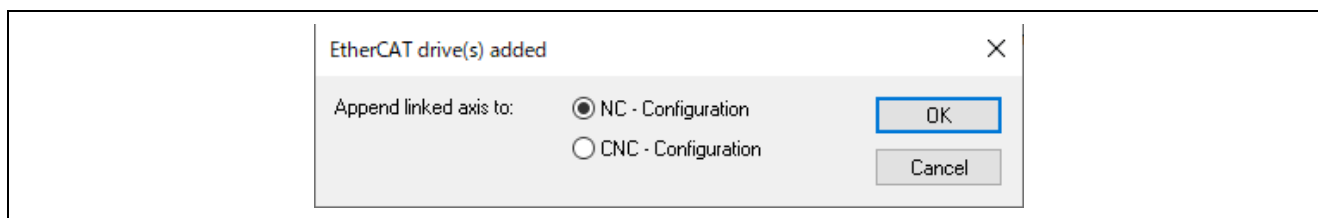
7. Apply the ESI file settings.
Select the device and remove it.



8. Scan the device again.
If the desired ESI file is displayed, it is correct operation.



Note). When scanning the device, the CiA402's ESI is used, so the axis configuration settings are displayed.
Select [NC-Configuration].



. And, when [Active Free Run] dialog box is displayed, click on [Yes].

Option A - Create ESI binary file from ESI XML and download.

1. SSC Tool → [Tool] → [EEPROM Programmer].
2. [FILE] → [OPEN] → Browse and select the ESI file.
3. [FILE] → [Save AS] → Select type as binary.
4. A binary file will be generated in the specified folder.
5. [Read from File] Select the ESI binary file → [Download].
6. Confirm the write status using [Upload] option.

After the data is replaced, restart the RZ/N2L (by turning it off and on, or resetting it) so that the new data is applied to the microcomputer. Execute [Restart TwinCAT System].

6.3 Sync Modes

The Slave Stack Code supports different modes of synchronization which are based on three physical signals: (PDI_) IRQ, Sync0 and Sync1.

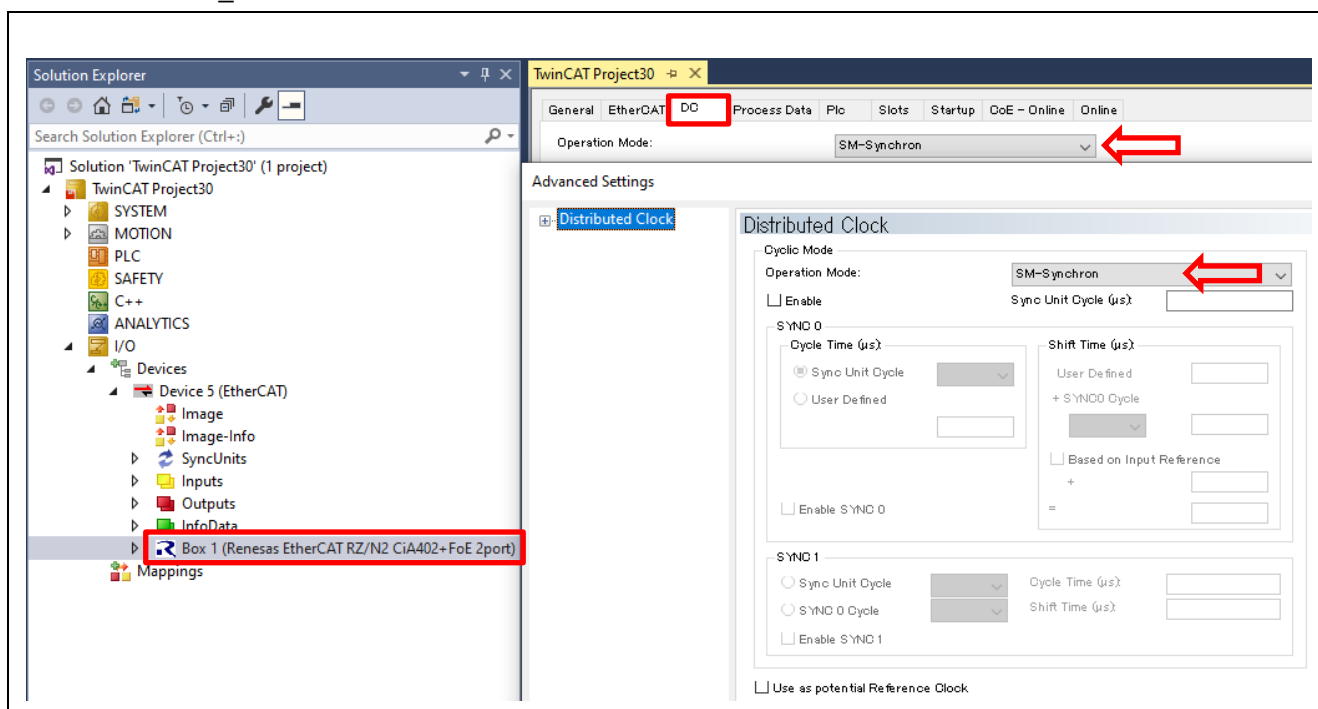
After setting the synchronous mode, please reflect the setting in [TwinCAT] → [Restart TwinCAT (Config Mode)] → [Reload Devices].

6.3.1 Free Run

In this mode there is no slave application synchronization, AL_EVENT_ENABLED and DC_SUPPORTED disabled.

6.3.2 Sync Manager Synchronization

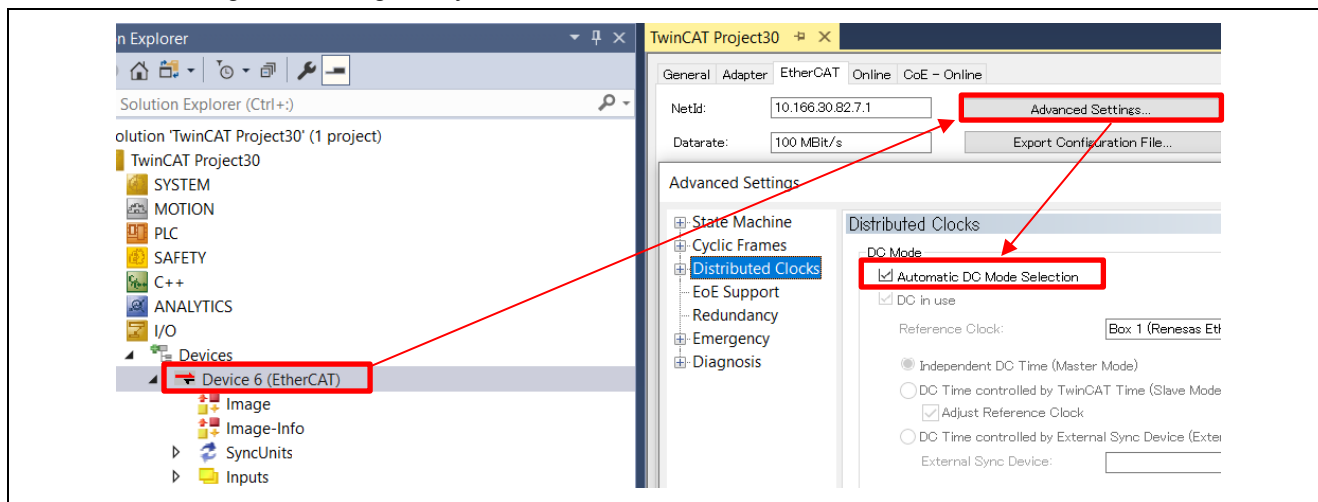
In this mode the slave application is executed as Sync Manager synchronous. AL_EVENT_ENABLED enabled and DC_SUPPORTED disabled.



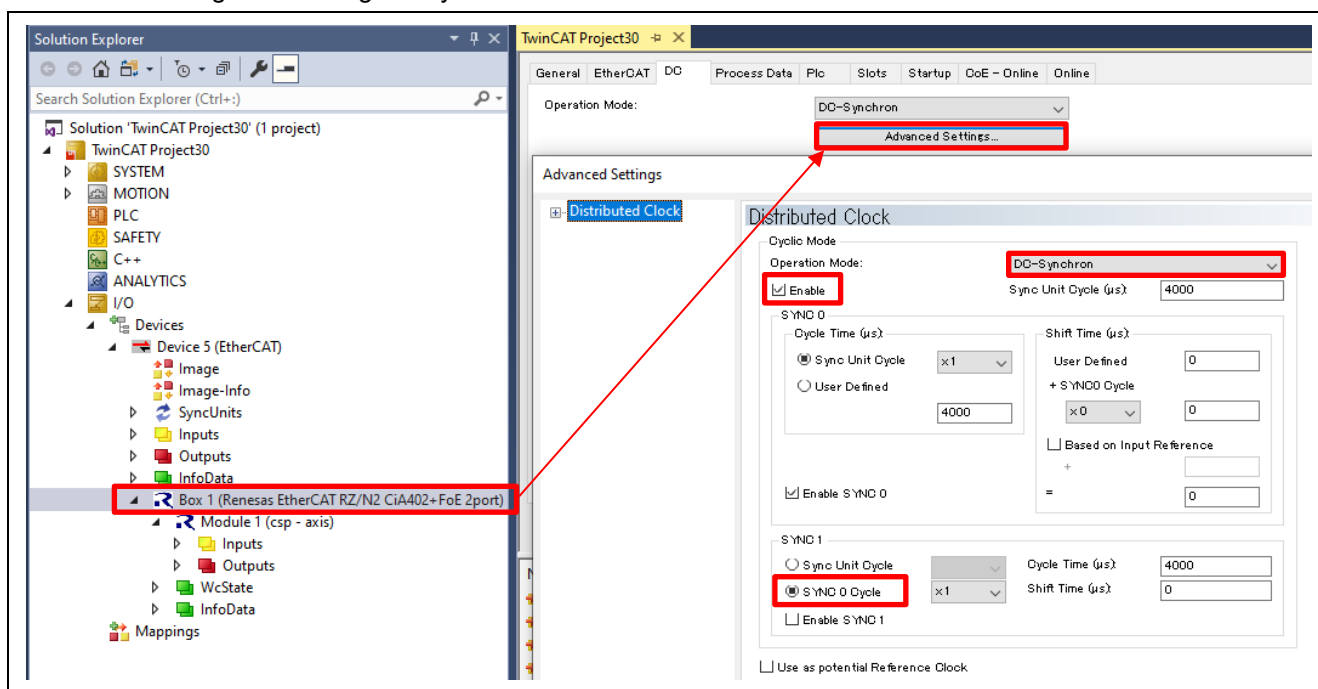
6.3.3 DC Synchronization

SyncManager/Sync0 & SyncManager/Sync0/Sync1 synchronous, both AL_EVENT_ENABLED and DC_SUPPORTED are enabled.

1. Master setting for enabling DC synchronization.



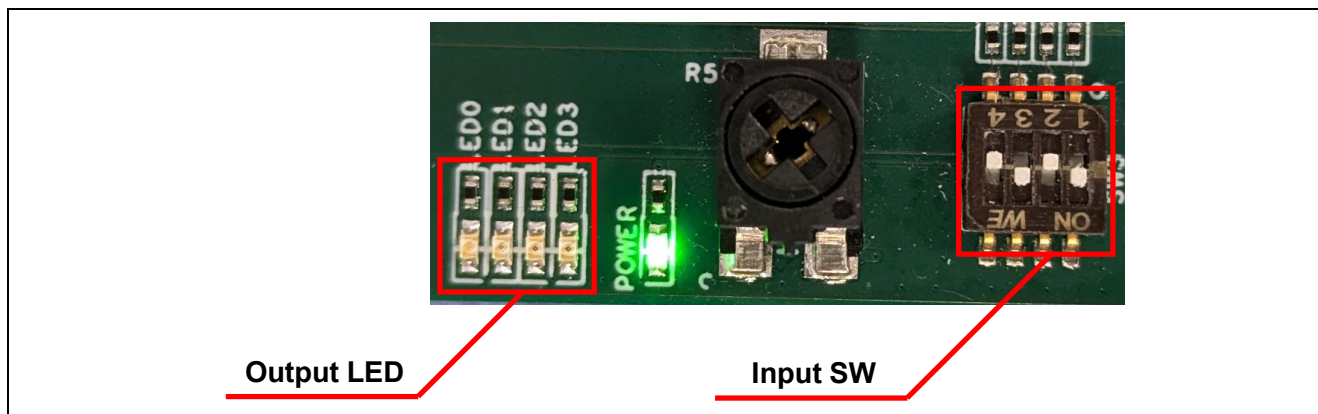
2. Slave setting for enabling DC synchronization.



Note: SYNC0 and SYNC1 are level triggered interrupt. It can cause synchronization issues due to multiple interrupts in a single pulse. To avoid this issue, pulse width can be reduced by changing the word 2 of EEPROM configuration value in ESI file. It sets the register 0x982 register of EtherCAT slave during power up. Also can be solved by waiting for the line to be low in the interrupt handler by reading the status register 0x98E.

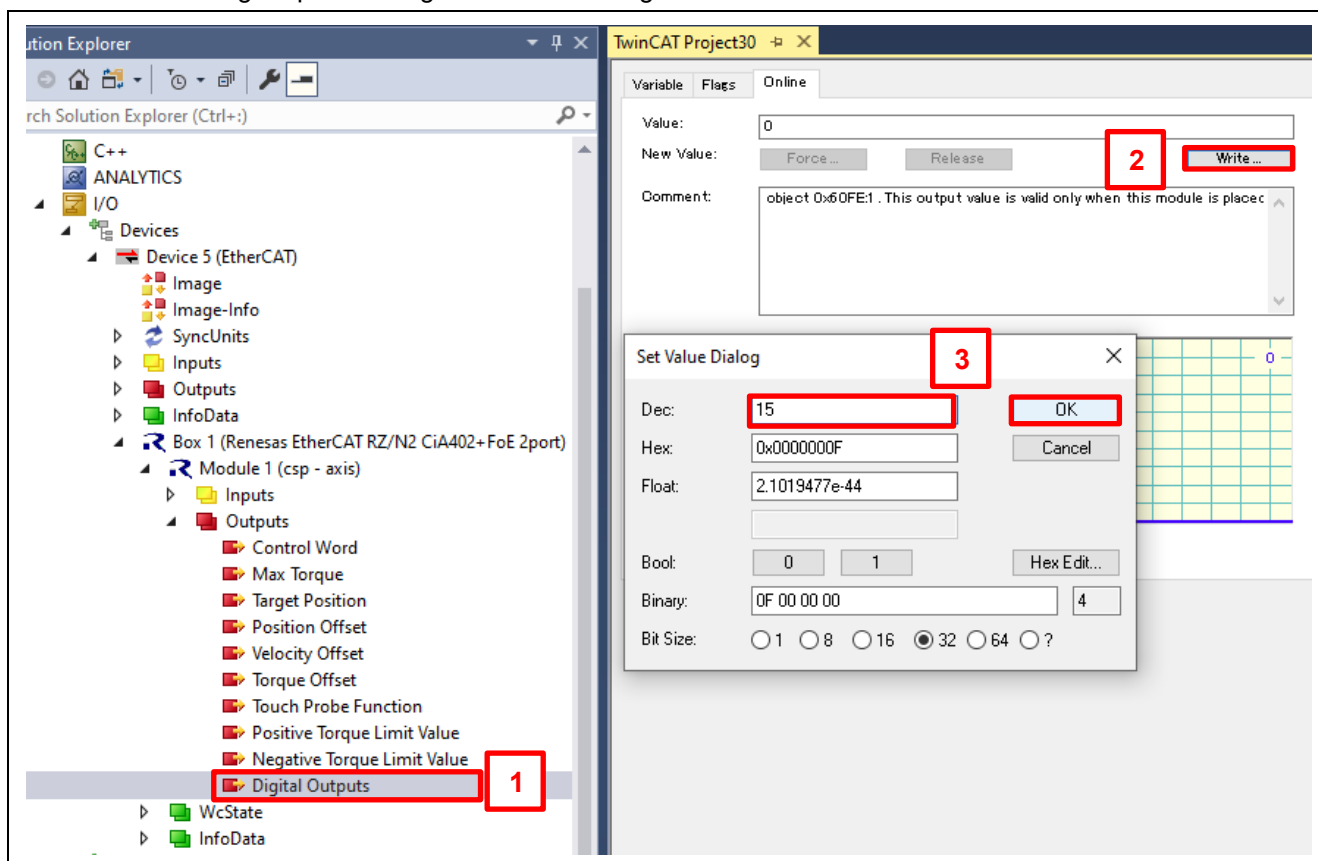
6.4 Testing the I/O Controller

I/O communication can be confirmed by the LED of the RSK board and the DIP SW.

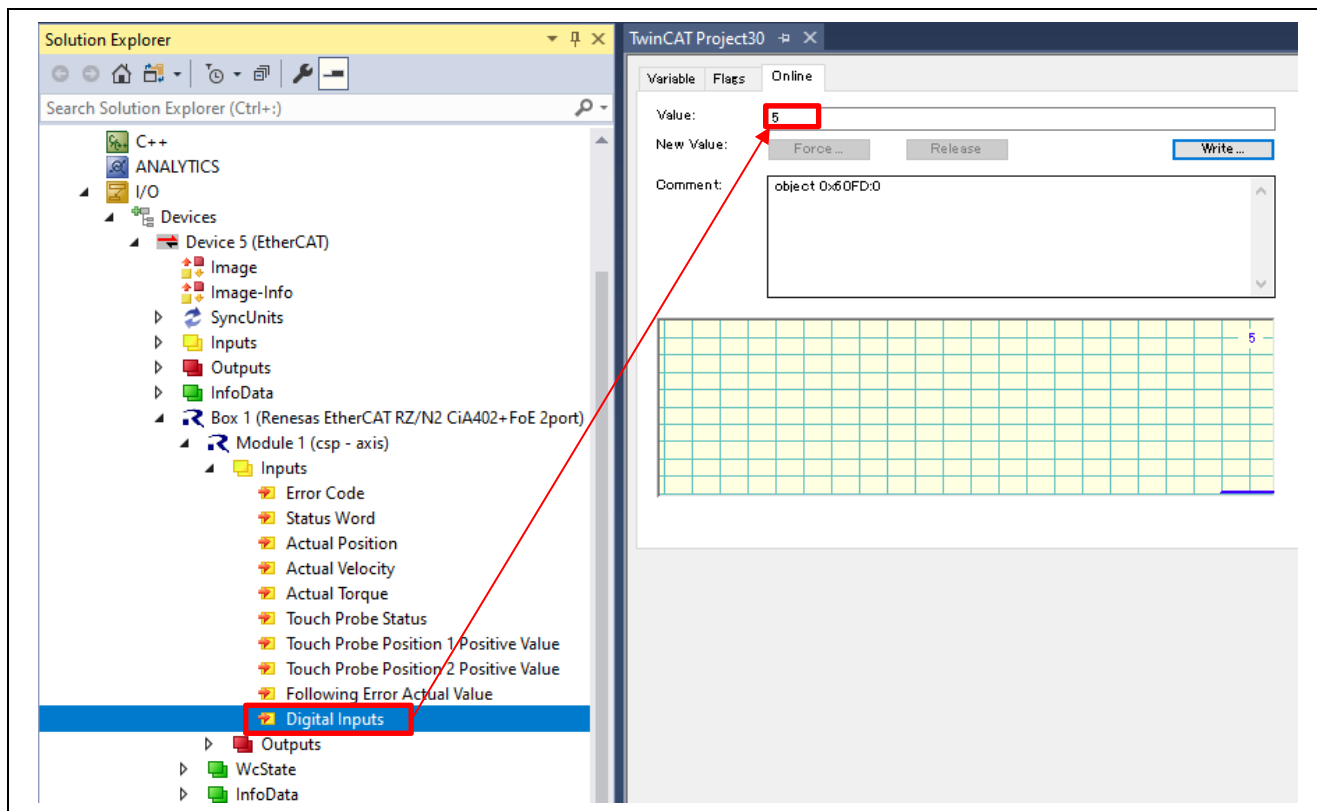


To confirm the I/O output, use TwinCAT3 to select [Digital Outputs] → [Online] → [Write] and enter the desired value.

LED1 and LED3 light up according to the 2-bit setting value.



To confirm the I/O input, use TwinCAT3 to select [Digital Inputs] → [Online] , 4-bit input value of Dip SW is displayed in Value.



6.5 CiA402 Drive Profile check

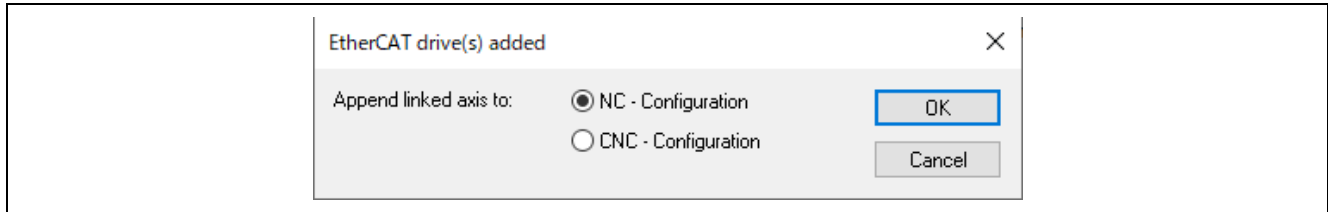
In this chapter, check the operation of the CiA402 drive profile.

6.5.1 Rescanning the Device

1. Press the [Restart TwinCAT (Config Mode)] button.
2. In the [Restart TwinCAT System in Config Mode] dialog box, click on [OK].
3. In the [Load I/O Devices] dialog box, click on [Yes].
4. In the [Active Free Run] dialog box, click on [Yes].

This will be OK if "Box 1" in the System Manager tree has turned to "Box 1 (RZ/N2 EtherCAT xxx)".

Note). When scanning the device, the CiA402's ESI is used, so the axis configuration settings are displayed.
Select [NC-Configuration]



6.5.2 Checking the Operation Mode

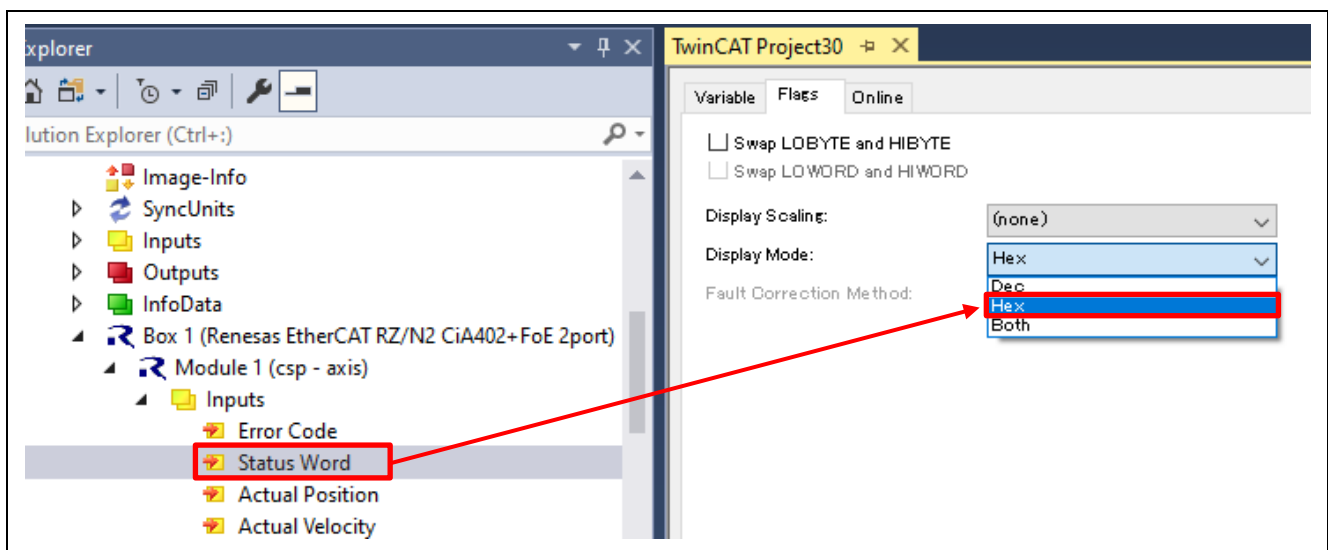
1. If double-click on [Box 1] in the System Manager tree, panel will be displayed on the right side of the screen.
2. Select the [Online] tab and check that "Current Status" has turned to "OP".
3. In the System Manager tree, expand + on the left side of "Box 1".

6.5.3 CiA402 State Transition

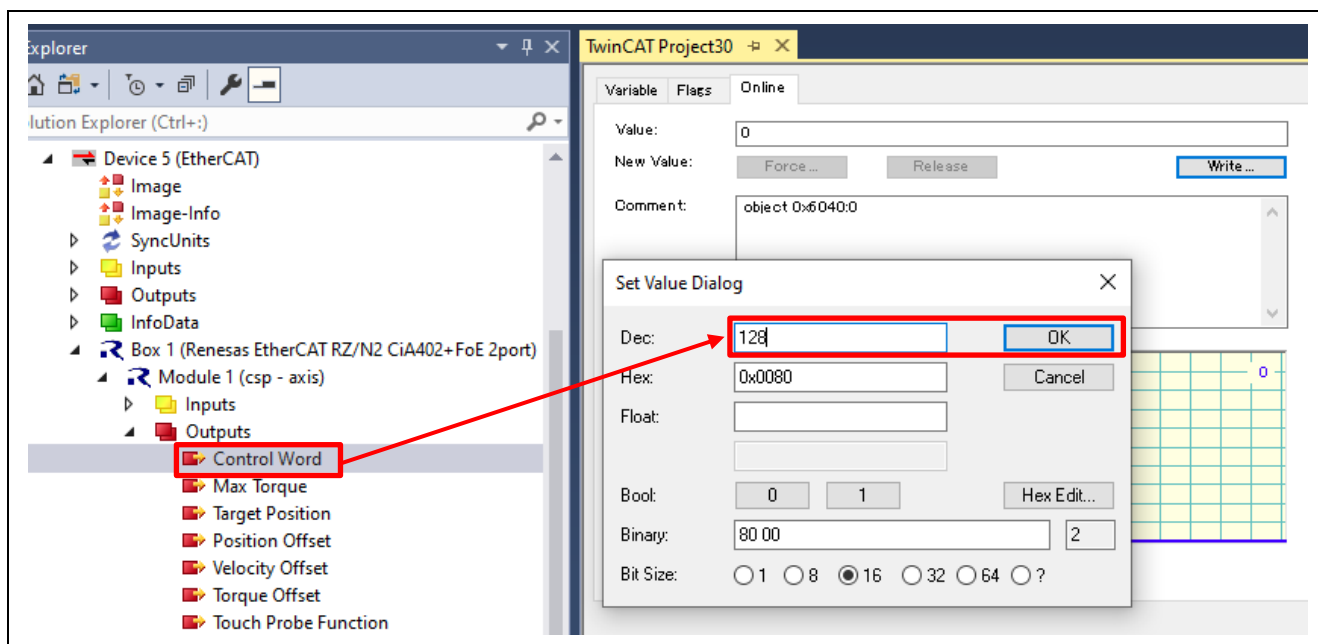
To check the operation in csp and csv modes, the state must be changed to "Operation Enabled" in both modes.

Change the state by setting a value in the "Control Word" object and check the state by confirming the value of the "Status Word" object.

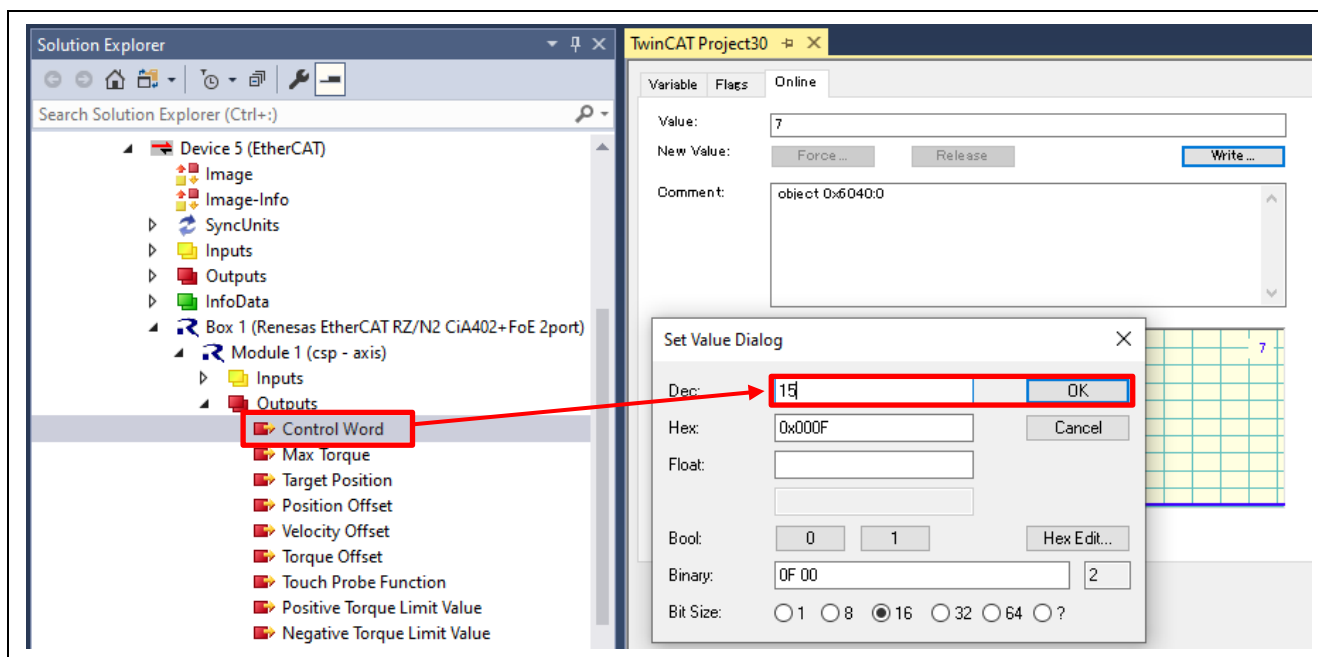
1. Select "Inputs" → "Status Word" in the Solution Explorer tree and then select the [Flags] tab on the right-side panel. value will then be displayed.
Change the display mode from [Dec] to [Hex]



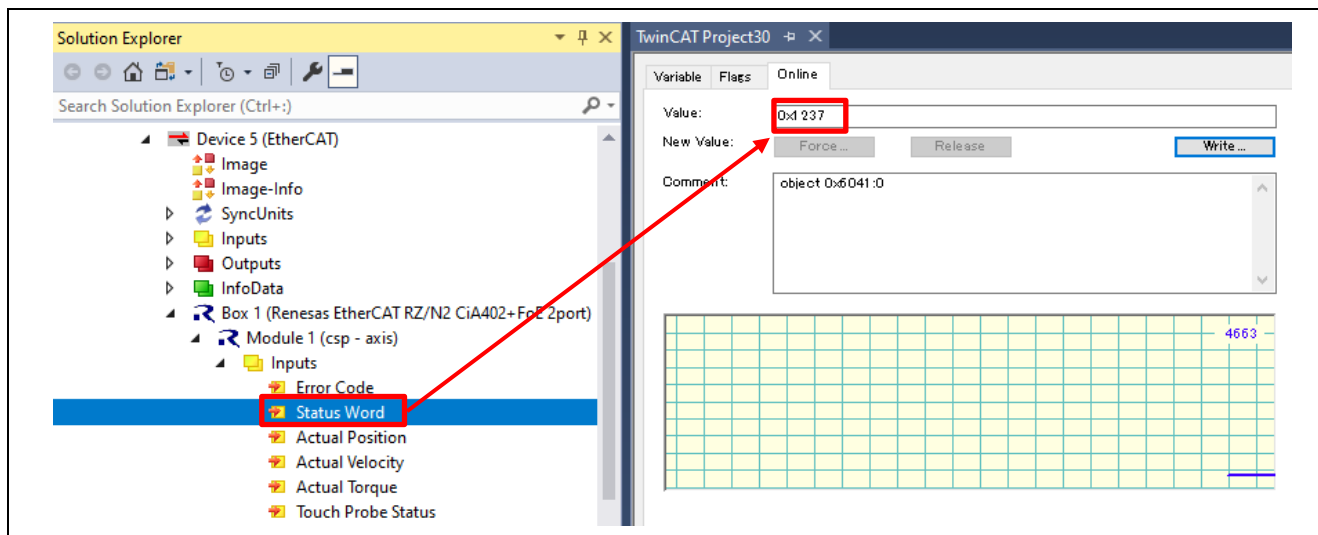
2. Check the operation of the profile. Initialize the control word by setting it to 128 (Dec).



3. Select "Outputs" → "Control Word" in the System Manager tree and then select the [Online] tab on the right-side panel. value will then be displayed.
Click on [Write] and set values from [7] to [15] in that order.



4. Select "Inputs" → "Status Word" in the System Manager tree and then select the [Online] tab on the right-side panel. value will then be displayed.
If the value is [0x1237], the state is "Operation Enabled". Proceed to the next step.
If the value is [0x1208], the state is "Fault" for some reason. Set "Control Word" to [128] once and return to step (1).



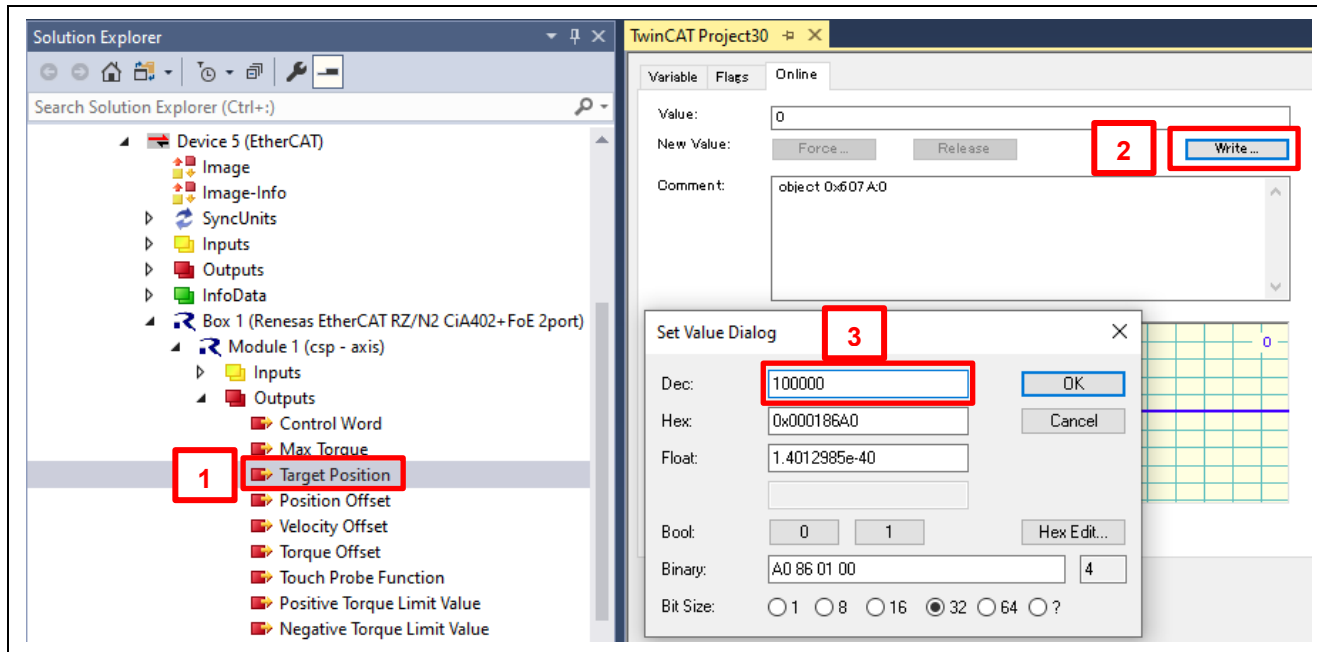
Note). Depending on PC, "abnormal state change (from 'OP' to 'SAFEOP') with ocde 0x1a, 'Synchronization error'." may be displayed on the TwinCAT3, there is no problem.

6.5.4 csp Mode

Check that "Module 1" is displayed as "Module 1 (csp-axis)".

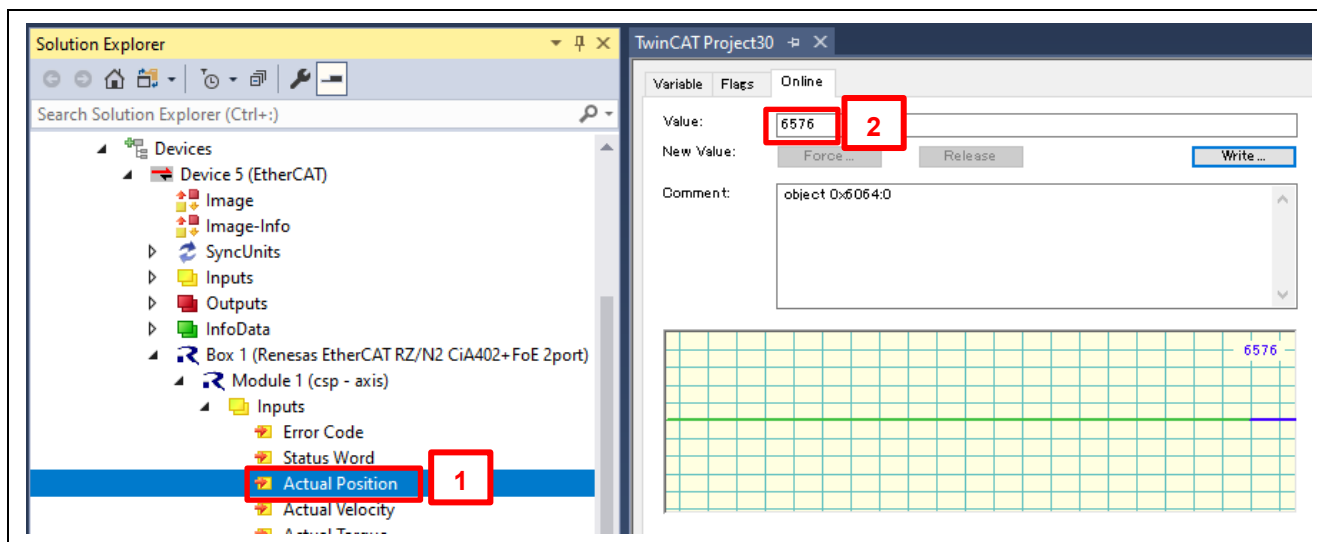
- (1) Select "Outputs" → "Target Position" in the System Manager tree and then select the [Online] tab on the right-side panel. A value will then be displayed.

Click on [Write] and set a desired value. As an example, set [100000] here.



- (2) Select "Inputs" → "Actual Position" in the System Manager tree and then select the [Online] tab on the right-side panel. A value will then be displayed.

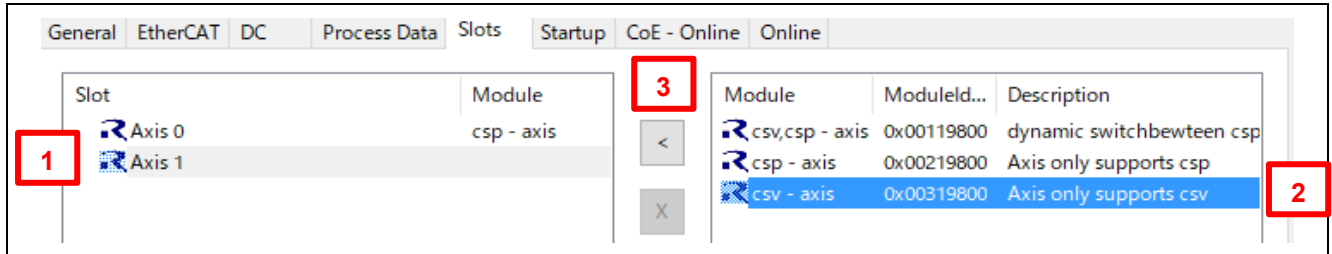
Check that the value set in "Target Position" allows incrementation up to [100000].



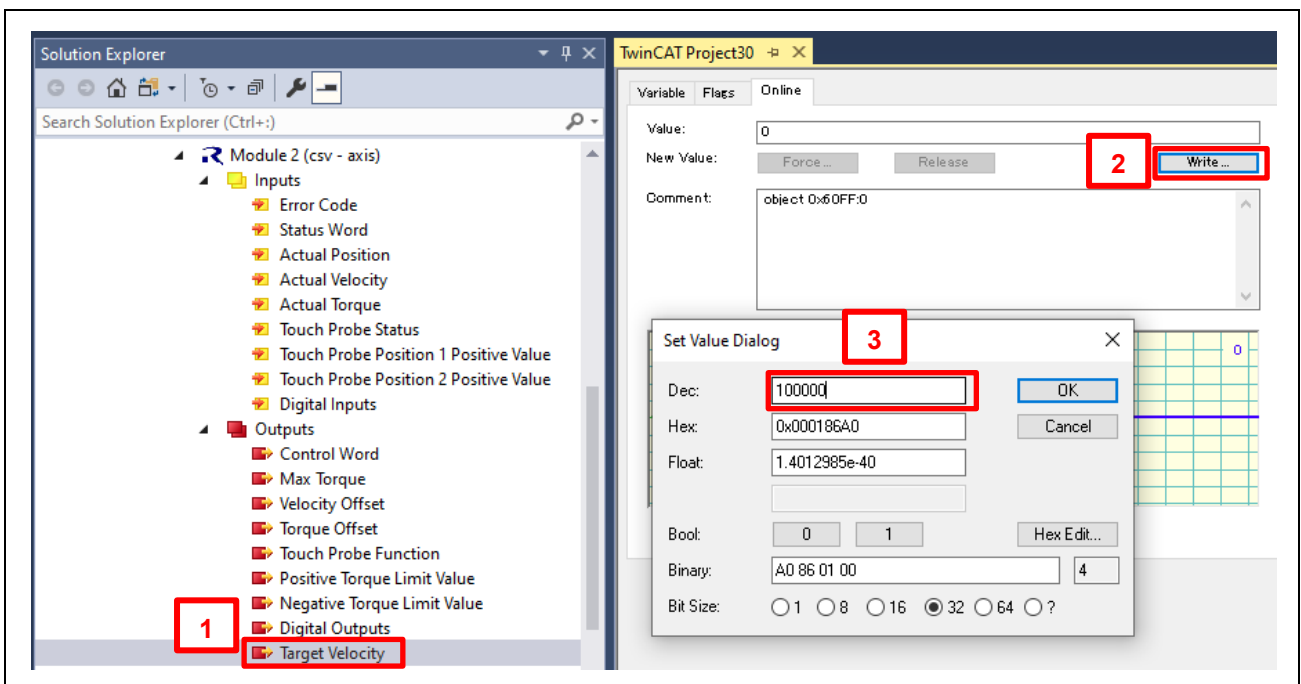
6.5.5 csv Mode

Change the operation mode

- (1) In the System Manager tree, select "Box 1 (RZ/N2 EtherCAT CiA402+FoE 2port)" and then select the [Slots] tab on the right-side panel.
- (2) If select "Axis 1" under "Slot" in the left frame of the tabbed page, modules which can be added will be displayed in the right frame of the page, so add "csv-axis".

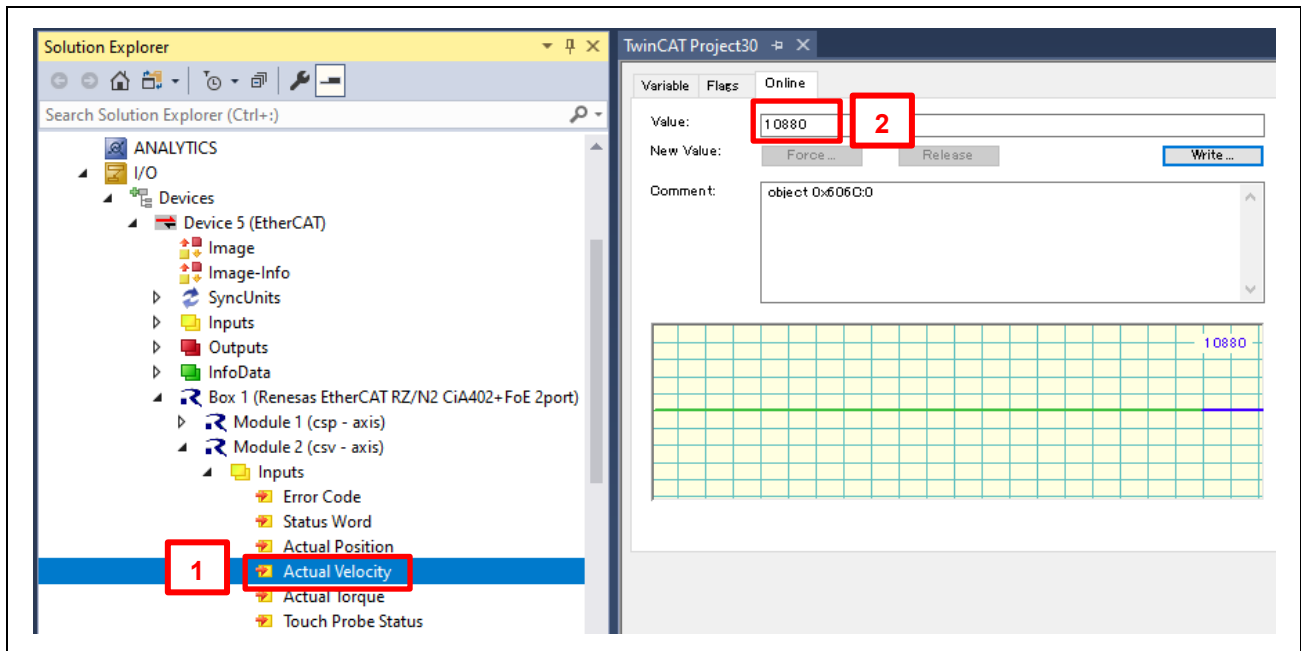


- (3) Check that "Module 2 (csv-axis)" has been added in the System Manager tree.
- (4) Execute "Reload Devices" to reflect the configuration.
- (5) Follow the procedure described in section 6.5.3 to change the state to "Operation Enabled".
- (6) Select "Outputs" → "Target Velocity" in the System Manager tree and then select the [Online] tab on the right-side panel. A value will then be displayed.
Click on [Write] and set a desired value. As an example, set [100000] here.



- (7) Select "Inputs" → "Actual Velocity" in the System Manager tree and then select the [Online] tab on the right-side panel. A value will then be displayed.

Check that the value set in "Target Velocity" allows incrementation up to [100000].



6.6 File over EtherCAT Profile check

In this chapter, check the operation of the Ethernet over EtherCAT(FoE) profile.
Follow the steps below to rewrite the firmware.

Note: When debugging, the operation of FoE will be failure.
Stop debugging, push the reset button on RSK board to reset RZ/N2L.

6.6.1 Confirm initial version

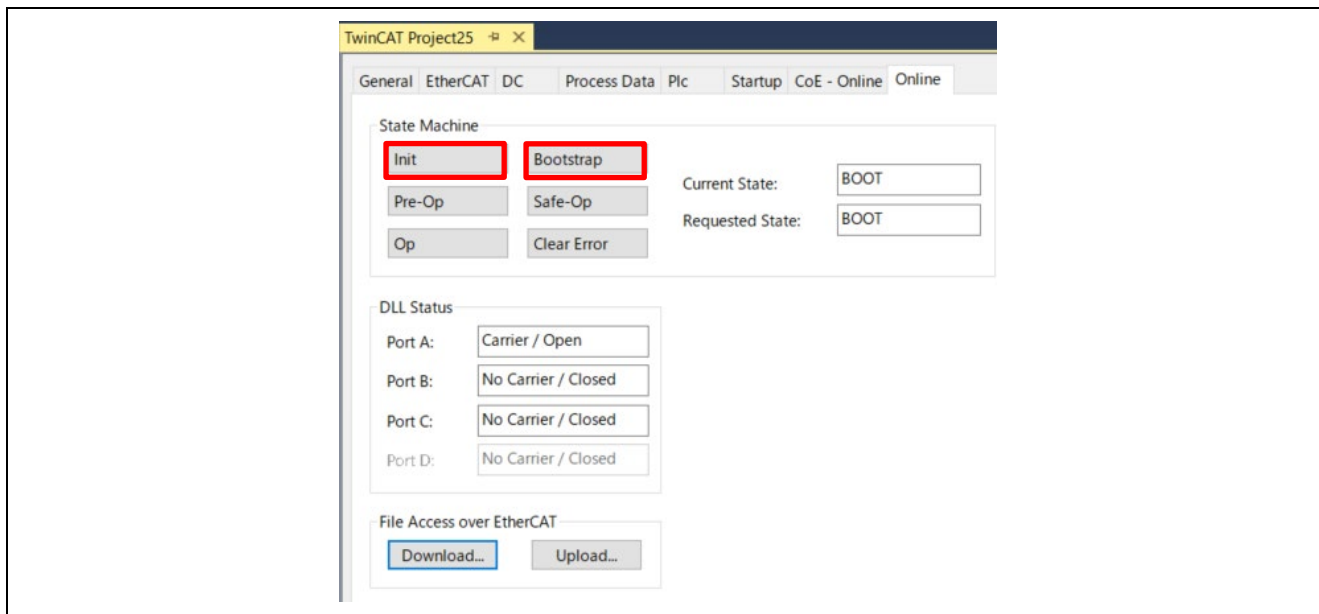
1. Confirm [Index 1018:0 Identify Object] on the [CoE-Online] tab.
2. Confirm [1018:03 Revision Number] is **0x00000300**. At this time, Bank0 is running.
Confirm [5000:00 Firmware Writable Bank] is **0x01**. At this time, you can do firmware update to BANK1.

The screenshot shows the TwinCAT Project30 software interface. On the left, the Solution Explorer displays the project structure, with 'Box 1 (Renesas EtherCAT RZ/N2 CiA402+FoE 2port)' selected under 'Devices'. On the right, the 'CoE-Online' tab is active, showing a table of indices and their values. A red arrow points from the selected device in the Solution Explorer to the 'CoE-Online' tab. The table lists various indices, with the 'Revision Number' (1018:03) and 'Firmware Writable Bank' (5000:00) highlighted with red boxes.

Index	Name	Flags	Value
1009	Manufacturer Hardware Version	RO	1.0
100A	Manufacturer Software Version	RO	2.0
1018:0	Identify Object	RO	> 4 <
1018:01	Vendor ID	RO	0x00000766 (1894)
1018:02	Product Code	RO	0x00000911 (2321)
1018:03	Revision Number	RO	0x00000300 (768)
1018:04	Serial Number	RO	0x00000000 (0)
10F1:0	Error Settings	RO	> 2 <
10F2	Timestamp Object	RWP	0xabd27c758ebf58c (7)
1600:0	csp/csv RxDPO Mapping	RO	> 5 <
1601:0	csp RxDPO Mapping	RO	> 10 <
1602:0	csp RxDPO Mapping	RO	> 9 <
1A00:0	csp/csv TxPDO Mapping	RO	> 5 <
1A01:0	csp TxPDO Mapping	RO	> 10 <
1A02:0	csp TxPDO Mapping	RO	> 9 <
1C00:0	Sync Manager Communication Type	RO	> 4 <
1C12:0	RxDPO Assign	RW	> 2 <
1C13:0	TxDPO Assign	RW	> 2 <
1C32:0	SM output parameter	RO	> 32 <
1C33:0	SM input parameter	RO	> 32 <
5000	Firmware Writable Bank	RO	0x01 (1)
603F	Error Code	RO P	0x0000 (0)
6040	Control Word	RWP	0x0000 (0)
6041	Status Word	RO P	0x0001 (1)

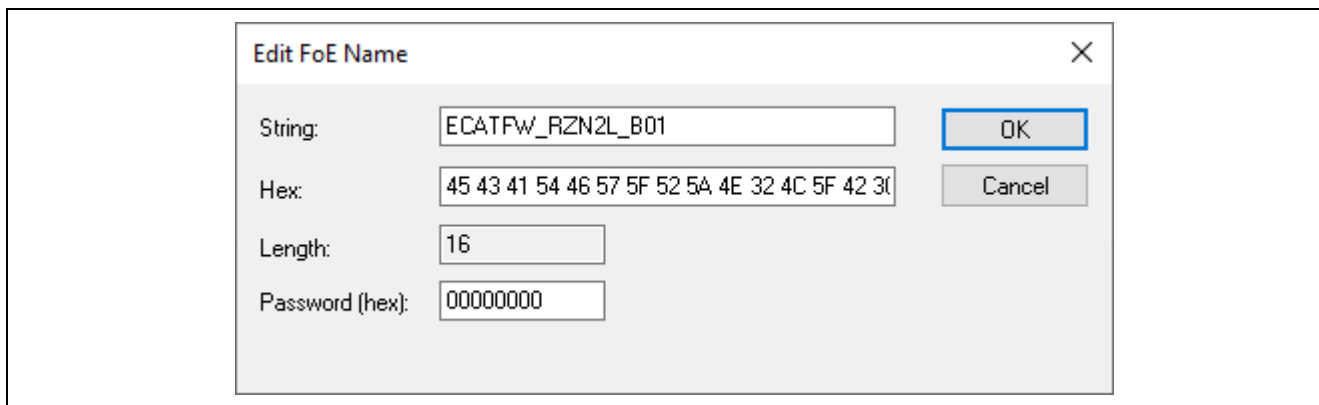
6.6.2 Firmware update

1. Select [Online] tab.
2. Press the [Init] → [Bootstrap] button to transition to boot mode.



3. In Click the Download button and select the following download file for BANK1.
e² studio:
[RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2\ecat_IO_CiA402_FoE\le2studio\
xSPI0\BANK1\ECAFW_RZN2L_B01.efw]
EWARM:
[RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2\ecat_IO_CiA402_FoE\lewarm\
xSPI0\BANK1\Exe\ECAFW_RZN2L_B01.efw]

4. Attributes of the update file are displayed, but if there are no changes, leave them.



5. After the download is complete, press the [Init] button.
RSK board will be software reset and the program will restart.

6.6.3 Check for updated version

1. Check the downloaded firmware.

Confirm that the [1018:03 Revision Number] is **0x00000301** on the [CoE-Online] tab. At this time, bank1 is running.

Confirm [5000:00 Firmware Writable Bank] is **0x00**. At this time, you can do wirmware updata to bank0.

Index	Name	Flags	Value
1009	Manufacturer Hardware Version	RO	1.0
100A	Manufacturer Software Version	RO	2.0
1018:0	Identity Object	RO	> 4 <
1018:01	Vendor ID	RO	0x00000766 (1894)
1018:02	Product Code	RO	0x00000911 (2321)
1018:03	Revision Number	RO	0x00000301 (769)
1018:04	Serial Number	RO	0x00000000 (0)
10F1:0	Error Settings	RO	> 2 <
10F8	Timestamp Object	RWP	0xabd2811c0c64b91 (773818767 ...)
1600:0	csp/csv RxDPO Mapping	RO	> 5 <
1601:0	csp RxDPO Mapping	RO	> 10 <
1602:0	csv RxDPO Mapping	RO	> 9 <
1A00:0	csp/csv TxPDO Mapping	RO	> 5 <
1A01:0	csp TxPDO Mapping	RO	> 10 <
1A02:0	csv TxPDO Mapping	RO	> 9 <
1C00:0	Sync Manager Communication Type	RO	> 4 <
1C12:0	RxDPO Assign	RW	> 2 <
1C13:0	TxDPO Assign	RW	> 2 <
1C32:0	SM output parameter	RO	> 32 <
1C33:0	SM input parameter	RO	> 32 <
5000	Firmware Writable Bank	RO	0x00 (0)
603F	Error Code	RO P	0x0000 (0)
6040	Control Word	RWP	0x0000 (0)

Note: When updating to bank0, use the following efw file.

e² studio:

[RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2l\ecat_IO_CiA402_FoE\e2studio\xSPI0\BANK0\ECAFW_RZN2L_B00.efw]

EWARM:

[RZN2L_EtherCAT_RSK_rev0200\project\rzn2l_rsk_rzn2l\ecat_IO_CiA402_FoE\ewarm\xSPI0\BANK0\Exe\ECAFW_RZN2L_B00.efw]

7. CiA402 Drive Profile

The CiA402 drive profile is a device profile for driving motors and motion control and mainly defines functional operations for servo drives, sine-wave inverters and stepping motor controllers. In this profile, the multiple operation modes and corresponding parameters are defined as an object dictionary. Also, Finite State Automaton (FSA) to define the internal and external behavior in every state is included. When changing the state, the result after transition is reflected in the status word object that shows the current state by specifying the state through the control word object. The control word and various command values (such as speed) are assigned to RxPDO, and the status word and various real values (such as position) are assigned to TxPDO. Please see the contents of the CiA402 standard for more details.

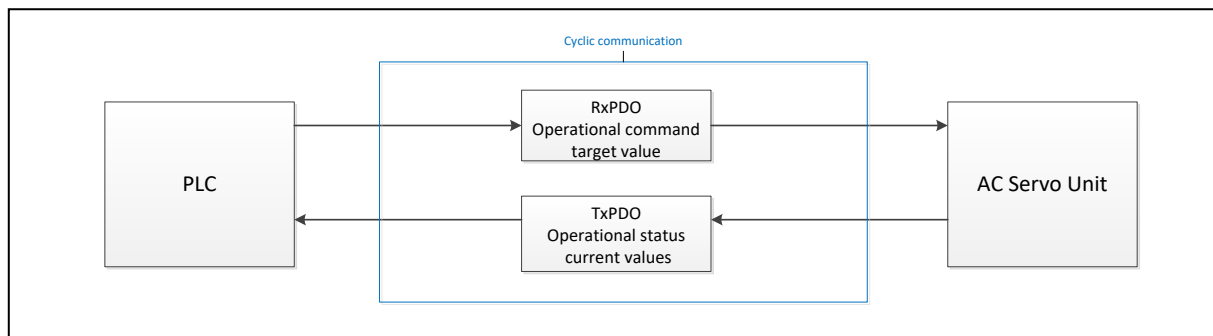


Figure 7.1 CiA402 Communication Flow

7.1 Operation Modes

In the application note, the following modes are supported from among the operation modes defined in the CiA402 standard.

Table 7.1 List of Supported Operation Modes

Operation Mode	Support
Profile position mode	No
Velocity mode (frequency converter)	No
Profile velocity mode	No
Profile torque mode	No
Homing mode	No
Interpolated position mode	No
Cyclic synchronous position mode	Yes
Cyclic synchronous velocity mode	Yes
Cyclic synchronous torque mode	No
Cyclic synchronous torque mode with commutation angle	No
Manufacturer specific mode	No

7.2 State Transition

In this application note, the following is supported as FSA defined in the CiA402 standard.

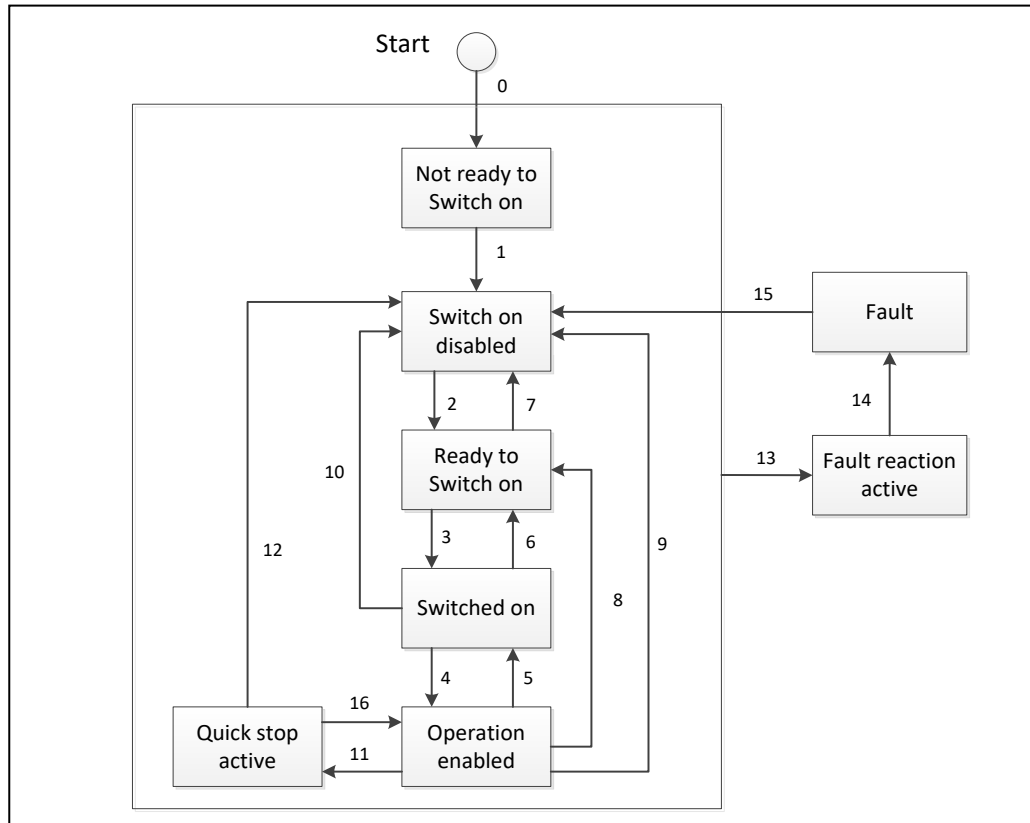


Figure 7.2 CiA402 State Transition Diagram

7.3 Object Dictionary [CiA402]

The following is the list of the object dictionaries related CiA402 and supported in this application note.

Table 7.2 List of Supported Object Dictionaries

Operation Mode	OBJECT Name	INDEX	Category	Access	Data Type	PDO Mapping
Cyclic synchronous position mode + Cyclic synchronous velocity mode	Position actual value	0x6064	Mandatory	ro	INT32	Yes
	Following error window	0x6065	Optional	rw	UINT32	No
	Following error time out	0x6066	Conditional	rw	UINT16	No
	Velocity actual value	0x606C	Conditional	ro	INT32	Yes
	Max torque	0x6072	Optional	rw	UINT16	Yes
	Torque actual value	0x6077	Conditional	ro	INT16	Yes
	Target position	0x607A	Optional	rw	INT32	Yes
	Position range limit	0x607B	Conditional	c,rw	INT32	Yes
	Software position limit	0x607D	Optional	c,rw	INT16	Yes
	Position offset	0x60B0	Optional	rw	INT32	Yes
	Velocity offset	0x60B1	Optional	rw	INT32	Yes
	Torque offset	0x60B2	Optional	rw	INT16	Yes
	Interpolation time period	0x60C2	Conditional	c,rw	UINT8	Yes
	Following error actual value	0x60F4	Optional	ro	INT32	Yes
	Target velocity	0x60FF	Conditional	rw	INT32	Yes

Function Group	OBJECT Name	INDEX	Category	Access	Data Type	PDO Mapping
Torque Limiting	Positive torque limit value	0x60E0	Conditional	rw	UINT16	Yes
	Negative torque limit value	0x60E1	Conditional	rw	UINT16	Yes
Homing	Home Offset	0x607C	Optional	rw	INT32	No
	Homing speeds	0x6099	Conditional	c,rw	UINT32	No
Touch Probe	Touch probe function	0x60B8	Optional	rw	UINT16	Yes
	Touch probe status	0x60B9	Optional	ro	UINT16	Yes
	Touch probe position 1 positive value	0x60BA	Optional	ro	INT32	Yes
	Touch probe position 1 negative value	0x60BB	Optional	ro	INT32	Yes
	Touch probe source	0x60D0	Conditional	c,rw	INT16	No
Gear ratio	Gear ratio	0x6091	Optional	c,rw	UINT32	No
Other object	OBJECT Name	INDEX	Category	Access	Data Type	PDO Mapping
Controlling the power drive system	Error code	0x603F	Optional	ro	UINT16	Yes
	Controlword	0x6040	Mandatory	rw	UINT16	Yes
	Statusword	0x6041	Mandatory	ro	UINT16	Yes
	Quick stop option code	0x605A	Optional	rw	INT16	No
	Shutdown option code	0x605B	Optional	rw	INT16	No
	Disable operation option code	0x605C	Optional	rw	INT16	No
	Halt option code	0x605D	Optional	rw	INT16	No
	Fault reaction option code	0x605E	Optional	rw	INT16	No
	Modes of operation	0x6060	Optional	rw	INT8	Yes
	Modes of operation display	0x6061	Optional	ro	INT8	Yes
	Supported drive modes	0x6502	Mandatory	ro	INT32	No
General object	Motor type	0x6402	Optional	rw	INT16	No
Position control function	Position demand value	0x6062	Optional	ro	INT32	No
	Position actual internal value	0x6063	Optional	ro	INT32	No
	Position window	0x6067	Optional	rw	UINT32	No
Optional application FE	Digital inputs	0x60FD	Optional	ro	UINT32	Yes
	Digital outputs	0x60FE	Optional	c,rw	UINT16	No, Yes

7.4 Implementing the Motor Control Program

According to the CiA402 standard from the list of CiA402 protocol stack I/F functions in Table 7.3, implement the motor control application. Each function links the number of each state transition of CiA402 FSA shown in Figure 7.2 and the corresponding function is called in case of state transition. In each function, describe the processing that calls the motor control program or the relevant processing of the main CPU.

Table 7.3 List of CiA402 Protocol Stack I/F Functions

CiA402_StateTransition1	
	<u>Description</u> This function is used when state transition 1 has occurred. Describe the operation in the case of the state transition.
	<u>Usage</u> #include "cia402appl.h"
	<u>Parameters</u> TCiA402Axis *pCiA402Axis
	<u>Return Value</u> 0 Normal end 1 Error
	<u>Remark</u> <u>Remark</u> In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition2	
	<u>Description</u> This function is used when state transition 2 has occurred. Describe the operation in the case of the state transition.
	<u>Usage</u> #include "cia402appl.h"
	<u>Parameters</u> TCiA402Axis *pCiA402Axis
	<u>Return Value</u> 0 Normal end 1 Error
	<u>Remark</u> In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition3	
<u>Description</u>	This function is used when state transition 3 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition4	
<u>Description</u>	This function is used when state transition 4 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition5	
<u>Description</u>	This function is used when state transition 5 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition6	
<u>Description</u>	This function is used when state transition 6 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition7	
<u>Description</u>	This function is used when state transition 7 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition8	
<u>Description</u>	This function is used when state transition 8 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition9	
<u>Description</u>	This function is used when state transition 9 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition10	
<u>Description</u>	This function is used when state transition 10 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition11	
<u>Description</u>	This function is used when state transition 11 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition12	
<u>Description</u>	This function is used when state transition 12 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_LocalError	
<u>Description</u>	This function is used when state transition 13 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	UINT16 ErrorCode
<u>Return Value</u>	None
<u>Remark</u>	If the error corresponding to state transition 13 occurs, call this function after processing required and saving data at error location.
CiA402_StateTransition14	
<u>Description</u>	This function is used when state transition 14 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition15	
<u>Description</u>	This function is used when state transition 15 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition16	
<u>Description</u>	This function is used when state transition 16 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
APPL_MOTOR_MotionControl_Main	
<u>Description</u>	Implement the motion control code when the state of CiA402 FSA is "Operation enabled". Describe the process for each mode of operation.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	At the initial state, this function is described in "main.c" and calls "CiA402_DummyMotionControl" function for reference.

8. The details of updating the Firmware

The sample program can be used to update the firmware of the slave as described below.

A slave vendor is able to provide an updating firmware file and password to a user, while the user is able to download the firmware to a slave by using the FoE from a master such as TwinCAT.

The file for updating the firmware has a checksum, which allows checking the validity of received data.

The updated firmware is written to a different area from that for the factory-default firmware in the serial flash ROM. After the update, the user application program in the form of the updated firmware is loaded to the ATCM to run through the sequence of booting.

If updating fails, restoration of the firmware written at the time of shipment is possible.

8.1 Procedure for Updating the Firmware

This section describes the procedure for updating the firmware for a slave and operations of the EtherCAT master and slave during the procedure. "Function" in the table shows which program is used to implement the corresponding slave operation.

Table 8.1 Procedure for Updating the Firmware

No	Master/User	Slave	Function		
			SSC	FW boot loader	FW updater
1	Request BOOT	Confirm BOOT	✓		
2	Download new slave FW	Download new slave FW			
		(1) Check filename			✓
		(2) Check password			✓
		(3) Compare checksum of receiving data			
		(4) Write file data to serial flash			✓
3		Update SII			✓
4	Request INIT	Reboot			
		(1) download new firmware to internal RAM		✓	
		(2) Start new FW		✓	
5	Request PREOP	Check if SII and firmware match	✓		
6		Confirm PREOP	✓		
7	User: Check firmware version				
8	Request SAFEOP	Confirm SAFEOP	✓		
9	Request OP	Confirm OP	✓		

1. Request the BOOT state.

Make the transition to the BOOT state for execution of the FoE.

2. Download new slave firmware.

Download the new updating firmware from the master.

The slave checks if (1) the filename and (2) the password are correct. If they are correct, it (3) compares checksum of receiving data with calculated checksum and (4) writes data to the serial flash ROM.

3. Update the EEPROM.

Write the revision number of the new firmware to the EEPROM.

4. Request the INIT state.

After the transition from the BOOT to the INIT state, the slave is rebooted, and (1) downloads the program code from the serial flash ROM to the internal ROM then (2) operates with the new firmware.

5. Request the PREOP state.

Check if the revision number in the EEPROM matches that of the firmware.

6. Confirm the PREOP state.

Confirm the transition to the PREOP state.

7. User: Check firmware version.

The user can check whether the firmware has been updated to the new version by reading the value at 0x100A through the CoE object.

The user can check the revision number by reading the value at 0x1018:03 (refer to chapter 11.3.1).

8. Request the SAFEOP state.

Make the transition to the SAFEOP state.

9. Request the OP state.

Make the transition to the OP state.

8.2 Hardware Configuration of the Sample Program

8.2.1 Booting from the Serial Flash ROM

To start the boot loader in the serial flash ROM, booting must be set to select xSPI0 boot mode (x1 boot serial flash).

8.2.2 Memory Map of Serial Flash ROM

Usage of the serial flash ROM is divided into three different areas.

Table 8.2 Classification of the Serial Flash ROM Areas

Address Range	Name (Size)	Description
6000_0000H to 6000_FFFFH	Boot loader parameter area (64 KB)	Parameter area for the boot loader to be referred to by the boot function of the RZ/T2 or RZ/N2 Note: BANK 0 and BANK 1 are referred to at the time of shipment and firmware updating, respectively.
6010_0000H to 601F_FFFFH	BANK 0 area (1MB)	Factory-default firmware area to be written by a serial flash ROM writer, ICE, etc. ※This area can also be written by firmware update using FoE.
6020_0000H to 602F_FFFFH	BANK 1 area (1MB)	Updating firmware area to be written by the FoE

Figure 8.1 Memory Map, shows the memory map of the serial flash ROM.

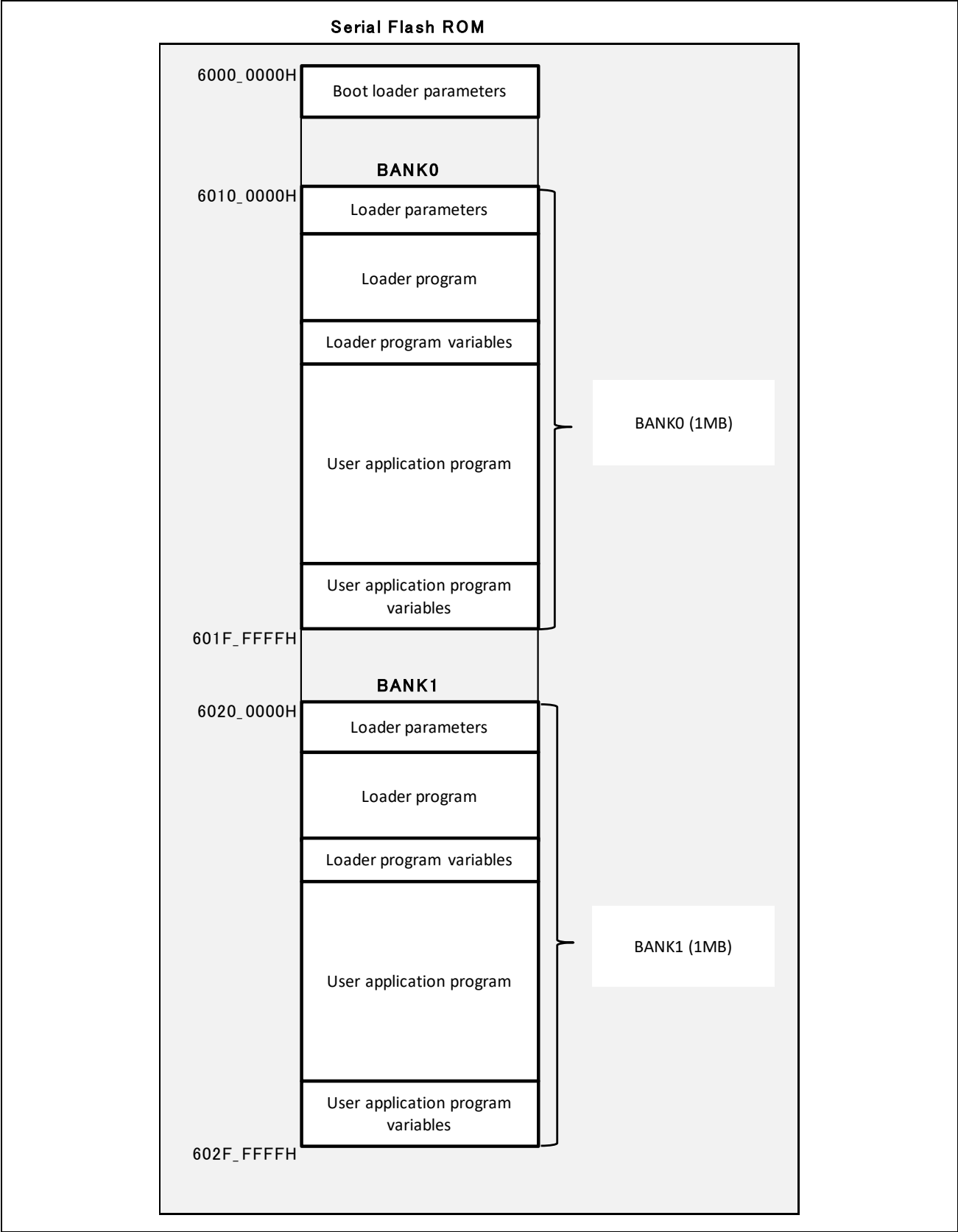


Figure 8.1 Memory Map

8.2.3 Overview of Bank 0 Boot Operations

This following describes operation for booting the factory-default firmware written to bank 0 through the procedure illustrated in Figure 8.2 BANK 0 Boot Operations.

The boot function of the RZ/T2 or RZ/N2

- (1) refers to the values in the boot loader parameter area,
- (2) transfers the loader program in bank 0 to the BTCM, and then.
- (3) hands processing to the loader program.

After initializing the various stack pointers, the loader program

- (4) transfers the loader program variables to the BTCM and makes settings for peripheral modules, etc.
It also refers to the values in the boot loader parameter area,
- (5) transfers the user application program to the System RAM, and then.
- (6) hands processing to the user application program.

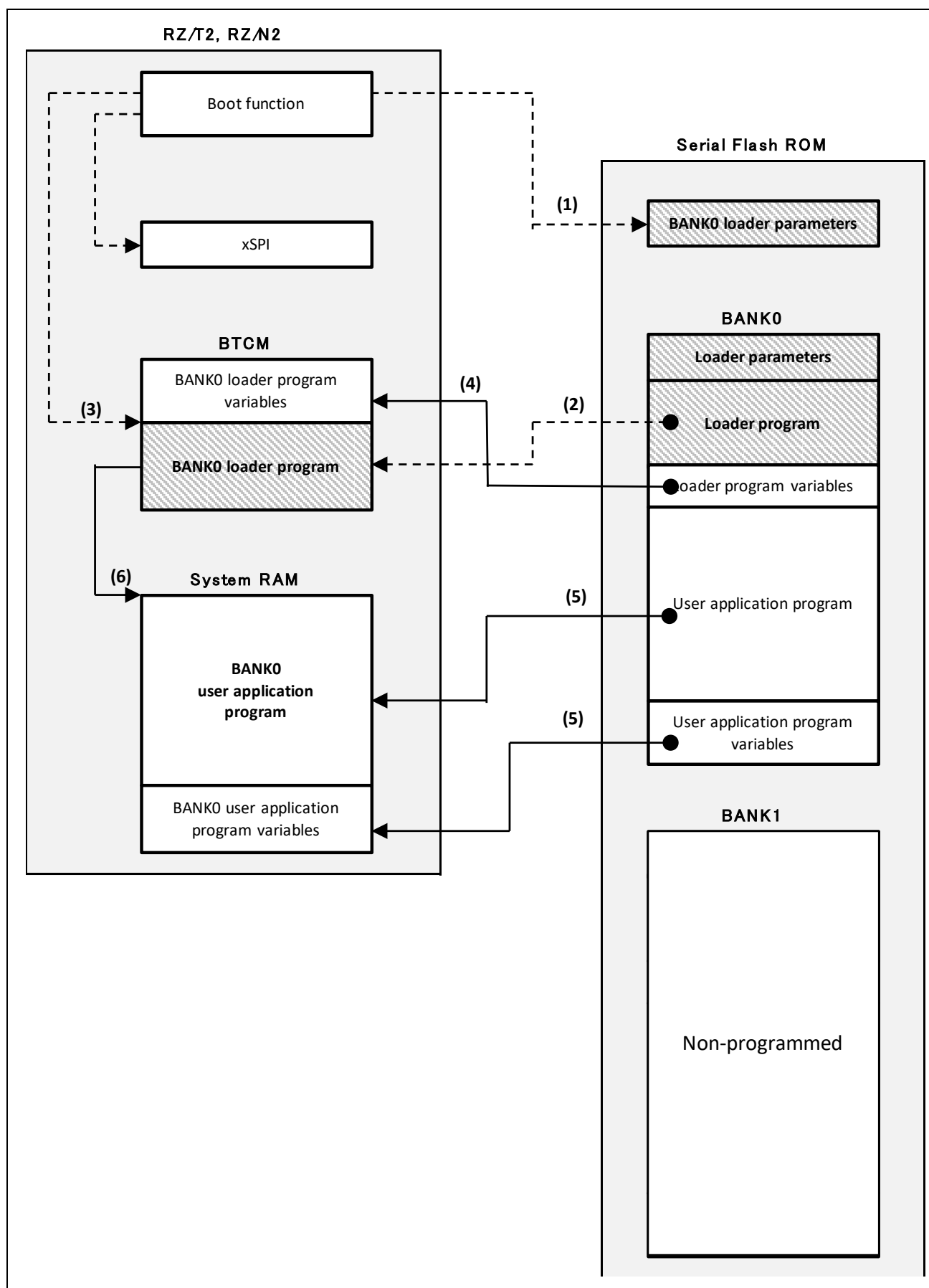


Figure 8.2 BANK 0 Boot Operations

8.2.4 Overview of Operation for Updating Firmware in Bank 1

The following describes operation for updating the firmware by using the FoE while a user application program is running from the System RAM, with the procedure illustrated in Figure 8.3, Operation for Updating the Firmware in Bank 1.

- (1) The master sends the filename and password of the binary file for the updating firmware at the opening of the FoE, so check if the prefix of the filename and the password are correct. If they are correct, reception of binary data starts.
- (2) Erase the boot loader parameter area at the beginning of the serial flash ROM.
- (3) If the firmware update is interrupted for any reason, copy the BANK0 loader parameters to the boot loader parameter area so that the factory firmware can be started.
- (4) One sector (64 KB) is erased from the address where bank 1 starts. During erasure, the busy status indicator is returned so that the master does not reach the timeout time.
- (5) Data are received on completion of the erasure. The checksum attached to the received data is compared with the calculated. The data are stored in the reception buffer allocated for the storage of user application program variables in the System RAM. The data in the reception buffer are written to the serial flash ROM every time one page (256 bytes) of data is accumulated. An ACK packet is returned to the master.
On completion of writing one sector of data, erase the next sector (Figure 8.3 -(5) in the figure).
- (6) Repeat step 3 until writing all receiving data ends.
- (7) Erase the boot loader parameter area at the start of the serial flash ROM.
- (8) Copy the bank 1 loader parameters to the boot loader parameter area (Figure 8.3-(8) in the figure).
- (9) Update the revision number in the SII memory to the revision number of the updating firmware.

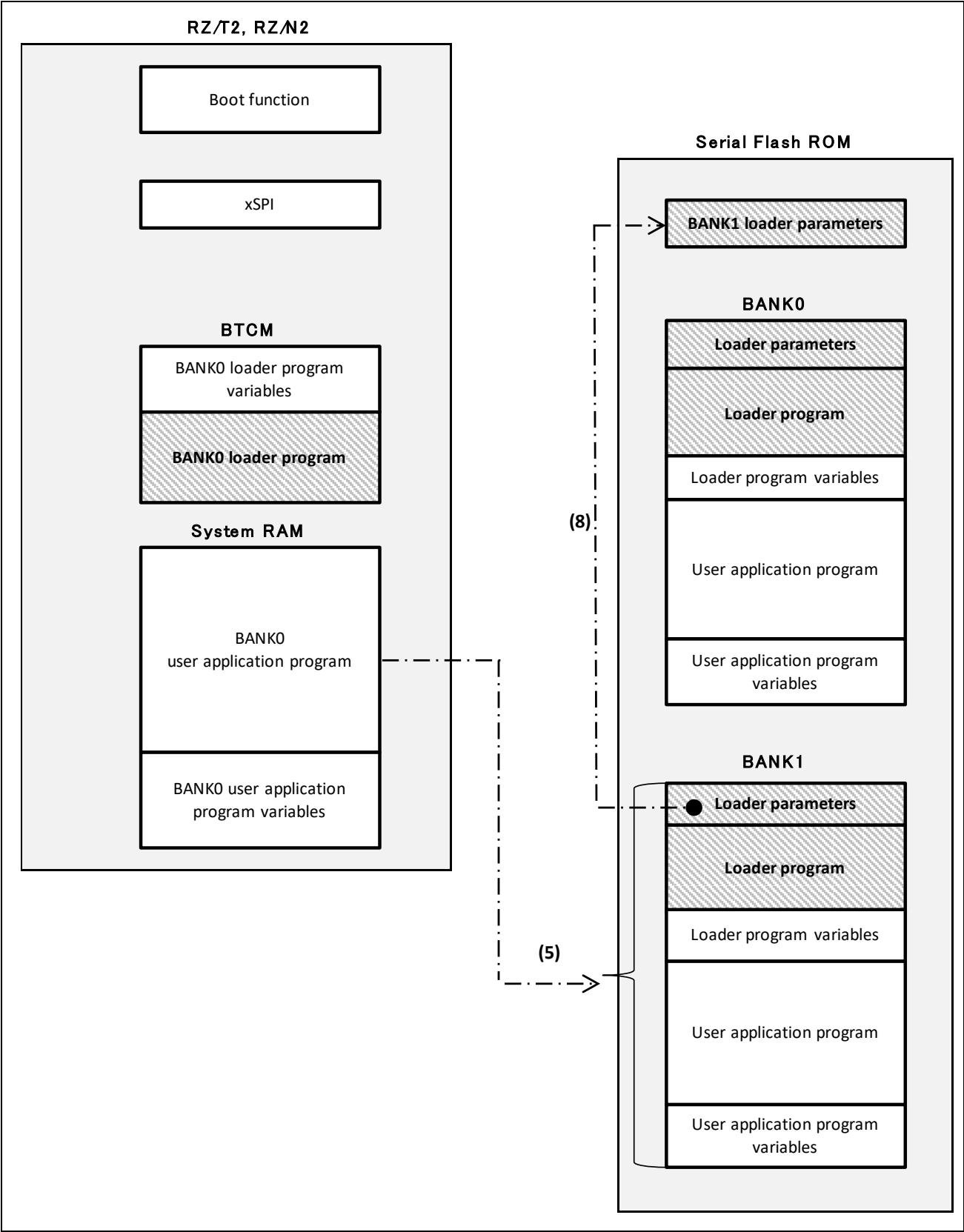


Figure 8.3 Operation for Updating Firmware in Bank 1

8.2.5 Overview of Operations to Reboot from Bank 1

The following describes operation from writing of the updating firmware to bank 1 until rebooting, with the procedure illustrated in Figure 8.4, Operations to Reboot from Bank 1.

After updating of the firmware in bank 1 has been completed normally,

The user application program operates rebooting by software reset.

The following describes operation for booting from bank after rebooting, with the procedure illustrated in Figure 8.4

- (1) The boot function refers to the values in the boot loader parameter area,
- (2) transfers the loader program in bank 1 to the BTCM, and then.
- (3) hands processing to the loader program.

After initializing the various stack pointers, the loader program

- (4) transfers the loader program variables to the BTCM and makes settings for peripheral modules, etc.
It also refers to the values in the boot loader parameter area,
- (5) transfers the user application program to the System RAM, and then.
- (6) hands processing to the user application program.

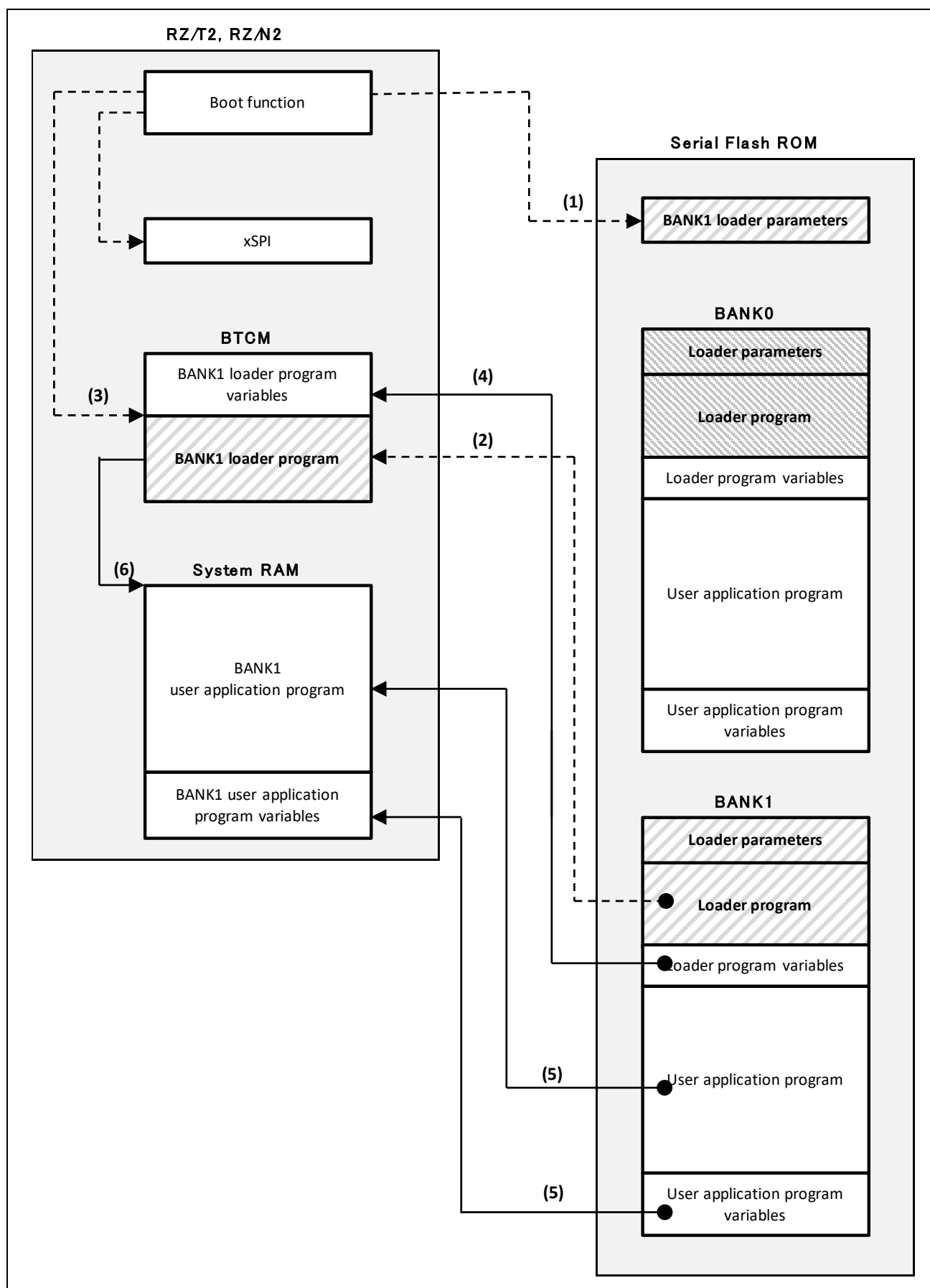


Figure 8.4 Operations to Reboot from Bank 1

8.2.6 Loader Parameters

Table 8.3 shows the loader parameter information for switching the bank area to be executed.

Table 8.3 Loader Parameter Information in xSPI boot mode

Parameter Name	Offset Address	Description
LDR_ADDR_NML	0000_0014H	Sets the address where the loader program starts. BANK0 : 0x6010_005C BANK1 : 0x6020_005C

8.3 Constants

Table 8.4 Constants Used in the Sample Program

Constant Name	Setting	Description
FW_UP_PAGE_SIZE	(256)	Page size of the serial flash ROM
FW_UP_LDRPRM_ADDR	(0x60000000)	Loader parameters address
FW_UP_LDRPRM_LDR_ADDR	(0x60000014)	LDR_ADDR_NML address
FW_UP_APPLI_ID_OFFSET	(0x6000004C)	Application identifies information placed address
FW_UP_BANK0_ADDR	(0x60100000)	Address where bank 0 starts
FW_UP_BANK1_ADDR	(0x60200000)	Address where bank 1 starts
FW_UP_MIRROR_OFFSET	(0x20000000)	xSPI0 Mirror space minus offset value
FW_UP_APPLI_SIZE	(0x00100000)	Bank area size (1MB)

8.4 Functions

The tables below list the functions related to the boot loader and updating of the FoE firmware.

Table 8.5 List of Functions Related to Updating of the FoE Firmware

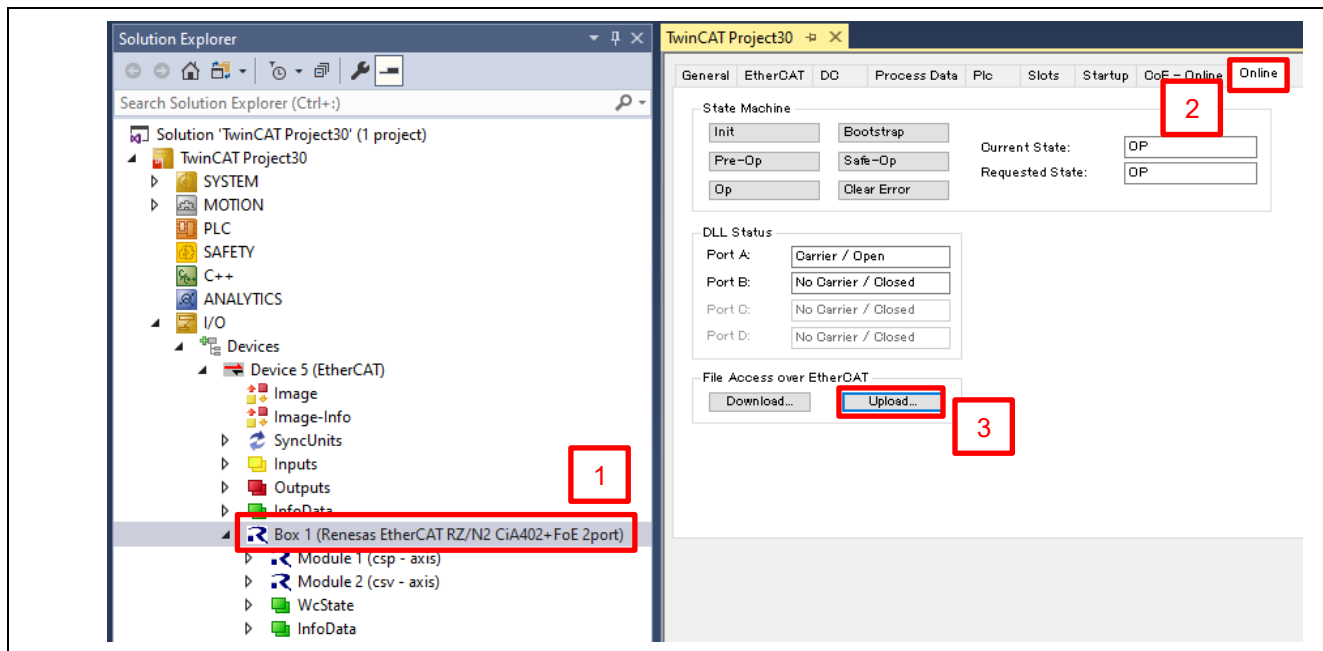
Function Name	Outline
BL_Start	Handles processing to start the transition from the INIT to the BOOT state.
BL_StartDownload	Handles processing to start downloading the FoE file data.
BL_Data	Handles processing to receive the FoE file data.
BL_Data_write	Handles processing to write the file data to the serial flash ROM.
BL_SetRebootFlag	Sets the reboot flag.
BL_CheckRebootFlag	Checks the reboot flag.
BL_Reboot	Reboot processing (BOOT -> INIT)

9. Appendix

9.1 Appendix A : File Access over EtherCAT FW Upload

This appendix describes FW upload using File Access over EtherCAT.

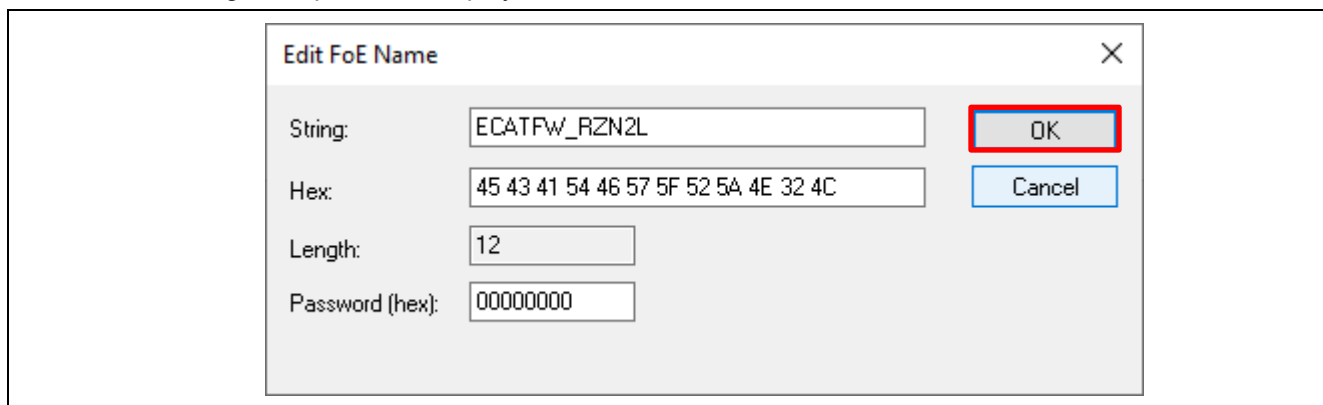
1. Select [Online] → [Upload].



2. Set filename and save. the file must start with "ECATFW_RZN2L."



3. File settings for upload are displayed. Press OK to save.



Note). The upload function is for checking the function, so the loaded file cannot be downloaded again.

9.2 Appendix B : Setting 3port

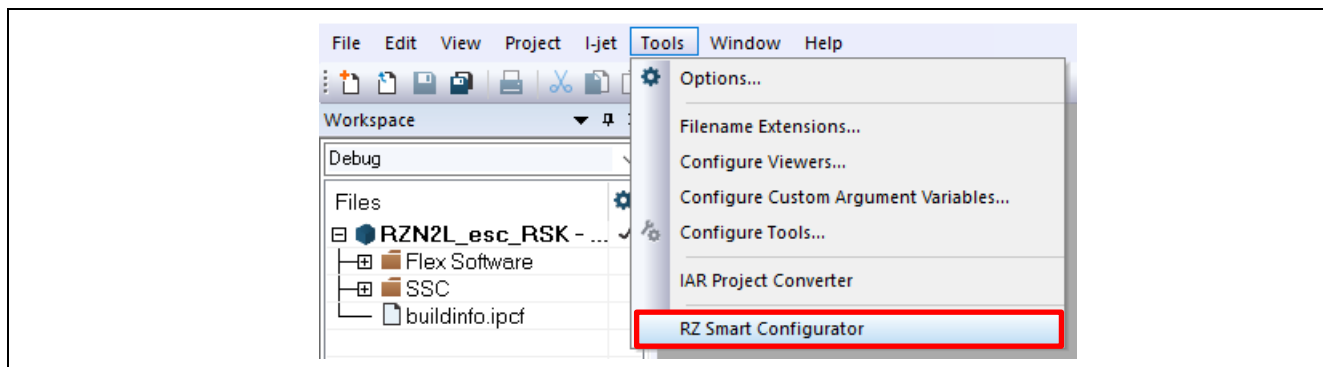
In the FSP2.0.0 compatible sample, it can change the settings from the smart configurator to change the PHY settings and the number of ports.

This appendix shows the addition of the 3rd port of EtherCAT.

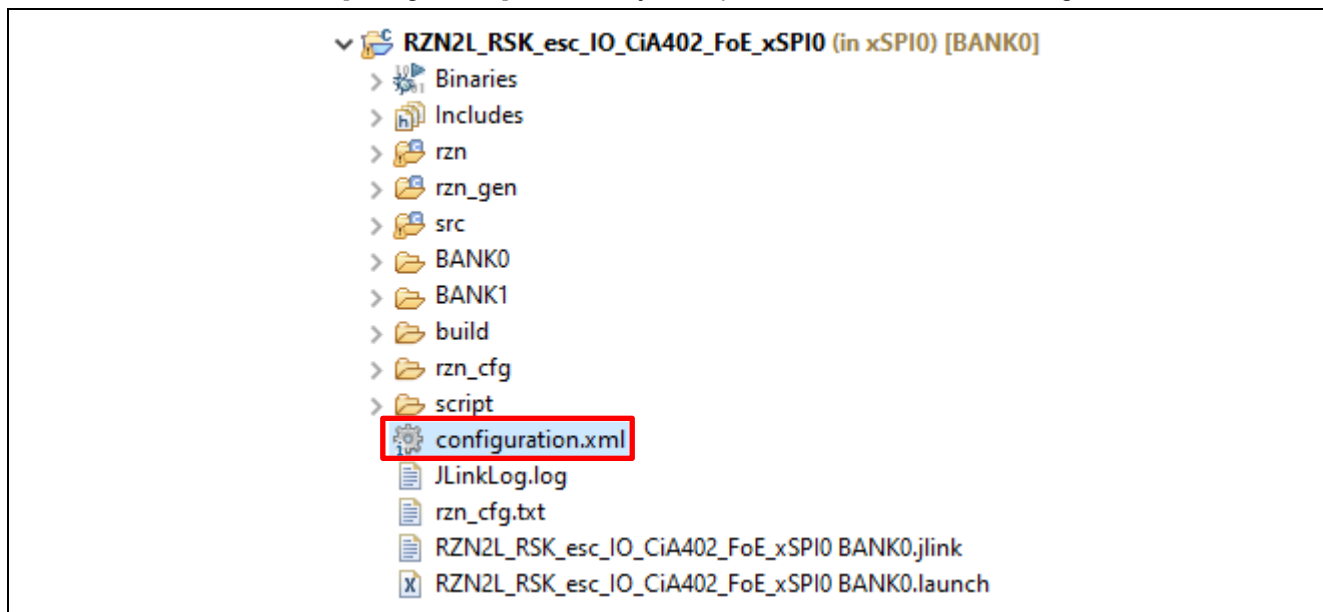
Display the stack diagram using FSP Configuration. Initially, it is set to 2 ports.

Method of starting smart configurator depends on the tool.

For EWARM, select [Tool]- [RZ smart configurator] to start smart configurator.



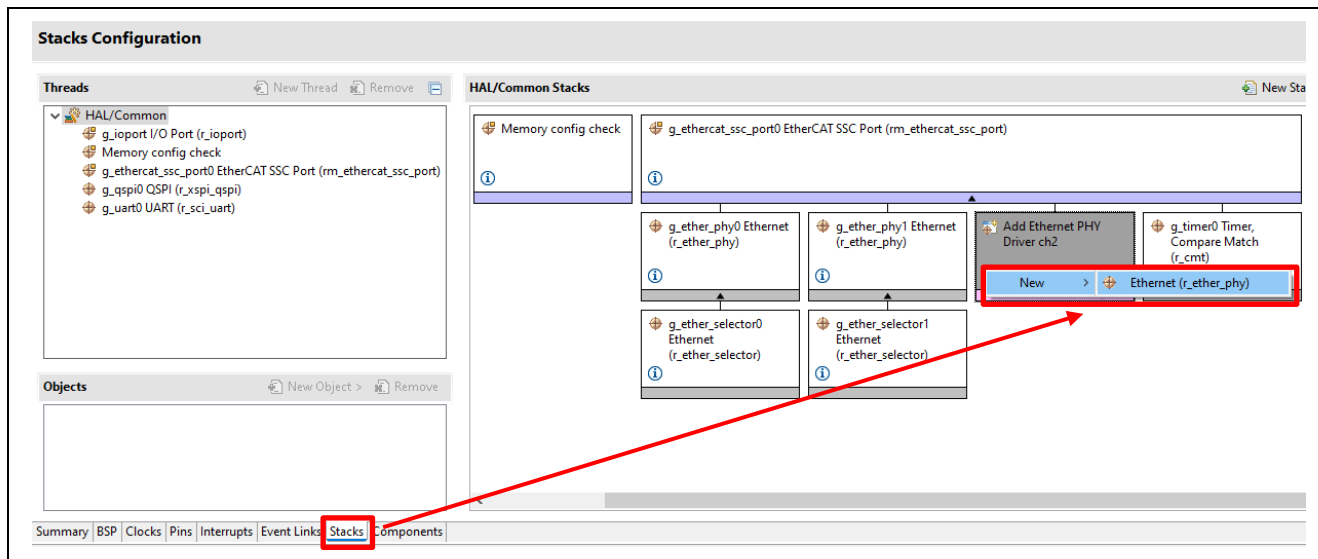
For e2studio, double-click [configuration] on the Project Explorer tab to start smart configurator.



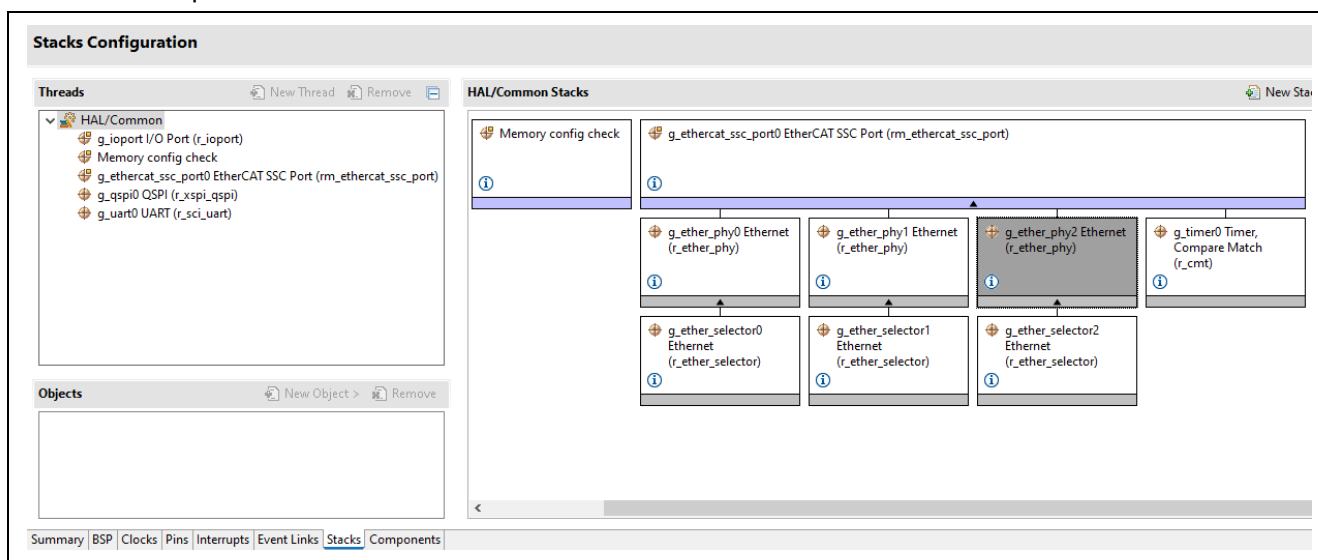
After startup, there is no difference in how to use smart configurator for each tool.

1. Add the 3rd port.

Add a new ether_PHY. [NEW] → [Ethernet Driver on r_rther_phy]



The third port was added.



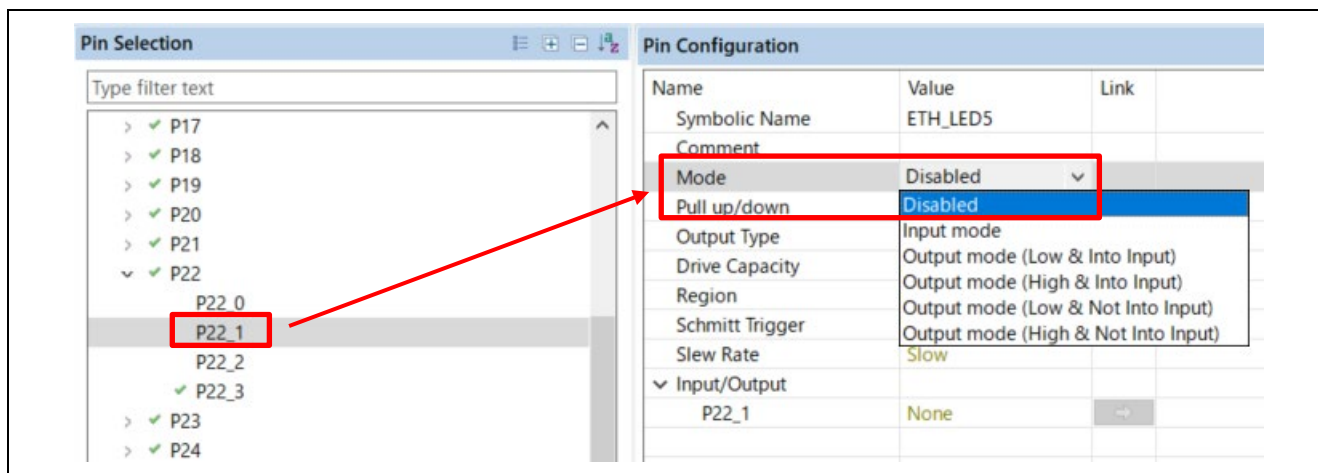
2. Set the 3rd port PHY
 - Set the 3rd port PHY from the PHY properties as follows.
 - Channel:2
 - PHY-LSI Address:2

Property	Value
Parameter Checking	Default (BSP)
VSC8541 Target	Enabled
KSZ9131 Target	Disabled
KSZ9031 Target	Disabled
KSZ8081 Target	Disabled
KSZ8041 Target	Disabled
User Own Target	Disabled
Module g_ether_phy2 Ethernet (r_ether_phy)	
Name	g_ether_phy2
Channel	2
PHY-LSI Address	2
PHY-LSI Reset Completion Timeout	0x00020000
Flow Control	Disable
Port Type	EtherCAT
Phy LSI type	Kit Component
Port Custom Init Function	NULL
Select MDIO type	GMAC
Auto Negotiation	ON
Speed	100M
Duplex	FULL
Reset Port	13
Reset Pin	4
Reset assert time	15000

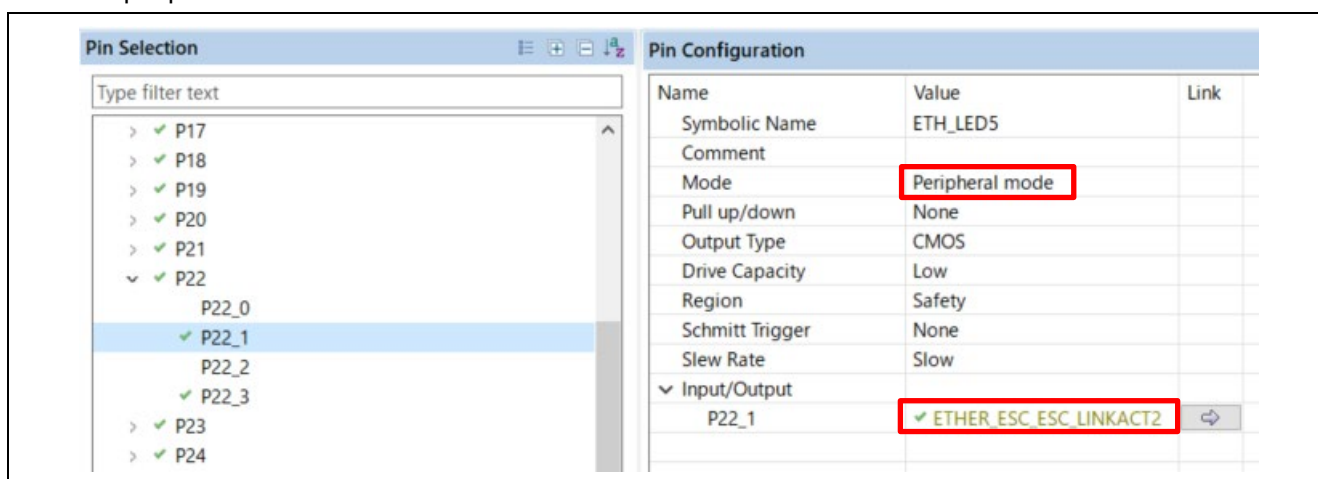
3. Set the ETHER_ESC
 - Set the ETHER_ESC from the Pin Selection as follows.
 - ESC_LINKACT2:P22_1
 - ESC_PHYLINK2:P00_5

Name	Value	Lock	Link
ESC_I2CDATA	P13_3		
ESC_IRQ	P17_0		
ESC_LATCH0	None		
ESC_LATCH1	None		
ESC_LEDERR	P20_3		
ESC_LED RUN	P20_2		
ESC_LEDSTER	None		
ESC_LINKACT0	P20_1		
ESC_LINKACT1	P20_4		
ESC_LINKACT2	P22_1		
ESC_MDC	None		
ESC_MDIO	None		
ESC_PHYLINK0	P10_4		
ESC_PHYLINK1	P05_5		
ESC_PHYLINK2	P00_5		
ESC_RESETOUT#	None		
ESC_SYNC0	P02_1		
ESC_SYNC1	None		

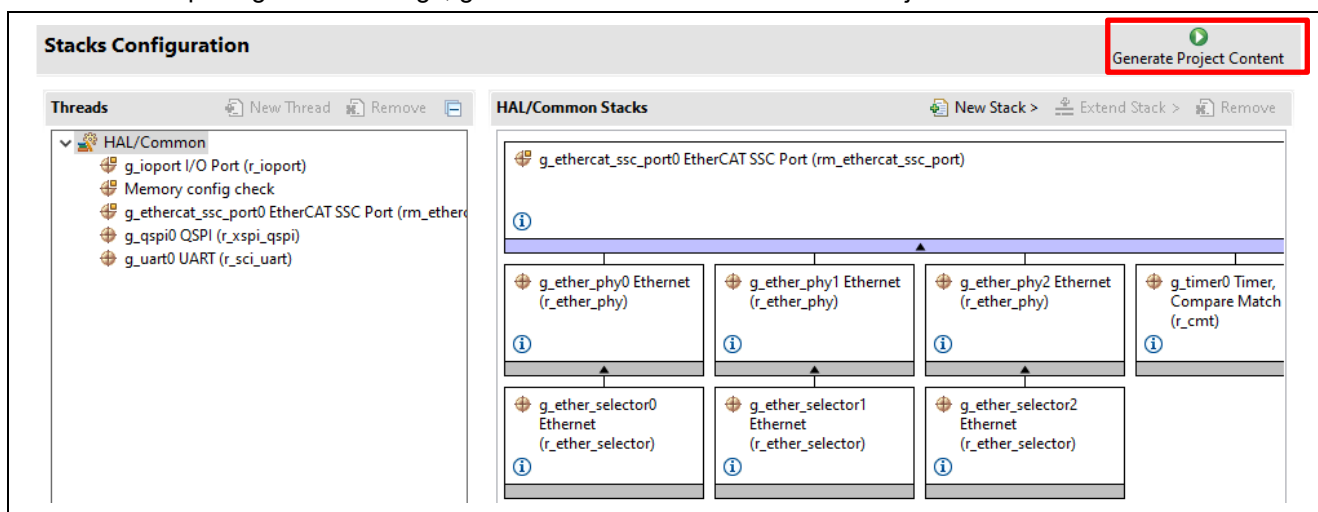
4. Change Pin setting attributes
Change the attribute of the Pin set on the 3rd port.
Set the Mode of P22_1 to Disabled.



When the Mode of P22_1 is set to Disabled, it is assigned to [ESC_LINKACT2] and the mode is set to the peripheral mode.



5. After completing all the settings, generate the code with "Generate Project Content".



9.3 Appendix C : ESC mode setting

In the FSP2.0.0 compatible sample, it can change the settings from the smart configurator to change the PHY settings and the number of ports.

This appendix shows how to change the PHY management interface from GMAC to ESC. Initially, it is set to GMAC.

Method of starting smart configurator depends on the tool.

1. Change PHY settings

Select each ether_phy stack. [Select MDIO type] → [ESC]

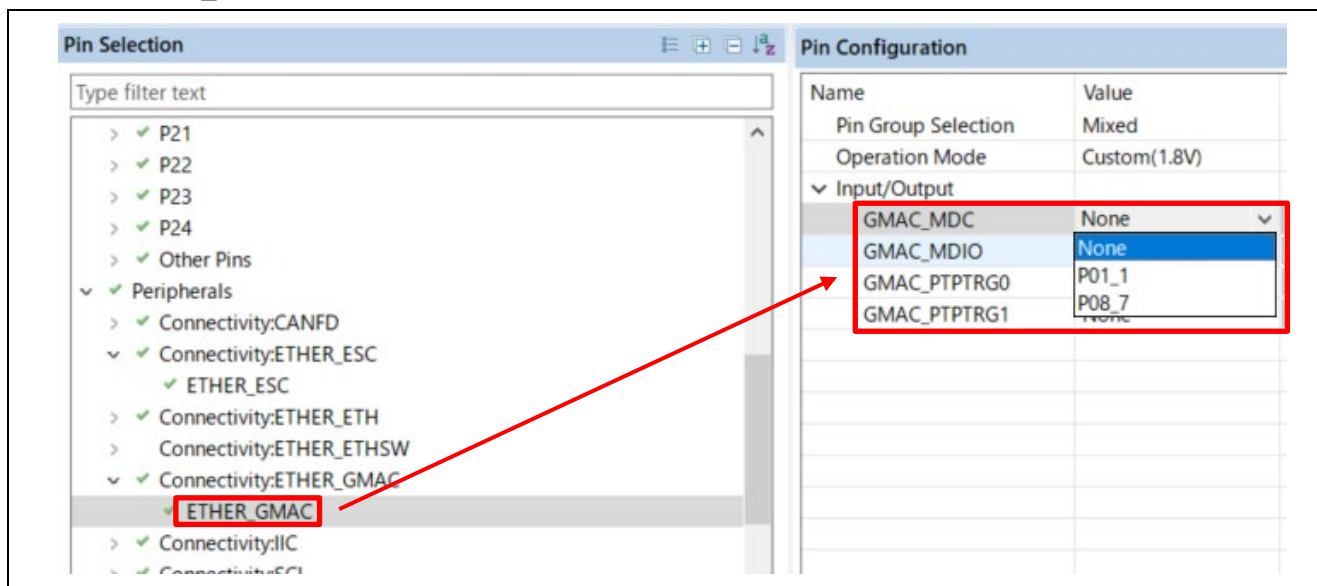
The screenshot displays the 'Stacks Configuration' interface. On the left, the 'Threads' pane shows the 'HAL/Common' thread containing components like 'g_ioport I/O Port (r_ioport)', 'Memory config check', 'g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)', 'g_qspi0 QSPI (r_xspi_qspi)', and 'g_uart0 UART (r_sci_uart)'. The 'Objects' pane is empty. The main workspace shows the 'HAL/Common Stacks' configuration. A stack named 'g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)' is expanded, revealing three Ethernet PHY components: 'g_ether_phy0 Ethernet (r_ether_phy)', 'g_ether_phy1 Ethernet (r_ether_phy)', and 'g_ether_phy2 Ethernet (r_ether_phy)'. Each PHY component is connected to a corresponding selector: 'g_ether_selector0 Ethernet (r_ether_selector)', 'g_ether_selector1 Ethernet (r_ether_selector)', and 'g_ether_selector2 Ethernet (r_ether_selector)'. A red arrow points from the 'g_ether_phy0' component to the 'Properties' pane at the bottom.

The 'Properties' pane shows the settings for 'g_ether_phy0 Ethernet (r_ether_phy)'. The 'Settings' tab is active, displaying a table of properties and their values:

Property	Value
Common	
Module g_ether_phy0 Ethernet (r_ether_phy)	
Name	g_ether_phy0
Channel	0
PHY-LSI Address	0
PHY-LSI Reset Completion Timeout	0x00020000
Flow Control	Disable
Port Type	EtherCAT
Phy LSI type	Kit Component
Port Custom Init Function	NULL
Select MDIO type	ESC
Auto Negotiation	GMAC
Speed	1000
Duplex	Full
Reset Port	13
Reset Pin	4
Reset assert time	15000

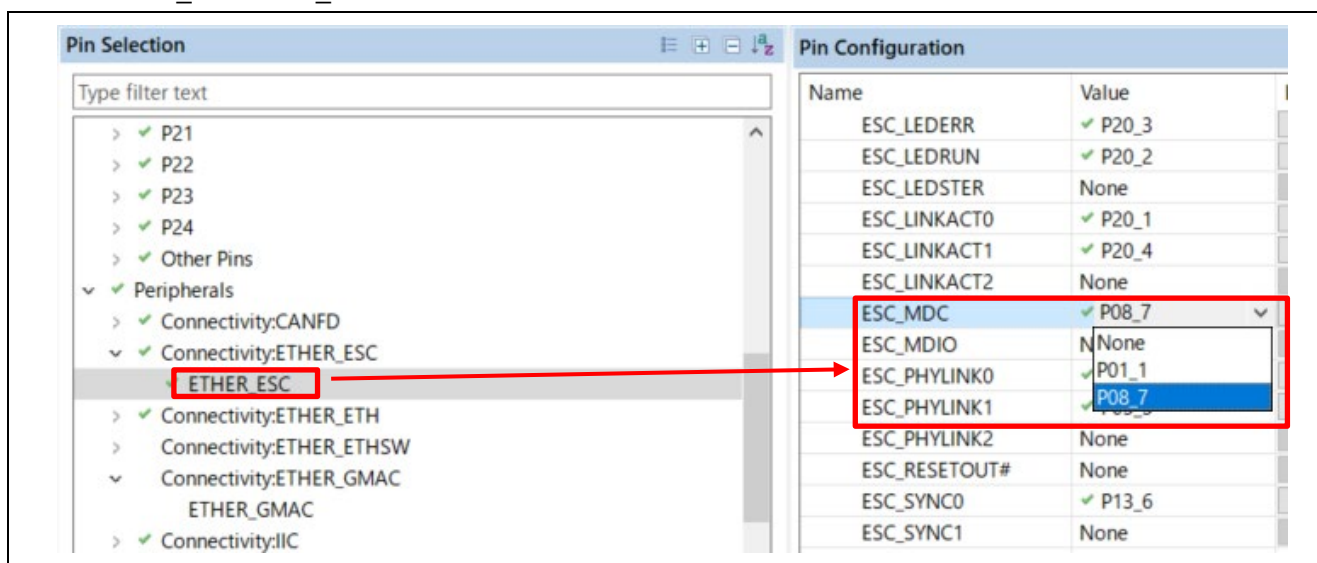
A red box highlights the 'Select MDIO type' property, and another red box highlights the 'ESC' value. A red arrow points from the 'g_ether_phy0' component in the stack configuration to the 'Select MDIO type' property in the properties pane.

2. Change PHY management interface
 - Change ETHER_GMAC setting
 - GMAC_MDC:None
 - GMAC_MDIO:None



Change ETHER_ESC setting

- ESC_MDC:P08_7
- ESC_MDIO:P09_0



3. After completing all the settings, generate the code with "Generate Project Content".

9.4 Appendix D : EEPROM capacity setting change

Use the smart configurator to change the EEPROM capacity size and generate the code.

32Kbit capacity EEPROM is mounted on boards other than serial numbers
"241783~241882, 243167~243266".

According to the EtherCAT specification, it is necessary to change the setting with 32 Kbit as the boundary.

1. Open "configuration.xml" in the e² studio project or Smart Configurator in the EWARM project.
Select SSC_port stack. [Properties] → [EEPROM Size]
2. Select 32Kbit Over or Under depending on the capacity of the EEPROM.

The screenshot displays the Smart Configurator's Stacks Configuration window. On the left, the 'Threads' pane shows a tree view of HAL/Common components, with 'g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)' selected. The 'Objects' pane is empty. The 'HAL/Common Stacks' pane on the right shows a stack diagram. At the top is 'g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)'. Below it are four components: 'g_ether_phy0 Ethernet (r_ether_phy)', 'g_ether_phy1 Ethernet (r_ether_phy)', 'g_ether_phy2 Ethernet (r_ether_phy)', and 'g_timer0 Timer, Compare Match (r_cmt)'. Below these are three components: 'g_ether_selector0 Ethernet (r_ether_selector)', 'g_ether_selector1 Ethernet (r_ether_selector)', and 'g_ether_selector2 Ethernet (r_ether_selector)'. A red box highlights the 'g_ethercat_ssc_port0' component. A red arrow points from this box to the 'Properties' tab in the bottom pane. The 'Properties' tab shows the 'EEPROM Size' property for 'g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)'. The property has three radio button options: 'Over 32Kbits', 'Under 32Kbits', and 'Over 32Kbits'. The 'Over 32Kbits' option is selected and highlighted with a red box.

3. Generate the code with "Generate Project Content".

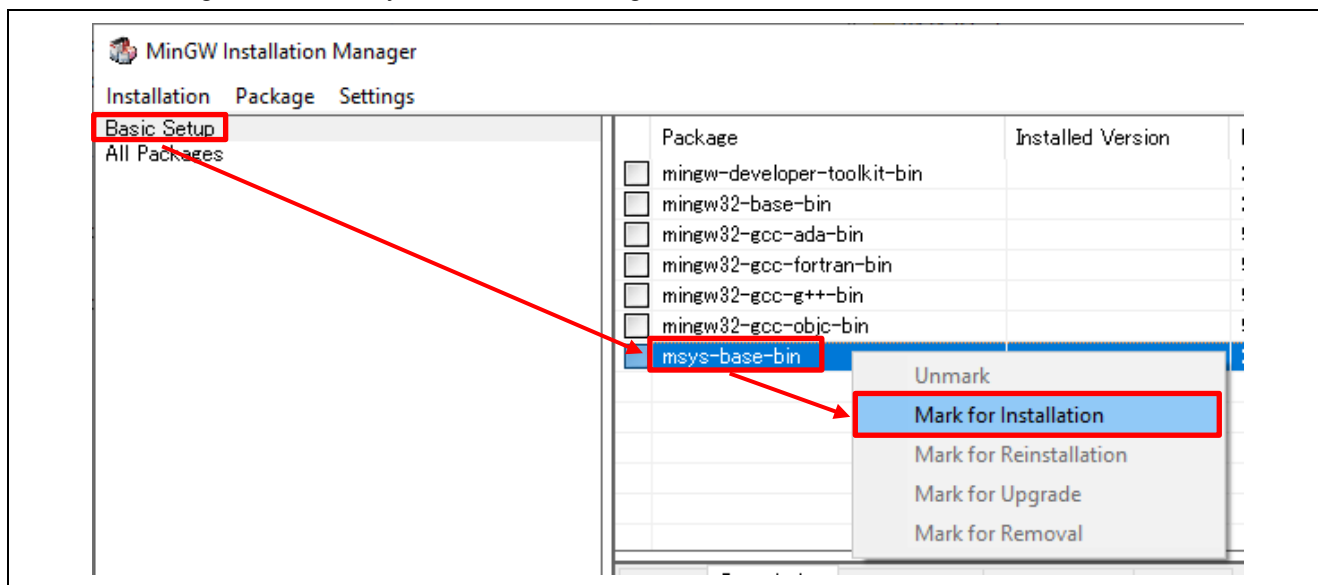
9.5 Appendix E : How to install patch

This section describes how to install patch in MinGW.

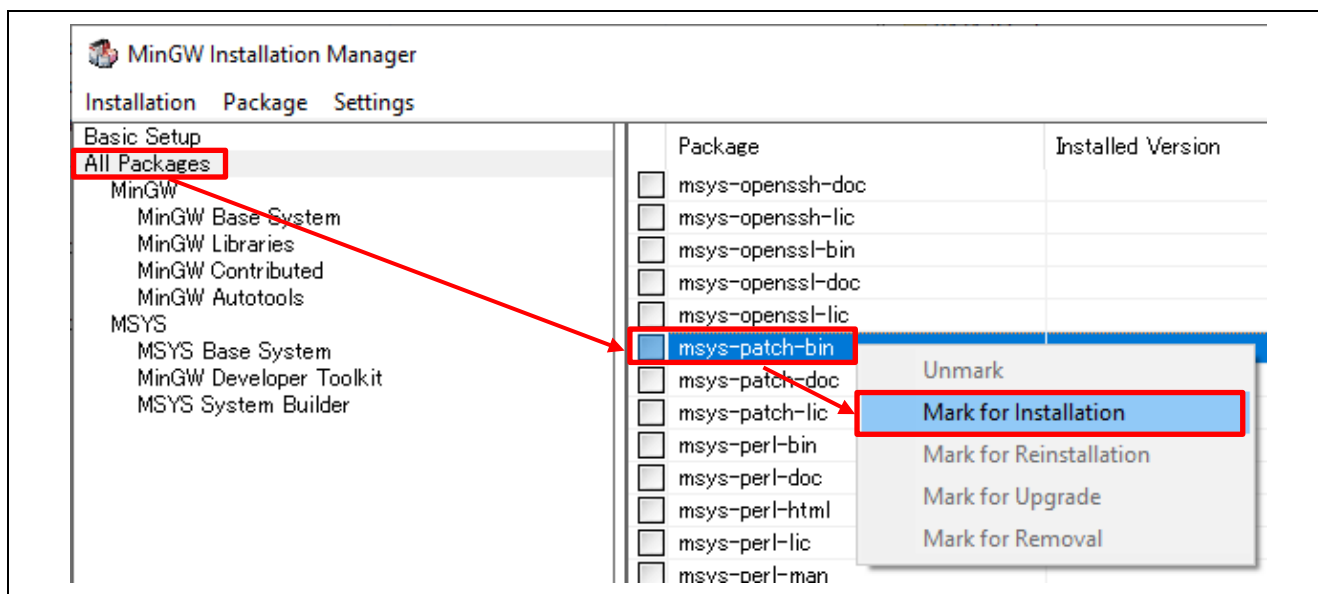
1. Download “mingw-get-setup.exe” from the following URL.

<https://osdn.net/projects/mingw/downloads/68260/mingw-get-setup.exe/>

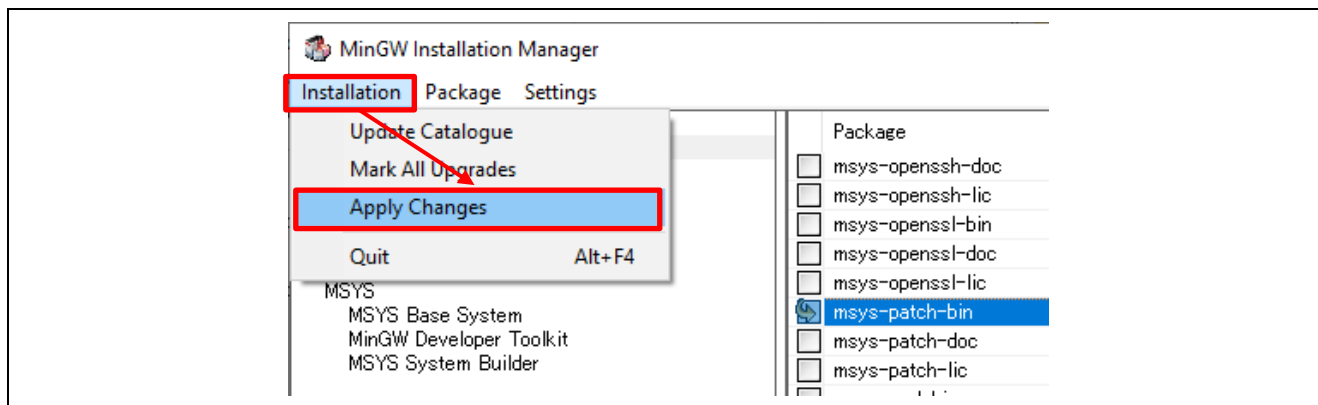
2. Execute “mingw-get-setup.exe”, install “Mingw-installation-manager” according to the dialog.
3. If it's completed and the Mingw-installation-manager window is displayed, select “Basic Setup” in the left window, right-click on “msys-base-bin” in the right window, and select “Mark for Installation”.



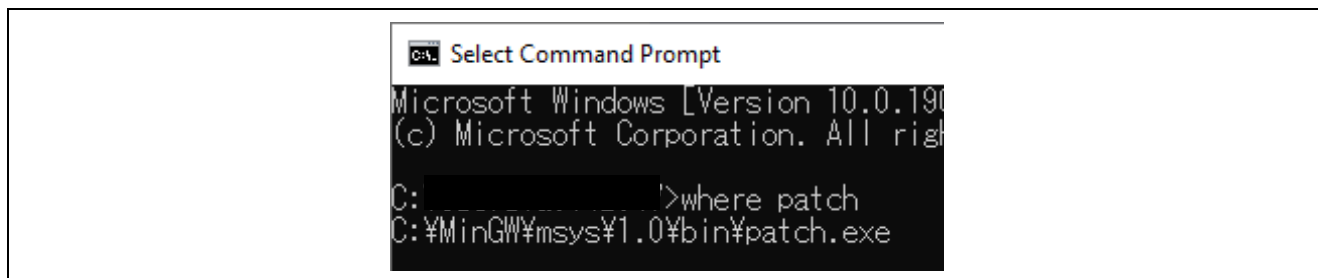
4. Select “All Packages” in the left window, right-click on “msys-patch-bin” in the right window, and select “Mark for Installation”.



5. Select “Apply Changes” in “Installation” in the above menubar.



6. “Schedule of Pending Actions” window is displayed, click “Apply” button.
7. If “All changes were applied successfully; you may now clone this dialogue.” is displayed, Installing patch.exe is succeeded.
8. Register the path to the installed patch.exe into system environment variables.
After registering, reboot your PC.
9. Start Command Prompt, enter “where patch”.
If the path to the installed patch.exe is displayed, there are no problem.



9.6 Appendix F : Semiconductor Device Profile [ETG.5003]

9.6.1 Common Device Profile [ETG.5003.1]

In the case of handling semiconductor devices with EtherCAT, it is necessary to support the device profile specified in the ETG5003 specifications.

The structure of ETG.5003 is as follows.

1. Common Device Profile (CDP) [ETG.5003.1]
2. Firmware update functionality [ETG.5003.2]
3. Specific Device Profile (SDP) [ETG.5003.2xxx]

Common Device Profile (CDP) specifies requirements that apply to all devices described in Specific Device Profile (SDP)

This sample program provides the object dictionary definition equivalent to Appendix A of the ETG document "ETG.5003 EtherCAT Semiconductor Device Profile - Part 1: Common Device Profile V1.1.0."

If you try executing this sample program with CDP, change the macro define "ETHERCAT_FEATURE" to "SEMI_DEVICE" in "src/ethercat/renesas/samplefeature.h"

The table 9.1 shows the details of the macro define "ETHERCAT_FEATURE."

Table 9.1 The details of the macro define "ETHERCAT_FEATURE."

ETHERCAT_FEATURE	Support application			
	IO	CiA402	FoE	CDP
CIA402_DRIVE	Support	Support	Support	Not support
SEMI_DEVICE	Support	Not support	Support	Support

When operating this sample program with CDP on TwinCAT, use "Renesas EtherCAT RZN2 FoE+CDP 2(3)port" to download into EEPROM.

This sample program provides only the framework of object dictionary definition. Separately consider and implement the settings and necessary processing.

For Common Device Profile Ver1.1.0, please refer to the following ETG.5003.1 standard.

If you have any questions regarding CDP, please contact the ETG Association.

ETG5003.1 standard

ETG5003-1 S (R) V1.1.0

EtherCAT Semiconductor Device Profile

Part1 Common Device Profile

9.6.2 Semi Test Record [ETG.7000.2-Annex5003-0001]

In this sample program, the program is implemented assuming that it corresponds to the following test items of Semi Test Record ETG.7000.2-Annex5003-0001.

1. Device Reset Command (Standard reset)
2. Dynamic PDO
3. Store Parameters

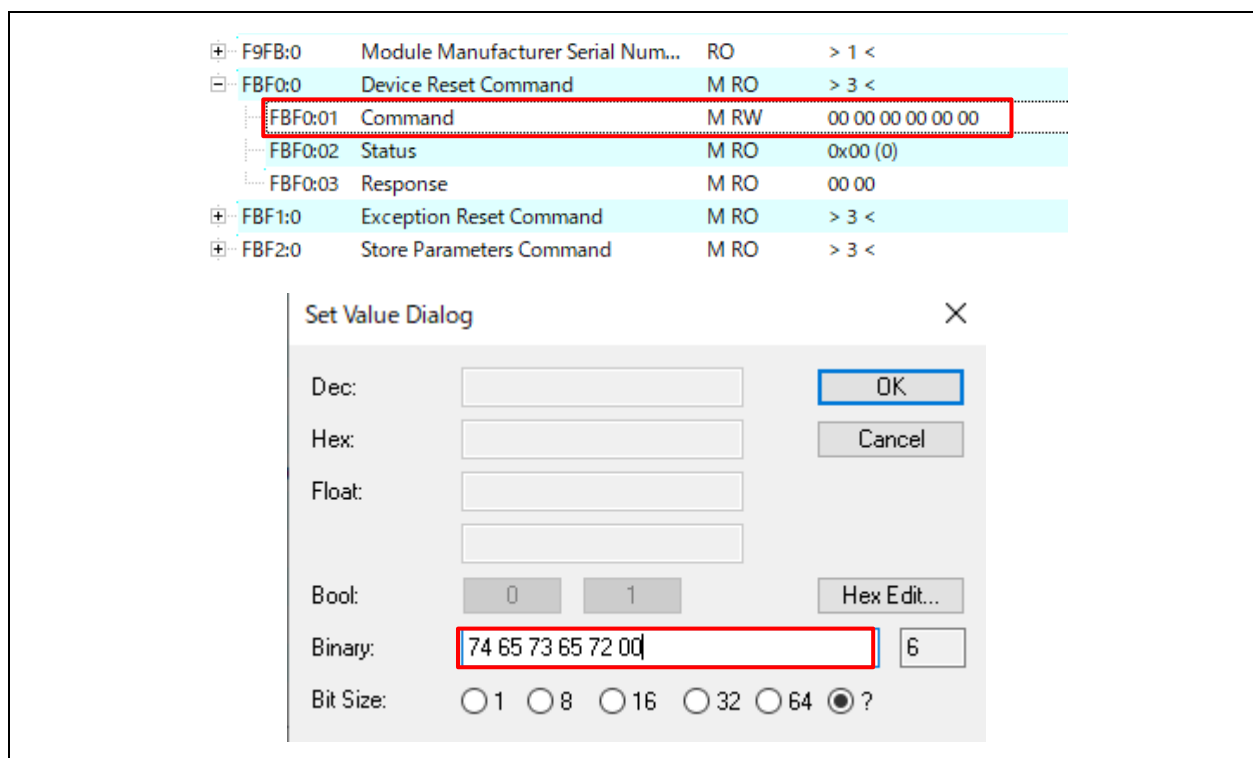
9.6.2.1 Device Reset Command (Standard reset)

When a specific value is entered in subindex 1 of the object 0xFBFB0, the evaluation board restarts.

Note: When debugging, this operation will be failure.
Stop debugging, push the reset button on RSK board to reset RZ/N2L.

- How to check

- (1) Connect the serial port of the evaluation board to the serial port of the PC and start terminal software such as Tera Term on the PC. In the serial settings of the terminal, set 115200 bps, 8-bit data, no parity, 1 stop bit, no flow control.
- (2) Perform "6. Connecting to TwinCAT3," Connect the evaluation board to TwinCAT.
- (3) Select the "Online" tab and make sure that the "Current Status" is set to "OP".
- (4) Select the "CoE - Online" tab, double-click on Index FBF0:01 "Command" and write "74 65 73 65 72 00" in "binary" to reset RZ/N2L.



- (5) If the "Execute Device_Reset()." and "RZ/N2L EtherCAT sample program starts on BANKx." output by serial communication are displayed in terminal software such as Tera Term, the evaluation board has been successfully restarted, Device Reset Command is no problem.

```
RZ/N2L EtherCAT sample program starts on BANK0.  
Execute Device_Reset().  
RZ/N2L EtherCAT sample program starts on BANK0.
```

9.6.2.2 Dynamic PDO

In this sample program, the settings related to PDO in the ESI file are shown in Table 9.2.

Table 9.2 PDO Settings for this Sample Program

PDO Items	setting
PdoAssign	true
PdoConfig	true

In addition, the PDO assignment/mapping object settings are shown in Table 9-3.

Table 9-3 PDO Assignment Mapping Object Settings

Object Name	Index	Access
RxPDO assign	0x1C12	RW only in the PreOP state
TxPDO assign	0x1C13	RW only in the PreOP state
Device User RxPDO-Map	0x17FF	RW only in the PreOP state
Device User TxPDO-Map	0x1BFF	RW only in the PreOP state

Therefore, objects 0x17FF "Device User RxPDO-Map" and 0x1BFF "Device User TxPDO-Map" that are not assigned by default can be assigned and mapped on the EtherCAT setting tool.

- How to set up

- (1) Perform "6. Connecting to TwinCAT3," Connect the evaluation board to TwinCAT.

- (2) Select the "Slots" tab, confirm the current slot.

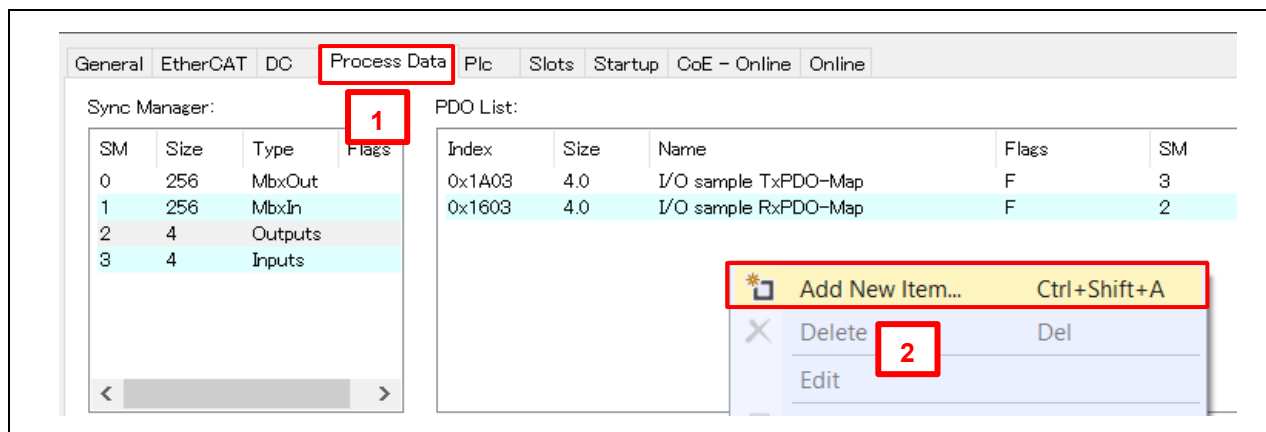
Select something in "Axis 0" of "Slot" of the left window, add it or change to it.

Also, "Axis 1" must be empty.

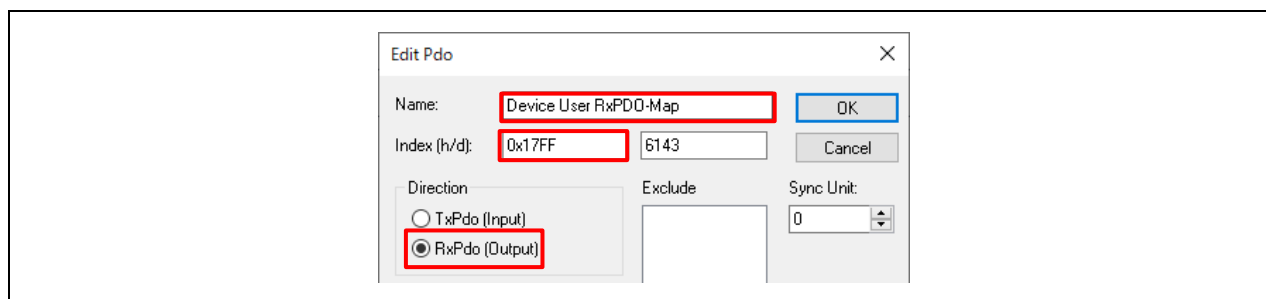
When you have added a module to "Slot" or changed it, restart TwinCAT by pressing [TwinCAT] → [Restart TwinCAT (Config Mode)] in the upper menu bar.

- (3) Select the "Online" tab and make sure that the "Current Status" is set to "OP".

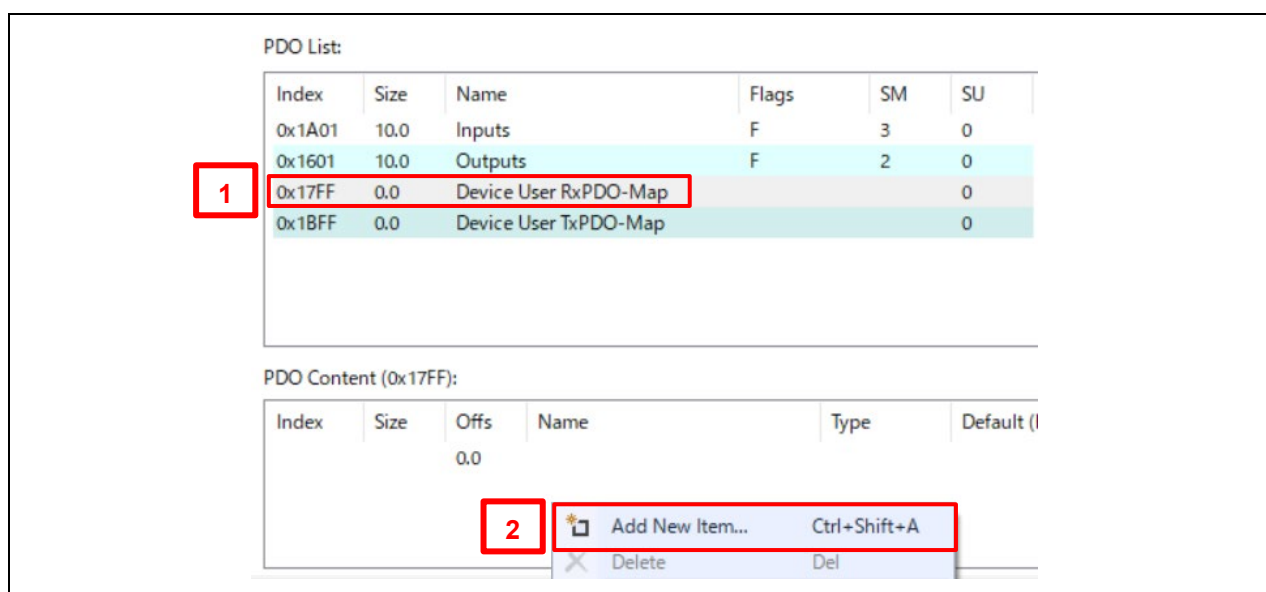
- (4) Select the "Process Data" tab and right-click in the "PDO List" area, select "Add New Item...".



- (5) In the "Edit Pdo" window, set the name to "Device User RxPDO-Map", the Index to "0x17FF", the Direction to "RxPdo" and press "OK".

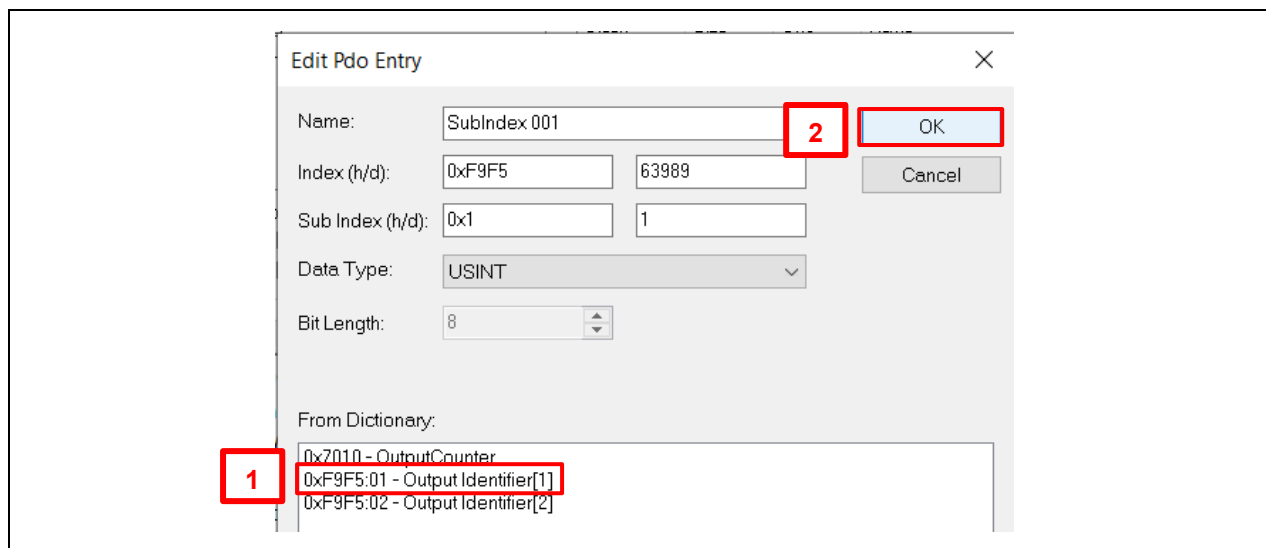


- (6) Right-click in the "PDO List" area again, select "Add New Item...", in the "Edit Pdo" window set the name to "Device User TxPDO-Map", the Index to "0x1BFF", the Direction to "TxPdo" and press "OK".
 "Device User RxPDO-Map" and "Device User TxPDO-Map" have now been added to the "PDO List".
- (7) Click "Device User RxPDO-Map" in the "PDO List" and right-click in the "PDO Content (0x17FF):" area, select "Add New Item...".



- (8) In the "Edit Pdo Entry" window, click "0xF9F5:01 – Output Identifier[1]" from "From Dictionary" and press OK.

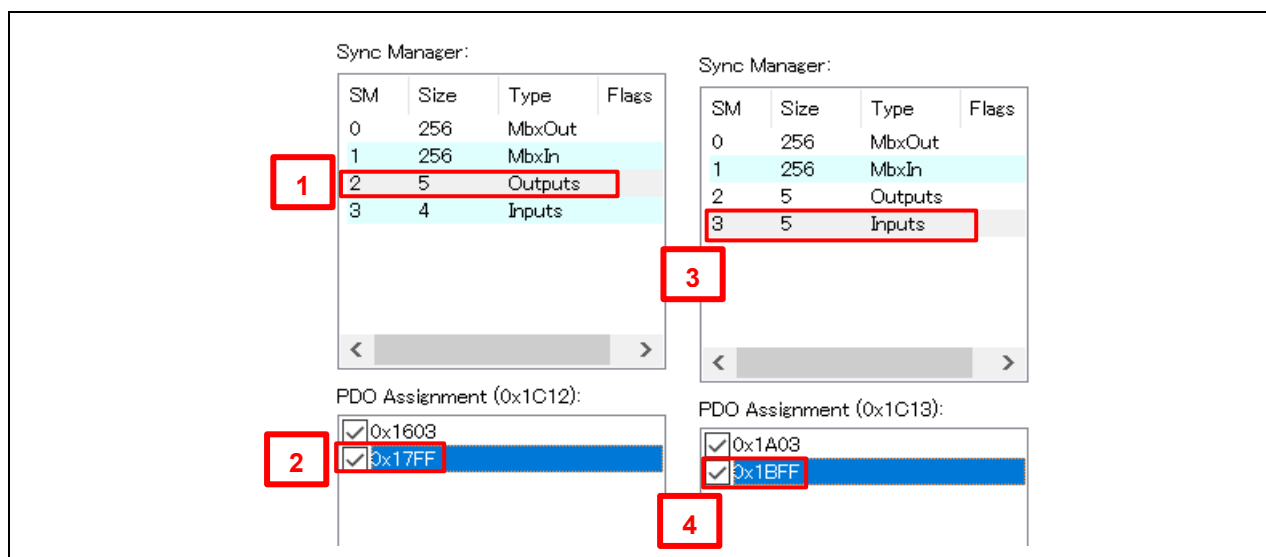
The Index 0xF9F5 is now mapped to the Index 0x17FF.



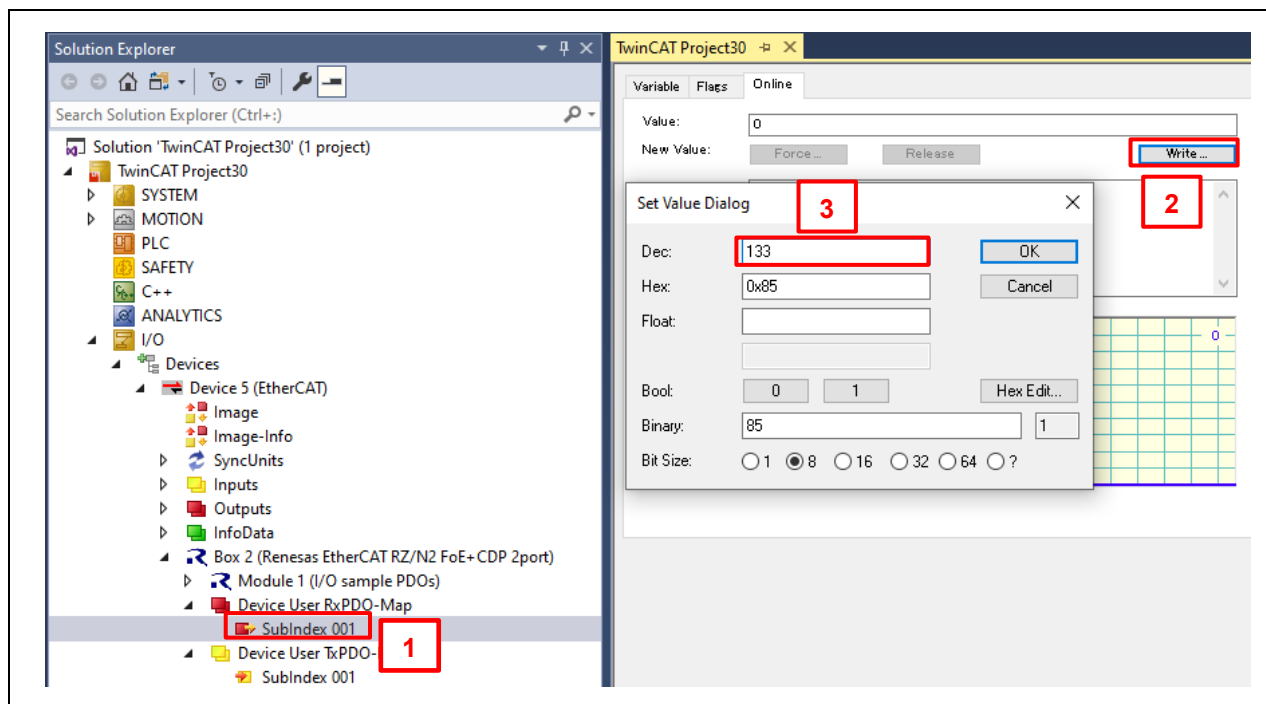
- (9) Similarly, click on "Device User TxPDO-Map" in the "PDO List", right-click on the "PDO Content (0x1BFF):" area, select "Add New Item...".

In the "Edit Pdo Entry" window, select "From Dictionary" to "0xF9F5:01 – Output Identifier[1]" and press OK. The Index 0xF9F5 is now mapped to the Index 0x1BFF.

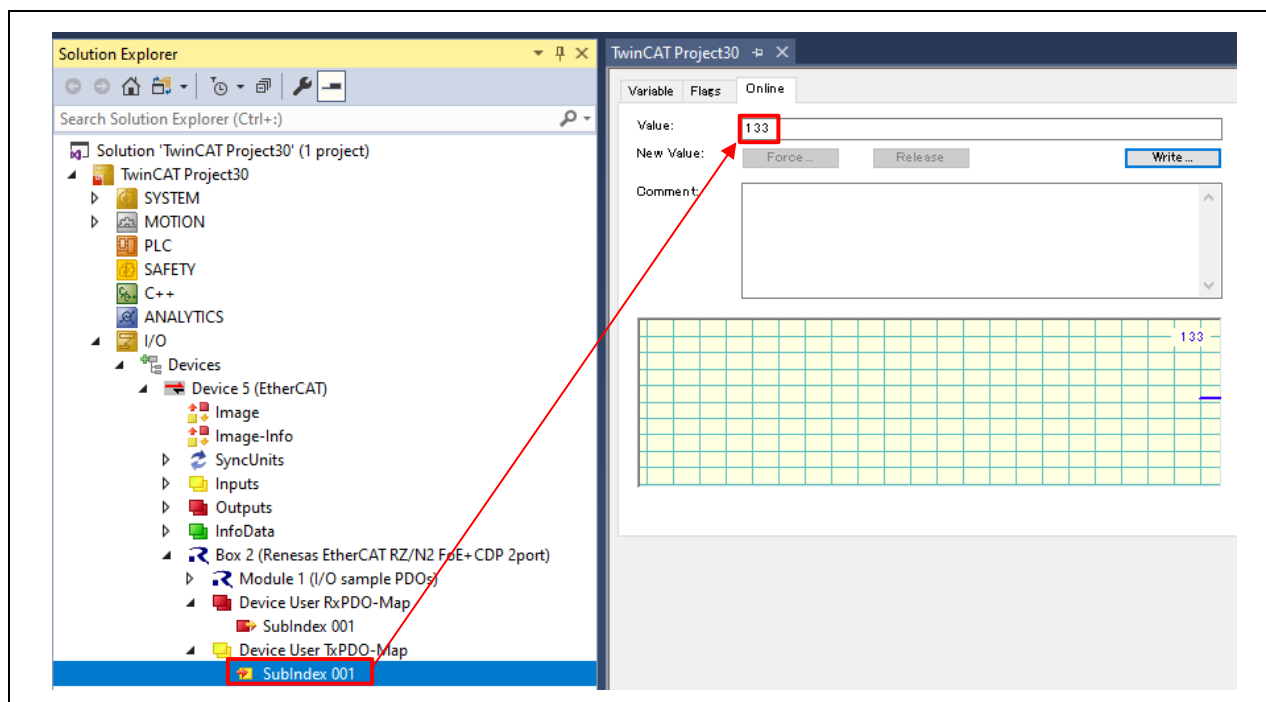
- (10) In the "Sync Manager" area, click on the row with the "SM" column of 2 and check the 0x17FF in the "PDO Assignment (0x1C12)" area.
- (11) Similarly, in the "Sync Manager" area, click on the row with the "SM" column of 3 and check the 0x1BFF in the "PDO Assignment (0x1C13)" area. You have now assigned an additional 0x17FF to the Index 0x1C12 and an additional 0x1BFF to the Index 0x1C13.



- (12) Restart TwinCAT by pressing [TwinCAT] → [Restart TwinCAT (Config Mode)] in the upper menu bar.
- (13) Expand [Device User RxPDO-Map] in [Box 1] and click [SubIndex 001].
- (14) Write an arbitrary value of 1~255 in Value.



- (15) Expand Device User TxPDO-Map and click SubIndex 001.
- (16) If the value of Value is the same as the value written in (13), it is normal.



9.6.2.3 Store Parameters

In this sample program, when the object value with the backup flag is changed by SDO communication, its value will be stored in the non-volatile memory of the evaluation board.

Table 9.4 shows the objects with the backup flag in this sample program.

Table 9.4 Objects with Backup Flags

Index	Name
0x10F0	Backup parameter handling
0xF3A1	Device Warming Mask
0xF3A2	Manufacture Warming Mask
0xF3A3	Device Error Mask
0xF3A4	Manufacture Error Mask
0xF3A5	Global Device Warming Mask
0xF3A6	Global Manufacture Warming Mask
0xF3A7	Global Device Error Mask
0xF3A8	Global Manufacture Error Mask

Table 9.5 shows the code flash memory area used in this sample program as a non-volatile memory area in which the values of objects with the Backup flag are stored.

Table 9.5 Non-volatile Memory Areas for Backup Objects

First address	Last address	Capacity
0x60F0_0000	0x60F0_1000	4KB

Table 9.6 shows the constants used by programs that store in non-volatile memory.

Table 9.6 Constants used in programs storing non-volatile memory (samplesemi.h)

Constant Name	Setting Values	Details
BACKUP_MEMORY_START_ADDRESS	Table 9.5 First Addresses	The beginning address of the non-volatile memory area for the Backup objects.
BACKUP_BYTESIZE	4096	Amount of non-volatile memory area for Backup objects.
Default_Data_Is_Not_Initialized	(0xFFFF)	The non-volatile memory area for the Backup object has not been initialized.
Default_Data_Is_Initialized	(0x5555)	The non-volatile memory area for the Backup object has been initialized.
NonVolatileWordOffset_0x10F0	(0x0002)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A1	(0x0008)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A2	(0x000c)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A3	(0x0010)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.

NonVolatileWordOffset_0xF3A4	(0x0014)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A5	(0x0018)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A6	(0x001c)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A7	(0x0020)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.
NonVolatileWordOffset_0xF3A8	(0x0024)	Number of offset words from BACKUP_MEMORY_START_ADDRESS.

This sample program uses the functions defined in 6.2.4.1 "Backup Parameter Support" of Application Note ET9300 (EtherCAT Slave Stack Code).

For more information, see Application Note ET9300 (EtherCAT Slave Stack Code) and samplesemi.c.

- How to check

- (1) Connect the serial port of the evaluation board to the serial port of the PC and start terminal software such as Tera Term on the PC. In the serial settings of the terminal, set 115200 bps, 8-bit data, no parity, 1 stop bit, no flow control.
- (2) Connect the evaluation board to TwinCAT 3 by following "5. Connecting to TwinCAT3."
- (3) Select the "Online" tab and make sure that "Current Status" is set to "OP".
- (4) Write the value to the Backup object. Here, we write "0xAAAAAAAA" at Index 0xF3A1:01.
- (5) If the writing is successful, "Index 0xF3A1 backup is success." will be displayed in terminal software such as Tera Term.



- (6) Reboot the RZ/N2L.
- (7) It is normal if the Index 0xF3A1:01 written in (4) is "0xAAAAAAAA".

Objects with the backup flag are stored in non-volatile memory and are read at startup.

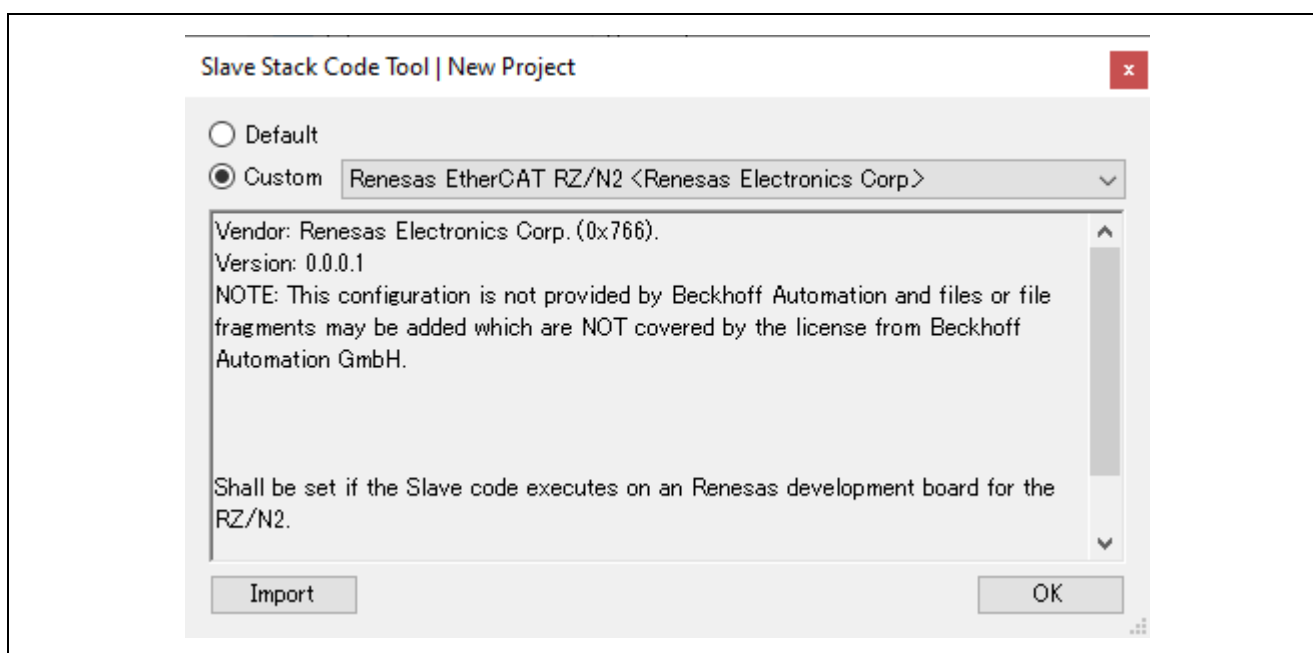
9.7 Appendix G : How to use the SSC tool configuration file

This section describes how to create the SSC project file by using the SSC tool configuration file.

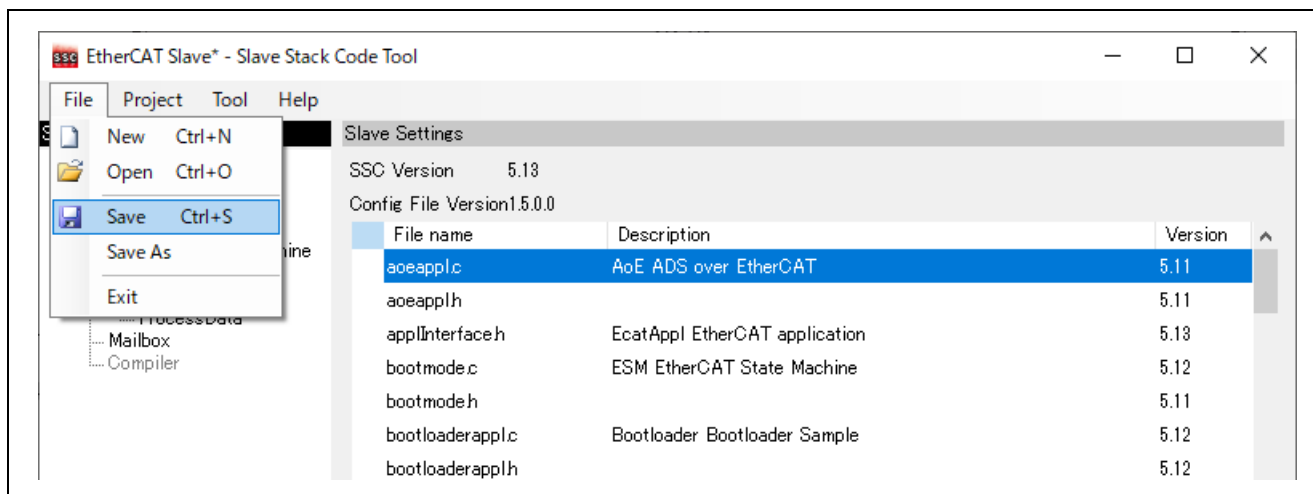
- (1) Start the SSC tool by clicking [EtherCAT Slave Stack Code] → [SSC Tool] in Windows start menu.
- (2) Click [import] in the [New Project] dialog, select the below SSC tool configuration file in this sample program, and click [OK].

**RZN2L_EtherCAT_RSK_rev0200\common\ecat_IO_CiA402_FoE\SSCconfig
\\Renesas_RZN2_config.xml**

- (3) Check the [Custom] checkbox, select “Renesas EtherCAT RZ/N2 <Renesas Electronics Corp>” in the list, and click [OK].



- (4) Click [File]→[Save], Save the SSC project file as any project file name.
(For example, “Renesas EtherCAT RZN2.esp”.)



9.8 Appendix H : Object dictionary

Table 9.7 shows all of the object dictionary defined in this sample program.

Table 9.7 Object dictionary

Index	ObjectCode	SI	DataType	Access	Object Name
0x1nnn Communication Area					
0x1000	VAR	-	UDINT	RO	Device Name
0x1001	VAR	-	USINT	RO	Error Register
0x1008	VAR	-	STRING(32)	RO	Manufacture Device Name
0x1009	VAR	-	STRING(3)	RO	Manufacturer Hardware Version
0x100A	VAR	-	STRING(3)	RO	Manufacturer Software Version
0x100B	VAR	-	STRING(3)	RO	Manufacturer Bootloader Version
0x1010	ARRAY	0	USINT	RO	Store parameters
		1	UDINT	RW	SubIndex 00x
0x1011	ARRAY	0	USINT	RO	Restore default parameters
		1	UDINT	RW	SubIndex 00x
0x1018	RECORD	0	USINT	RO	Identity Object
		1	UDINT	RO	Vendor ID
		2	UDINT	RO	Product Code
		3	UDINT	RO	Revision Number
		4	UDINT	RO	Serial Number
0x10F0	RECORD	0	USINT	RO	Backup parameter handling
		1	UDINT	RO	Checksum
		2	BOOL	RW	Backup Parameter Changed
0x10F1	RECORD	0	USINT	RO	Error Settings
		1	UDINT	RO	Local Error Reaction
		2	UINT	RW	Sync Error Counter Limit
0x10F8	VAR	-	ULINT	RO	Timestamp Object
0x1600	RECORD	0	USINT	RO	csp/csv RxPDO Mapping
		1-5	UDINT	RO	SubIndex 00x
0x1601	RECORD	0	USINT	RO	csp RxPDO Mapping
		1-3	UDINT	RO	SubIndex 00x
0x1602	RECORD	0	USINT	RO	csv RxPDO Mapping
		1-3	UDINT	RO	SubIndex 00x
0x17FF	RECORD	0	USINT	RW	Device User RxPDO-Map
		1	UDINT	RW	SubIndex 00x
0x1A00	RECORD	0	USINT	RO	csp/csv TxPDO Mapping
		1-5	UDINT	RO	SubIndex 00x

Index	ObjectCode	SI	Data Type	Access	Object Name
0x1A01	RECORD	0	USINT	RO	csp TxPDO Mapping
		1-3	UDINT	RO	SubIndex 00x
0x1A02	RECORD	0	USINT	RO	csv TxPDO Mapping
		1-4	UDINT	RO	SubIndex 00x
0x1BFF	RECORD	0	USINT	RW	Device User TxPDO-Map
		1	UDINT	RW	SubIndex 00x
0x1C00	ARRAY	0	USINT	RO	Sync Manager Communication Type
		1-4	USINT	RO	SubIndex 00x
0x1C12	ARRAY	0	USINT	RO *	RxPDO Assign
		1-2	UINT	RO *	SubIndex 00x
0x1C13	ARRAY	0	USINT	RO *	TxPDO Assign
		1-2	UINT	RO *	SubIndex 00x
0x1C32	RECORD	0	USINT	RO	SM output parameter
		1	UINT	RO *	Synchronization Type
		2	UDINT	RO	Cycle Time
		4	UINT	RO	Synchronization Types supported
		5	UDINT	RO	Minimum Cycle Time
		6	UDINT	RO	Calc and Copy Time
		8	UINT	RW	Get Cycle Time
		9	UDINT	RO	Delay Time
		10	UDINT	RW	Sync0 Cycle Time
		11	UINT	RO	SM-Event Missed
		12	UINT	RO	Cycle Time Too Small
		13	UINT	RO	Shift Time Too Short Counter
		32	BOOL	RO	Sync Error
0x1C33	RECORD	0	USINT	RO	SM input parameter
		1	UINT	RO *	Synchronization Type
		2	UDINT	RO	Cycle Time
		4	UINT	RO	Synchronization Types supported
		5	UDINT	RO	Minimum Cycle Time
		6	UDINT	RO	Calc and Copy Time
		8	UINT	RW	Get Cycle Time
		9	UDINT	RO	Delay Time
		10	UDINT	RW	Sync0 Cycle Time
		11	UINT	RO	SM-Event Missed
		12	UINT	RO	Cycle Time Too Small
		13	UINT	RO	Shift Time Too Short Counter

		32	BOOL	RO	Sync Error
Index	ObjectCode	SI	Data Type	Access	Object Name
0x5nnn Manufacturer Area					
0x5000	VAR	-	USINT	RO	Firmware Writable Bank
0x6nnn CiA402 Drive Profile Area					
0x603F	VAR	-	UINT	RO	Error Code
0x6040	VAR	-	UINT	RW	Control Word
0x6041	VAR	-	UINT	RO	Status Word
0x605A	VAR	-	INT	RW	Quick stop option code
0x605B	VAR	-	INT	RW	Shutdown option code
0x605C	VAR	-	INT	RW	Disable operation option code
0x605D	VAR	-	INT	RW	Halt option code
0x605E	VAR	-	INT	RW	Fault reaction option code
0x6060	VAR	-	SINT	RW	Modes of Operation
0x6061	VAR	-	SINT	RO	Modes of operation display
0x6062	VAR	-	DINT	RO	Position demand value
0x6063	VAR	-	DINT	RO	Position actual internal value
0x6064	VAR	-	DINT	RO	Position actual value
0x6065	VAR	-	UDINT	RW	Following error window
0x6066	VAR	-	UINT	RW	Following error time out
0x6067	VAR	-	UDINT	RW	Position window
0x606C	VAR	-	DINT	RO	Velocity actual value
0x6072	VAR	-	UINT	RW	Max Torque
0x6077	VAR	-	INT	RO	Torque actual value
0x607A	VAR	-	DINT	RW	Target position
0x607B	RECORD	0	USINT	RO	Position range limit
		1	DINT	RW	Min position range limit
		2	DINT	RW	Max position range limit
0x607C	VAR	-	DINT	RW	Home Offset
0x607D	RECORD	0	USINT	RO	Software position limit
		1	DINT	RW	Min position limit
		2	DINT	RW	Max position limit
0x6091	RECORD	0	USINT	RO	Gear ratio
		1	UDINT	RW	Motor revolutions
		2	UDINT	RW	Shaft revolutions
0x6099	RECORD	0	USINT	RO	Homing speeds
		1	UDINT	RW	Speed during search for switch
		2	UDINT	RW	Speed during search for zero

Index	ObjectCode	SI	Data Type	Access	Object Name
0x60B0	VAR	-	DINT	RW	Position offset
0x60B1	VAR	-	DINT	RW	Velocity offset
0x60B2	VAR	-	INT	RW	Torque offset
0x60B8	VAR	-	UINT	RW	Touch probe function
0x60B9	VAR	-	UINT	RO	Touch probe status
0x60BA	VAR	-	DINT	RO	Touch probe position 1 positive value
0x60BB	VAR	-	DINT	RO	Touch probe position 1 negative value
0x60C2	RECORD	0	USINT	RO	Interpolation time period
		1	USINT	RW	Interpolation time period value
		2	SINT	RW	Interpolation time index
0x60D0	RECORD	0	USINT	RO	Touch probe source
		1	INT	RW	Touch probe 1 source
0x60E0	VAR	-	UINT	RW	Positive torque limit value
0x60E1	VAR	-	UINT	RW	Negative torque limit value
0x60F4	VAR	-	DINT	RO	Following error actual value
0x60FD	VAR	-	UDINT	RO	Digital inputs
0x60FE	RECORD	0	USINT	RO	Digital outputs
		1	UDINT	RW	Physical outputs
		2	UDINT	RW	Bit mask
0x60FF	VAR	-	DINT	RW	Target velocity
0x6402	VAR	-	UINT	RW	Motor Type
0x6502	VAR	-	UDINT	RO	Supported drive modes
0xFnnn Common Device Profile Area					
0xF000	RECORD	0	USINT	RO	Semiconductor Device Profile
		1	UINT	RO	Index Distance
		2	UINT	RO	Maximum Number of Modules
0xF010	ARRAY	0	USINT	RO	Module Profile List
		1	UDINT	RO	SubIndex 00x
0xF020	ARRAY	0	USINT	RO	Configured Address List
		1	UDINT	RO	SubIndex 00x
0xF030	ARRAY	0	USINT	RO	Configured Module Ident List
		1	UDINT	RO	SubIndex 00x
0xF050	ARRAY	0	USINT	RO	Detected Module Ident List
		1	UDINT	RO	SubIndex 00x
0xF380	VAR	-	USINT	RO	Active Exception Status
0xF381	ARRAY	0	USINT	RO	Active Device Warming Details
		1	UDINT	RO	SubIndex 00x

Index	ObjectCode	SI	Data Type	Access	Object Name
0xF382	ARRAY	0	USINT	RO	Active Manufacture Warming Details
		1	UDINT	RO	SubIndex 00x
0xF383	ARRAY	0	USINT	RO	Active Device Error Details
		1	UDINT	RO	SubIndex 00x
0xF384	ARRAY	0	USINT	RO	Active Manufacture Error Details
		1	UDINT	RO	SubIndex 00x
0xF385	RECORD	0	USINT	RO	Active Global Device Warming Details
		1	UDINT	RO	SubIndex 001
0xF386	RECORD	0	USINT	RO	Active Global Manufacture Warming Details
		1	UDINT	RO	SubIndex 001
0xF387	RECORD	0	USINT	RO	Active Global Device Error Details
		1	UDINT	RO	SubIndex 001
0xF388	RECORD	0	USINT	RO	Active Global Manufacture Error Details
		1	UDINT	RO	SubIndex 001
0xF390	VAR	-	USINT	RO	Latched Exception Status
0xF391	ARRAY	0	USINT	RO	Latched Device Warming Details
		1	UDINT	RO	SubIndex 00x
0xF392	ARRAY	0	USINT	RO	Latched Manufacture Warming Details
		1	UDINT	RO	SubIndex 00x
0xF393	ARRAY	0	USINT	RO	Latched Device Error Details
		1	UDINT	RO	SubIndex 00x
0xF394	ARRAY	0	USINT	RO	Latched Manufacture Error Details
		1	UDINT	RO	SubIndex 00x
0xF395	RECORD	0	USINT	RO	Latched Global Device Warming Details
		1	UDINT	RO	SubIndex 001
0xF396	RECORD	0	USINT	RO	Latched Global Manufacture Warming Details
		1	UDINT	RO	SubIndex 001
0xF397	RECORD	0	USINT	RO	Latched Global Device Error Details
		1	UDINT	RO	SubIndex 001
0xF398	RECORD	0	USINT	RO	Latched Global Manufacture Error Details
		1	UDINT	RO	SubIndex 001
0xF3A1	ARRAY	0	USINT	RO	Device Warming Mask
		1	UDINT	RW	SubIndex 00x
0xF3A2	ARRAY	0	USINT	RO	Manufacture Warming Mask
		1	UDINT	RW	SubIndex 00x
0xF3A3	ARRAY	0	USINT	RO	Device Error Mask
		1	UDINT	RW	SubIndex 00x

Index	ObjectCode	SI	Data Type	Access	Object Name
0xF3A4	ARRAY	0	USINT	RO	Manufacture Error Mask
		1	UDINT	RW	SubIndex 00x
0xF3A5	RECORD	0	USINT	RO	Global Device Warming Mask
		1	UDINT	RW	SubIndex 001
0xF3A6	RECORD	0	USINT	RO	Global Manufacture Warming Mask
		1	UDINT	RW	SubIndex 001
0xF3A7	RECORD	0	USINT	RO	Global Device Error Mask
		1	UDINT	RW	SubIndex 001
0xF3A8	RECORD	0	USINT	RO	Global Manufacture Error Mask
		1	UDINT	RW	SubIndex 001
0xF6F0	ARRAY	0	USINT	RO	Input Latch Local Timestamp
		1	UDINT	RO	SubIndex 00x
0xF6F1	ARRAY	0	USINT	RO	Input Latch ESC Timestamp (32-bit)
		1	UDINT	RO	SubIndex 00x
0xF6F2	ARRAY	0	USINT	RO	Input Latch ESC Timestamp (64-bit)
		1	ULINT	RO	SubIndex 00x
0xF9F0	VAR	-	STRING(8)	RO	Manufacturer Serial Number
0xF9F1	ARRAY	0	USINT	RO	CDP Function Generation Number
		1	UDINT	RO	SubIndex 00x
0xF9F2	ARRAY	0	USINT	RO	SDP Function Generation Number
		1	UDINT	RO	SubIndex 00x
0xF9F3	VAR	-	STRING(7)	RO	Vendor Name
0xF9F4	ARRAY	0	USINT	RO	Semiconductor SDP Device Name
		1	STRING(8)	RO	SubIndex 00x
0xF9F5	ARRAY	0	USINT	RO	Output Identifier
		1	USINT	RW	SubIndex 00x
0xF9F6	VAR	-	UDINT	RO	Time since power on
0xF9F7	VAR	-	UDINT	RO	Total time powered
0xF9F8	VAR	-	UDINT	RO	Firmware Update Functional Generation Number
0xF9F9	ARRAY	0	USINT	RO	Module Manufacturer Hardware Version
		1	STRING (8)	RO	SubIndex 00x
0xF9FA	ARRAY	0	USINT	RO	Module Manufacturer Software Version
		1	STRING (8)	RO	SubIndex 00x
0xF9FB	ARRAY	0	USINT	RO	Module Manufacturer Serial Number
		1	STRING (8)	RO	SubIndex 00x

Index	ObjectCode	SI	DataType	Access	Object Name
0xFBF0	RECORD	0	USINT	RO	Device Reset Command
		1	ARRAY [0..5] OF BYTE	RW	Command
		2	USINT	RO	Status
		3	ARRAY [0..1] OF BYTE	RO	Response
0xFBF1	RECORD	0	USINT	RO	Exception Reset Command
		1	ARRAY [0..4] OF BYTE	RW	Command
		2	USINT	RO	Status
		3	ARRAY [0..1] OF BYTE	RO	Response
0xFBF2	RECORD	0	USINT	RO	Store Parameters Command
		1	ARRAY [0..3] OF BYTE	RW	Command
		2	USINT	RO	Status
		3	ARRAY [0..1] OF BYTE	RO	Response
0xFBF3	RECORD	0	USINT	RO	Calculate Checksum Command
		1	ARRAY [0..3] OF BYTE	RW	Command
		2	USINT	RO	Status
		3	ARRAY [0..7] OF BYTE	RO	Response
0xFBF4	RECORD	0	USINT	RO	Load Parameters Command
		1	ARRAY [0..3] OF BYTE	RW	Command
		2	USINT	RO	Status
		3	ARRAY [0..1] OF BYTE	RO	Response

* This is RW only if its status is PreOP.

※ In case this sample program is used for customer's products, refer to ETG specification documents and, separately consider and implement the settings and necessary processing for each object.

10. Limitations

1. When executing the EtherCAT sample program for the first time, be sure to set GMAC_MDC and GMAC_MDIO in Smart Configurator.
Write the EEPROM data with this configuration. It is not possible to set ESC_MDC and ESC_MDIO without writing the EEPROM data.
2. This sample program does not support NOR Flash boot mode.
3. When run the EtherCAT sample program for the first time, be sure to set the number of ports to 2.
Write EEPROM data during running in a 2-port configuration. It is not possible to configure 3 ports without writing EEPROM data.

Revision History

Rev.	Date	Description	
		Page	Summary
2.00	Jul 12, 2024	-	First edition issued

General Precautions in the Handling of Microprocessing Unit and Microcontroller Unit Products

The following usage notes are applicable to all Microprocessing unit and Microcontroller unit products from Renesas. For detailed usage notes on the products covered by this document, refer to the relevant sections of the document as well as any technical updates that have been issued for the products.

1. Precaution against Electrostatic Discharge (ESD)

A strong electrical field, when exposed to a CMOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop the generation of static electricity as much as possible, and quickly dissipate it when it occurs. Environmental control must be adequate. When it is dry, a humidifier should be used. This is recommended to avoid using insulators that can easily build up static electricity.

Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors must be grounded. The operator must also be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions must be taken for printed circuit boards with mounted semiconductor devices.

2. Processing at power-on

The state of the product is undefined at the time when power is supplied. The states of internal circuits in the LSI are indeterminate and the states of register settings and pins are undefined at the time when power is supplied. In a finished product where the reset signal is applied to the external reset pin, the states of pins are not guaranteed from the time when power is supplied until the reset process is completed. In a similar way, the states of pins in a product that is reset by an on-chip power-on reset function are not guaranteed from the time when power is supplied until the power reaches the level at which resetting is specified.

3. Input of signal during power-off state

Do not input signals or an I/O pull-up power supply while the device is powered off. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Follow the guideline for input signal during power-off state as described in your product documentation.

4. Handling of unused pins

Handle unused pins in accordance with the directions given under handling of unused pins in the manual. The input pins of CMOS products are generally in the high-impedance state. In operation with an unused pin in the open-circuit state, extra electromagnetic noise is induced in the vicinity of the LSI, an associated shoot-through current flows internally, and malfunctions occur due to the false recognition of the pin state as an input signal become possible.

5. Clock signals

After applying a reset, only release the reset line after the operating clock signal becomes stable. When switching the clock signal during program execution, wait until the target clock signal is stabilized. When the clock signal is generated with an external resonator or from an external oscillator during a reset, ensure that the reset line is only released after full stabilization of the clock signal. Additionally, when switching to a clock signal produced with an external resonator or by an external oscillator while program execution is in progress, wait until the target clock signal is stable.

6. Voltage application waveform at input pin

Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between V_{IL} (Max.) and V_{IH} (Min.) due to noise, for example, the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between V_{IL} (Max.) and V_{IH} (Min.).

7. Prohibition of access to reserved addresses

Access to reserved addresses is prohibited. The reserved addresses are provided for possible future expansion of functions. Do not access these addresses as the correct operation of the LSI is not guaranteed.

8. Differences between products

Before changing from one product to another, for example to a product with a different part number, confirm that the change will not lead to problems. The characteristics of a microprocessing unit or microcontroller unit products in the same group but having a different part number might differ in terms of internal memory capacity, layout pattern, and other factors, which can affect the ranges of electrical characteristics, such as characteristic values, operating margins, immunity to noise, and amount of radiated noise. When changing to a product with a different part number, implement a system-evaluation test for the given product.

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